

Thank you for purchasing this Esco Laminar Flow Clean Bench. Please read this manual thoroughly to familiarize yourself with the many unique features and exciting innovations we have built into your new equipment. Esco provides many other resources at our website, www.escoglobal.com, to complement this manual and help you enjoy many years of productive and safe use of your Esco products.



User & Service Manual

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Esco Airstream® Gen 3 Laminar Flow Clean Bench
(LHG, LHS, LVG, LVS)
User & Service Manual Version F – Released July 2019

Airstream® Gen 3

Horizontal & Vertical Laminar Flow
Clean Benches

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MANUAL REVISION HISTORY

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Warranty Terms and Conditions

Esco products come with a limited warranty. The warranty period will vary depending on the product purchased, beginning on the date of shipment from any Esco international warehousing location. To determine which warranty applies to your product, refer to the appendix below.

Esco's limited warranty covers defects in materials and workmanship. Esco's liability under this limited warranty shall be, at our option, to repair or replace any defective parts of the equipment, provided if proven to the satisfaction of Esco that these parts were defective at the time of being sold, and that all defective parts shall be returned, properly identified with a Return Authorization.

This limited warranty covers parts only, and not transportation/insurance charges.

This limited warranty does not cover:

- Freight or installation (inside delivery handling) damage. If your product was damaged in transit, you must file a claim directly with the freight carrier.
- Products with missing or defaced serial numbers.
- Products for which Esco has not received payment.
- Problems that result from:
 - External causes such as accident, abuse, misuse, problems with electrical power, improper operating environmental conditions.
 - Servicing not authorized by Esco.
 - Usage that is not in accordance with product instructions.
 - Failure to follow the product instructions.
 - Failure to perform preventive maintenance.
 - Problems caused by using accessories, parts, or components not supplied by Esco.
 - Damage by fire, floods, or acts of God.
 - Customer modifications to the product
- Consumables
- Esco is not liable for any damage incurred on the objects used on or stored in Esco equipment. If the objects are highly valuable, user is advised to have in place independent external preventive measures such as connection to a centralized alarm system.

Factory installed, customer specified equipment or accessories are warranted only to the extent guaranteed by the original manufacturer. The customer agrees that in relation to these products purchased through Esco, our limited warranty shall not apply and the original manufacturer's warranty shall be the sole warranty in respect of these products. The customer shall utilize that warranty for the support of such products and in any event not look to Esco for such warranty support.

Esco encourages all users to register their equipment online at www.escoglobal.com/warranty or complete the warranty registration form included with each product.

ALL EXPRESS AND IMPLIED WARRANTIES FOR THE PRODUCT, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES AND CONDITIONS OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE LIMITED IN TIME TO THE TERM OF THIS LIMITED WARRANTY. NO WARRANTIES, WHETHER EXPRESS OR IMPLIED, WILL APPLY AFTER THE LIMITED WARRANTY PERIOD HAS EXPIRED. ESCO DOES NOT ACCEPT LIABILITY BEYOND THE REMEDIES PROVIDED FOR IN THIS LIMITED WARRANTY OR FOR SPECIAL, INDIRECT, CONSEQUENTIAL OR INCIDENTAL DAMAGES, INCLUDING, WITHOUT LIMITATION, ANY LIABILITY FOR THIRD-PARTY CLAIMS AGAINST YOU FOR DAMAGES, FOR PRODUCTS NOT BEING AVAILABLE FOR USE, OR FOR LOST WORK. ESCO'S LIABILITY WILL BE NO MORE THAN THE AMOUNT YOU PAID FOR THE PRODUCT THAT IS THE SUBJECT OF A CLAIM. THIS IS THE MAXIMUM AMOUNT FOR WHICH ESCO IS RESPONSIBLE.

These Terms and Conditions shall be governed by and construed in accordance with the laws of Singapore and shall be subject to the exclusive jurisdiction of the courts of Singapore.

Technical Support, Warranty Service Contacts

Toll-free USA and Canada: 1-877-479-3726

Singapore: +65 65420833

Global Email Helpdesk: support@escoglobal.com

Visit <http://www.escoglobal.com/> to talk to a Live Support Representative

Distributors are encouraged to visit the Distributor Intranet for self-help materials.

Product Appendix, Warranty Listings

Laminar Flow Cabinets, ULPA-Filtered Cabinets (except Streamline® brand)	The warranty periods for LFC may vary by country. Contact your local distributor for specific warranty details.
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The warranty period starts two months from the date your equipment is shipped from Esco facility for international distributors. This allows shipping time so the warranty will go into effect at approximately the same time the equipment is delivered to the user. The warranty protection extends to any subsequent owner during the warranty period. Distributors who stock Esco equipment are allowed an additional four months for delivery and installation, providing the product is registered with Esco. User can register product online at www.escoglobal.com/warranty or complete the warranty registration form included with each product.

Policy updated on 1st January 2015 (This limited warranty policy applies to products purchased on and after 1st January 2015)

Introduction

1. Products Covered

Esco Vertical Laminar Flow Clean Benches						
Model	Electrical Rating	0.9 m (3 ft.)	1.2 m (4 ft.)	1.5 m (5 ft.)	1.8 m (6 ft.)	2.4 m (8 ft.)
Glass Side Wall	230 V AC, 50/60Hz, 1Φ	LVG-3AG-F8	LVG-4AG-F8 LVG-4CG-F8 LVG-4AG-S8	LVG-5AG-F8	LVG-6AG-F8 LVG-6CG-F8 LVG-6AG-S8	
	115 V AC, 50/60Hz, 1Φ	LVG-3AG-F9 LVG-3AS-F9	LVG-4AG-F9 LVG-4CG-F9 LVG-4AS-F9 LVG-4AG-S9	LVG-5AG-F9 LVG-5CG-F9 LVG-5AS-F9	LVG-6AG-F9 LVG-6CG-F9 LVG-6AS-F9 LVG-6AG-S9	
Stainless Steel Side Wall	230 V AC, 50/60Hz, 1Φ	LVS-3AG-F8	LVS-4AG-F8 LVS-4CG-F8 LVS-4AG-S8	LVS-5AG-F8	LVS-6AG-F8 LVS-6CG-F8 LVS-6AG-S8	LVS-8CG-F8
	115 V AC, 50/60Hz, 1Φ	LVS-3AG-F9 LVS-3AS-F9	LVS-4AG-F9 LVS-4CG-F9 LVS-4AS-F9 LVS-4CS-F9 LVS-4AG-S9	LVS-5AG-F9 LVS-5CG-F9 LVS-5AS-F9 LVS-5CS-F9	LVS-6AG-F9 LVS-6CG-F9 LVS-6AS-F9 LVS-6CS-F9 LVS-6AG-S9	LVS-8CG-F9 LVS-8CS-F9

Esco Horizontal Laminar Flow Clean Benches						
Model	Electrical Rating	0.9 m (3 ft.)	1.2 m (4 ft.)	1.5 m (5 ft.)	1.8 m (6 ft.)	2.4 m (8 ft.)
Glass Side Wall	230 V AC, 50/60Hz, 1Φ	LHG-3AG-F8 LHG-3BG-F8 LHG-3CG-F8	LHG-4AG-F8 LHG-4BG-F8 LHG-4CG-F8	LHG-5AG-F8 LHG-5BG-F8	LHG-6AG-F8 LHG-6BG-F8 LHG-6CG-F8	
	115 V AC, 50/60Hz, 1Φ	LHG-3AG-F9 LHG-3BG-F9 LHG-3CG-F9 LHG-3BS-F9	LHG-4AG-F9 LHG-4BG-F9 LHG-4CG-F9 LHG-4BS-F9	LHG-5AG-F9 LHG-5BG-F9 LHG-5BS-F9	LHG-6AG-F9 LHG-6BG-F9 LHG-6CG-F9 LHG-6BS-F9	
Stain Steel Side Wall	230 V AC, 50/60Hz, 1Φ	LHS-3AG-F8 LHS-3BG-F8	LHS-4AG-F8 LHS-4BG-F8 LHS-4CG-F8	LHS-5AG-F8 LHS-5BG-F8	LHS-6AG-F8 LHS-6BG-F8 LHS-6CG-F8	LHS-8CG-F8
	115 V AC, 50/60Hz, 1Φ	LHS-3AG-F9 LHS-3BG-F9 LHS-3BS-F9	LHS-4AG-F9 LHS-4BG-F9 LHS-4CG-F9 LHS-4BS-F9 LHS-4CS-F9	LHS-5AG-F9 LHS-5BG-F9 LHS-5BS-F9	LHS-6AG-F9 LHS-6BG-F9 LHS-6CG-F9 LHS-6BS-F9 LHS-6CS-F9	LHS-8CG-F9 LHS-8CS-F9

Guide to Platform Laminar Flow Cabinet Models														
1 st Placeholder	2 nd Placeholder	3 rd Placeholder	4 th Placeholder	5 th Placeholder	6 th Placeholder	7 th Placeholder	8 th Placeholder							
Production Line	Flow		Side Wall		Width		Height		Control		Window		Electrical	
Laminar Flow	L	Horizontal	H	Glass	G	3 Feet	3	2 Feet	A	Switch*	S	Fixed	F	230 VAC 50/60Hz
		Vertical	V	Stainless Steel	S	4 Feet	4	2.5 Feet	B	Sentinel Gold	G	Sliding**	S	115 VAC 50/60Hz
						5 Feet	5	3 Feet	C					
						6 Feet	6							
						7 Feet	7							
						8 Feet	8							
*Switch Control models available only in the USA														
**Sliding Sash window option is only available for Vertical Laminar Flow cabinets with 2 ft. internal height and 4 and 6 ft of width.														

2. Safety Warning

- Anyone working with, on or around this equipment should read this manual. Failure to read, understand and comply with the instructions given in this manual may result in damage to the unit, injury to operating personnel, and/or poor equipment performance.
- Any internal adjustment, modification or maintenance to this equipment must be undertaken by qualified service personnel.
- The use of any hazardous materials in this equipment is prohibited. Laminar flow cabinets are intended for product protection ONLY.
- Explosive or inflammable substances should never be used in the cabinet unless adequate risk assessment has been carried out.
- Before you process, you should thoroughly understand the installation procedures and take note of the environmental/electrical requirements.

- In this manual, important safety-related points will be marked with the symbol.



- If the equipment is used in a manner not specified by this manual, the protection provided by this equipment may be impaired.

3. Document Management

We recommend that you keep this manual, along with the factory test report close to the cabinet for easy reference by the cabinet operator and qualified maintenance personnel.

If you require replacements for any of the provided documentation (including factory test reports) you can request copies from Esco Customer Services*. Please provide the following information when making requests for replacement documents:

- Company (Organization) Name
- Product Brand and Model
- Product Serial Number
- Documents requested

* There may be a nominal charge for this service.

4. Limitation of Liability

The disposal and / or emission of substances used in connection with this equipment may be governed by various local regulations. Familiarization and compliance with any such regulations are the sole responsibility of the users. Esco's liability is limited with respect to user compliance with such regulations.

5. European Union Directive on WEEE and RoHS

The European Union has issued two directives:

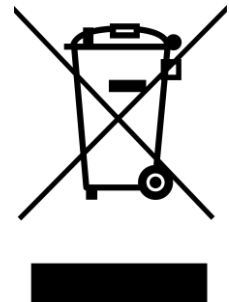
- **Directive 2002/96/EC on Waste Electrical and Electronic Equipment (WEEE)**

This product is required to comply with the European Union's Waste Electrical & Electronic Equipment (WEEE) Directive 2002/96/EC. It is marked with the following symbol:

Esco sells products through distributors throughout Europe. Contact your local Esco distributor for recycling/disposal.

Recommended method of disposal is according to The Federal, State, and Local Government regulations

- **Directive 2002/95/EC on Restriction on the use of Hazardous Substances (RoHS)**



With respect to the directive on RoHS, please note that this hood falls under category 8 (medical devices) and category 9 (monitoring and control instruments) and is therefore exempted from requirement to comply with the provisions of this directive.

6. Symbols

Information in this manual may be prefaced with the following symbols. They are provided to help you identify important operational, safety, maintenance or conformance issues.



Safety First: Important safety reminders



Note: Important reminders and helpful tips



Electrical Hazard: Danger of electric shock



Turn Off and Disconnect From Main Supply Before Proceeding: Do not perform this operation while the unit is operational

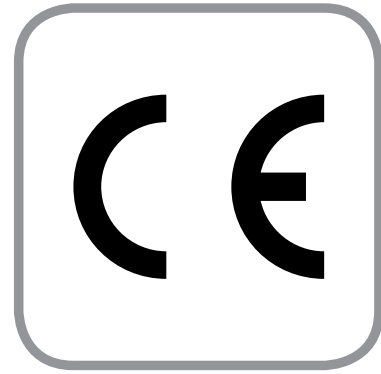


Approved Service Engineer Only: Operation to be performed only by approved engineers

Certification of Conformation

In accordance to EN ISO/IEC 17050-1:2010

We, Esco Micro Pte Ltd
of 21 Changi South Street 1
Singapore, 486777
Tel: +65 6542 0833
Fax: +65 6542 6920



declare on our sole responsibility that the product:

Category : Horizontal Laminar Flow Clean Benches
Brand : Airstream
Model : LHG-3AG-F8, LHS-3AG-F8, LHG-4AG-F8, LHS-4AG-F8, LHG-5AG-F8, LHS-5AG-F8,
LHG-6AG-F8, LHS-6AG-F8, LHG-3BG-F8, LHS-3BG-F8, LHG-4BG-F8, LHS-4BG-F8,
LHG-5BG-F8, LHS-5BG-F8, LHG-6BG-F8, LHS-6BG-F8, LHG-3CG-F8, LHG-4CG-F8,
LHS-4CG-F8, LHG-6CG-F8, LHS-6CG-F8, LHS-8CG-F8

in accordance with the following directives:

2014/35/EU : The Low Voltage Directive and its amending directives
2014/30/EU : The Electromagnetic Compatibility Directive and its amending directives
2011/65/EU : The RoHS in Electrical and Electronic Equipment Directive and its amending directives

has been designed to comply with the requirement of the following Harmonized Standard:

Low Voltage : EN 61010-1:2010
EMC : EN 61326-1:2006 Class B

More information may be obtained from Esco's authorized distributors located within the European Union. A list of these parties and their contact information is available on request from Esco.

A handwritten signature in black ink that reads 'Lim Lay Yew'.

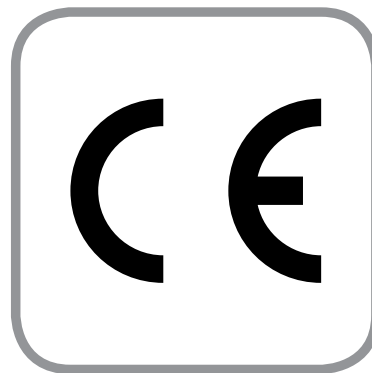
Lim Lay Yew
Director, Esco

This Declaration of Conformity is only applicable for 230V AC 50 Hz units

Certification of Conformation

In accordance to EN ISO/IEC 17050-1:2010

We, Esco Micro Pte Ltd
of 21 Changi South Street 1
Singapore, 486777
Tel: +65 6542 0833
Fax: +65 6542 6920



declare on our sole responsibility that the product:

Category : Vertical Laminar Flow Clean Benches
Brand : Airstream
Model : LVG-3AG-F8, LVS-3AG-F8, LVG-4AG-F8, LVS-4AG-F8, LVG-5AG-F8, LVS-5AG-F8,
LVG-6AG-F8, LVS-6AG-F8, LVG-4CG-F8, LVS-4CG-F8, LVG-6CG-F8, LVS-6CG-F8,
LVS-8CG-F8, LVG-4AG-S8, LVS-4AG-S8, LVG-6AG-S8, LVS-6AG-S8

in accordance with the following directives:

2014/35/EU : The Low Voltage Directive and its amending directives
2014/30/EU : The Electromagnetic Compatibility Directive and its amending directives
2011/65/EU : The RoHS in Electrical and Electronic Equipment Directive and its amending directives

has been designed to comply with the requirement of the following Harmonized Standard:

Low Voltage : EN 61010-1:2010
EMC : EN 61326-1:2006 Class B

More information may be obtained from Esco's authorized distributors located within the European Union. A list of these parties and their contact information is available on request from Esco.

A handwritten signature in black ink that reads 'Lim Lay Yew'.

Lim Lay Yew
Director, Esco

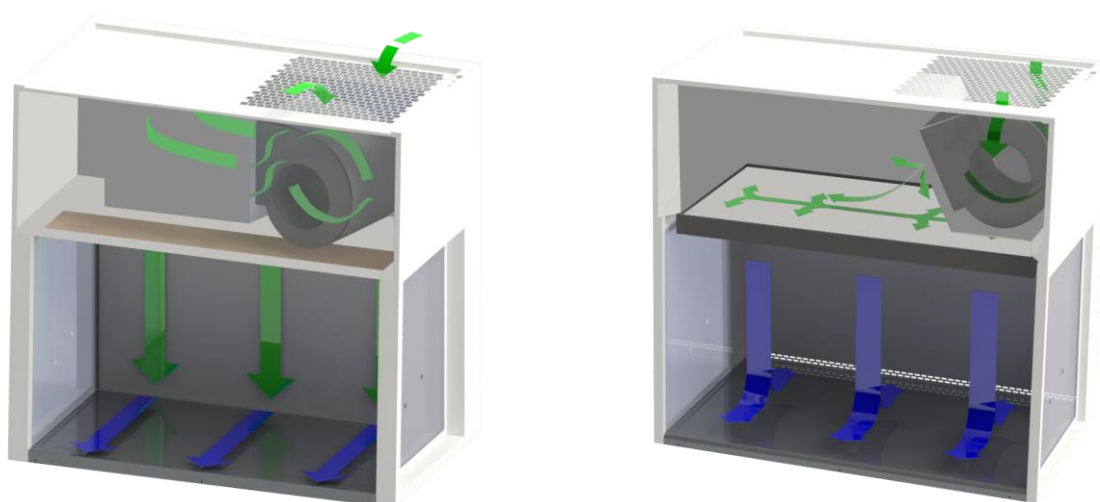
This Declaration of Conformity is only applicable for 230V AC 50 Hz unit

Chapter 1 – Product Information

1.1 About Laminar Flow Clean Benches (LFC)

Laminar flow clean benches or cabinets provide a controlled and clean environment inside a laboratory. It is a type of unidirectional clean air device which provides a higher quality of air to critical areas where products or processes are open to contamination. Since this can only protect products or processes, it is not applicable and should NEVER be used with hazardous samples.

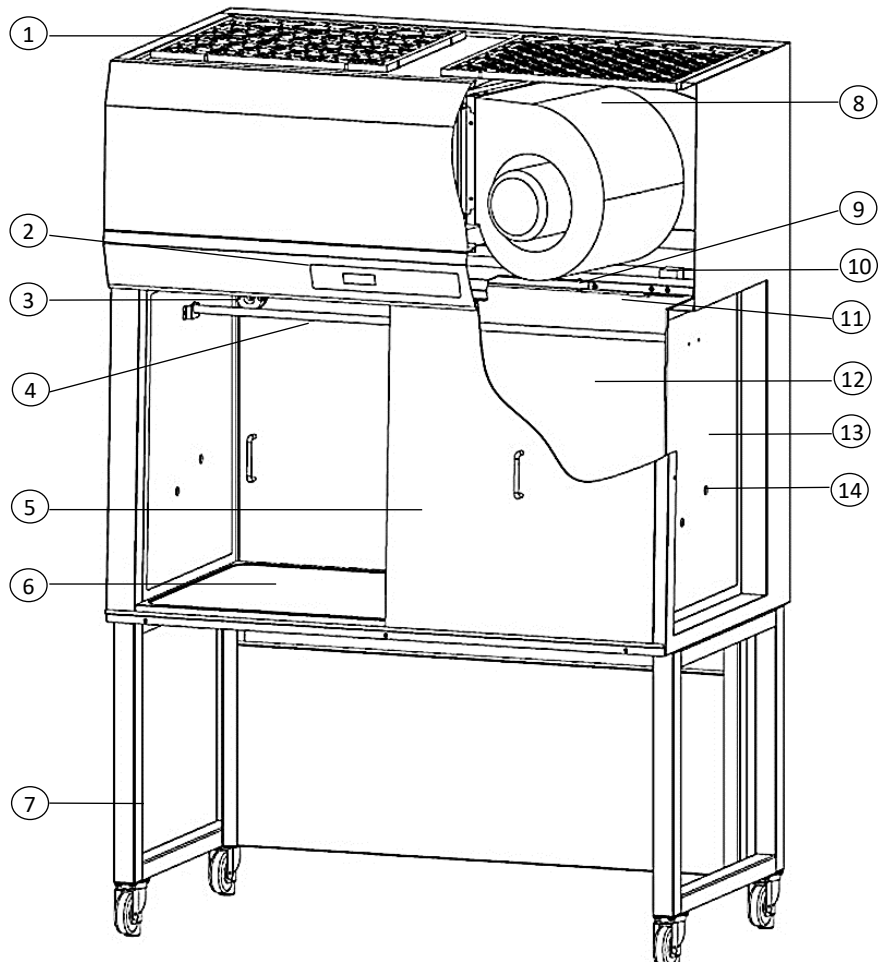
Clean benches can be classified into two types, depending on the direction of the laminar air flow: Horizontal and Vertical. Horizontal-type clean benches have a horizontal air flow, with the HEPA filter located at the back of the cabinet. Vertical-type clean benches have a vertical air flow, striking the work surface and the HEPA filter located at the top part of the cabinet.



Direction of air flow inside a horizontal (left diagram) and vertical (right diagram) clean bench.

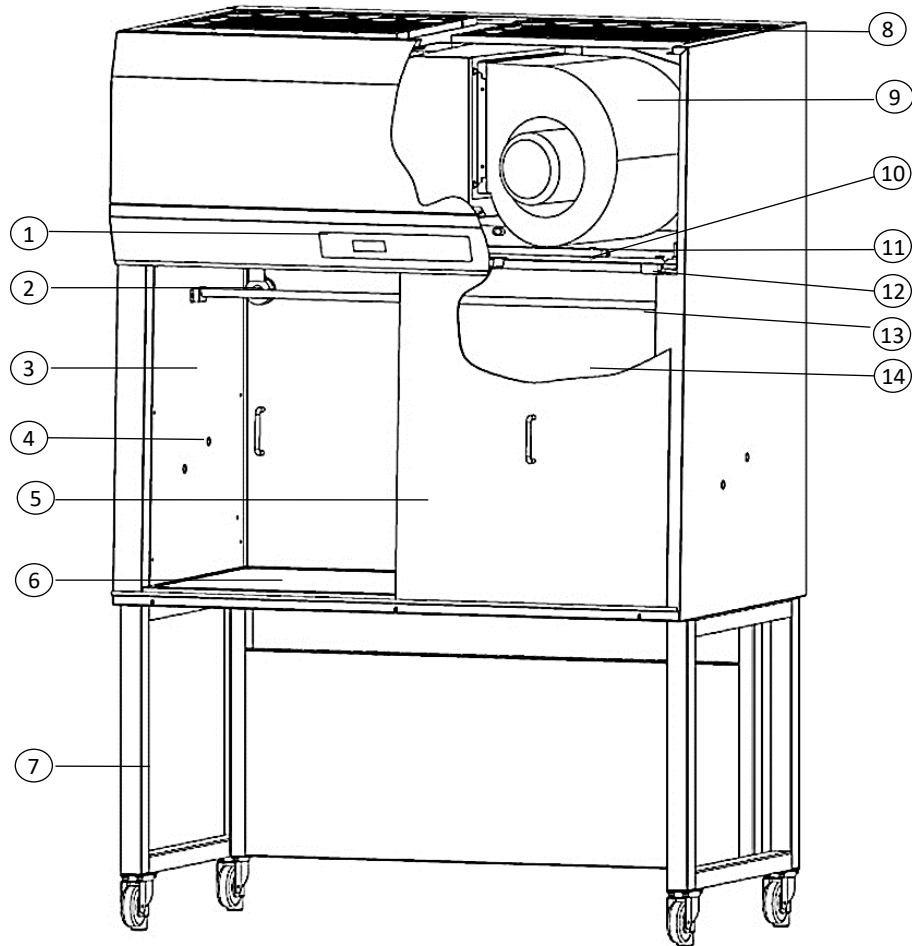
1.2 Quick View

1.2.1 Horizontal Laminar Flow Clean Bench with Glass Side Wall (LHG)



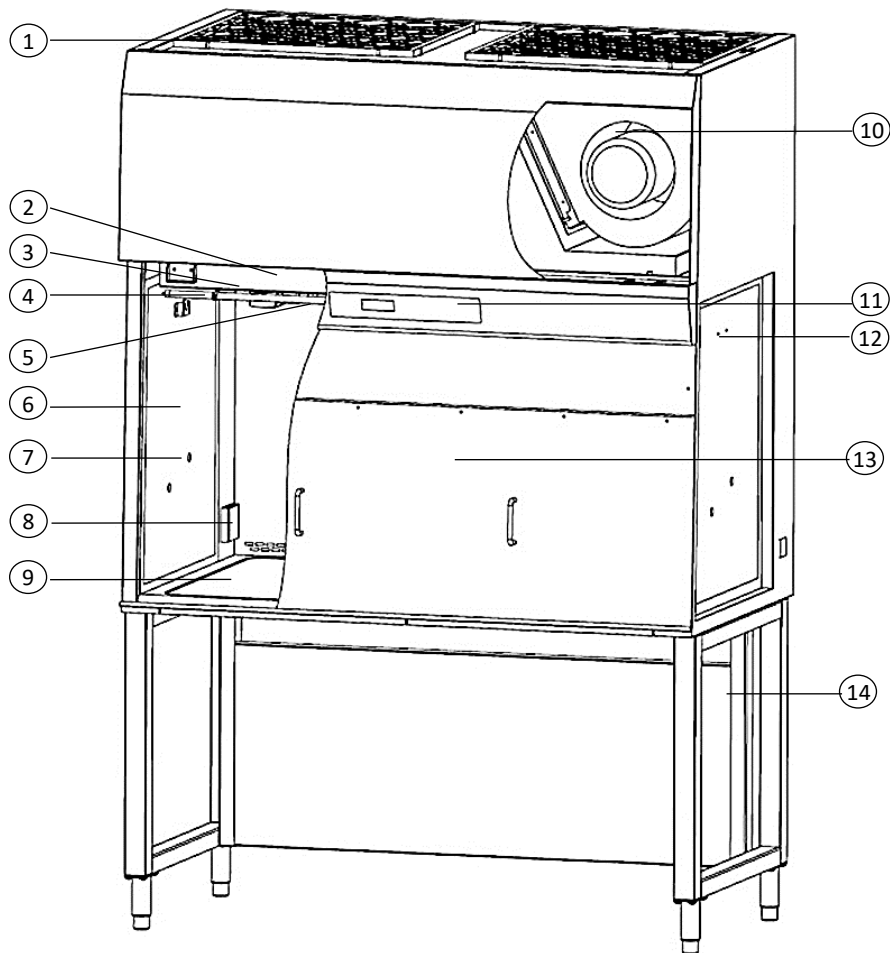
- | | |
|--|--|
| 1. Pre-filter | 8. ECM Blower |
| 2. Control System | 9. Fluorescent Lamp |
| 3. Donut-shaped Airflow Sensor | 10. Optional Electrical Outlet |
| 4. IV Bar Retrofit Kit Provision | 11. UV Light Retrofit Kit Provision |
| 5. Optional Front Cover for UV Protection | 12. ULPA Filter |
| 6. Spill-retaining Stainless Steel Worktop | 13. Transparent Tempered Glass Side Panels |
| 7. Support Stand | 14. Service Fixture Retrofit Kit Provision |

1.2.2 Horizontal Laminar Flow Clean Bench with Stainless Steel Side Wall (LHS)



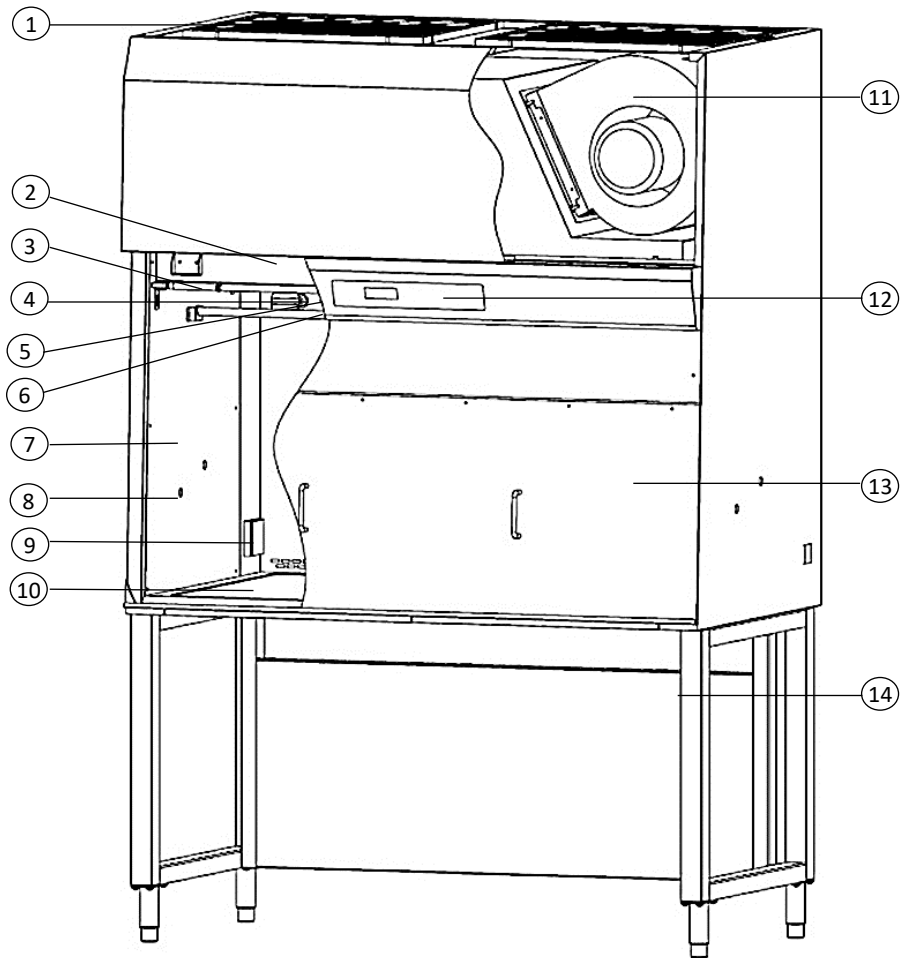
- | | |
|--|-------------------------------------|
| 1. Control System | 8. Pre-filter |
| 2. Donut-shaped Airflow Sensor | 9. ECM Blower |
| 3. Stainless Steel Side Panels | 10. Fluorescent Lamp |
| 4. Service Fixture Retrofit Kit Provision | 11. Optional Electrical Outlet |
| 5. Optional Front Cover for UV Protection | 12. UV Light Retrofit Kit Provision |
| 6. Spill-retaining Stainless Steel Worktop | 13. IV Bar Retrofit Kit Provision |
| 7. Support Stand | 14. ULPA Filter |

1.2.3 Vertical Laminar Flow Clean Bench with Glass Side Wall (LVG)



- | | |
|---|--|
| 1. Pre-filter | 8. Optional Electrical Outlet |
| 2. ULPA Filter | 9. Spill-retaining Stainless Steel Worktop |
| 3. Donut-Shaped Airflow Sensor | 10. ECM Blower |
| 4. Fluorescent Lamp | 11. Control System |
| 5. UV Light Retrofit Kit Provision | 12. IV Bar Retrofit Kit Provision |
| 6. Transparent Tempered Glass Side Panels | 13. (Optional) Front Cover for
Fixed Sash (LV_-F variant) /
(Optional) Manual Sliding Sash (LV_-S variant) |
| 7. Service Fixture Retrofit Kit Provision | 14. Support Stand |

1.2.4 Vertical Laminar Flow Clean Bench with Stainless Steel Side Wall (LVS)



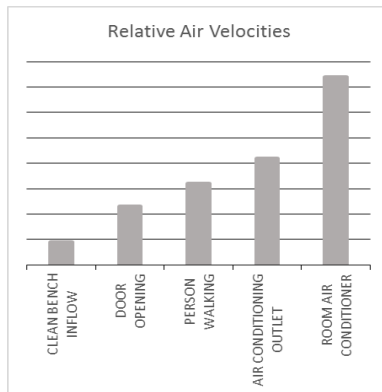
- | | |
|------------------------------------|--|
| 1. Pre-filter | 8. Service Fixture Retrofit Kit Provision |
| 2. ULPA Filter | 9. Optional Electrical Outlet |
| 3. Fluorescent Lamp | 10. Spill-retaining Stainless Steel Worktop |
| 4. Donut-Shaped Airflow Sensor | 11. ECM Blower |
| 5. UV Light Retrofit Kit Provision | 12. Sentinel Control System |
| 6. Stainless Steel Side Panels | 13. (Optional) Front Cover for Fixed Sash (LV_-F variant) / (Optional) Manual Sliding Sash (LV_-S variant) |
| 7. IV Bar Retrofit Kit Provision | 14. Support Stand |

Chapter 2 – Installation

2.1 General Requirement

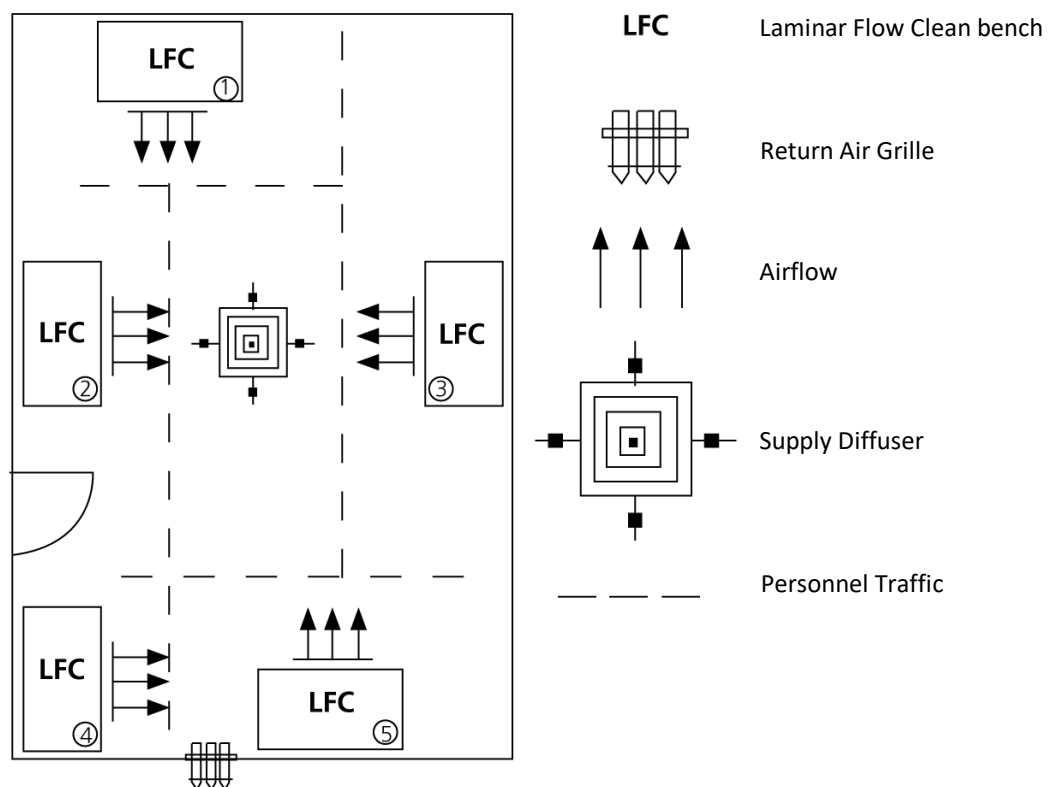
2.1.1 Location Requirements

The clean bench needs to be sited in a location that does not compromise the performance of the unit.



As seen in the chart, your cabinet's internal airflow velocity is relatively low compared to the airflow disturbances potentially caused by opening a door, a person walking by or a direct exposure to an air-conditioning outlet. These external airflow disturbances can affect the containment of the clean bench. Therefore, the clean bench should be located as far away as possible from sources of airflow disturbance and in an orientation which optimally shields the cabinet's airflow from all external airflow disturbances. There should be adequate SOP in place to minimize events that will affect the performance of the cabinets.

Diagram below illustrates various possible influences that can be induced by a room's design and ventilation system on the laminar flow clean bench's airflow. Please note that diagram below does not depict any typical installation. In fact it is NOT recommended that so many clean benches be placed in a small room, or in close proximity to each other.



1. LFC 1 is quite appropriately located with respect to avoidance of excessive air movements from surrounding areas.
2. LFC 2 is too close to the doorway and its airflow could be influenced by the supply diffuser.
3. The airflow of LFC 3 could also be influenced by the supply diffuser.
4. LFC 4 is too close to the doorway.
5. LFC 5 is suitably located provided that the adjacent return air grille does not influence clean bench airflow.

2.2 Preparing For Installation

2.2.1 Environmental Requirements

- Indoor use only.
- Altitude of up to 2,000 meter (6,600 feet).
- Relative humidity between 20% – 90%.
- Main supply voltage fluctuations up to $\pm$ of the nominal voltage.
- Transient voltages up to levels of Overvoltage Category II.
- Temperature between 18°C – 30°C (65°F – 86°F).
- Pollution Degree 2.0

Pollution degree describes the amount of conductive pollutants present in an operating environment. In pollution degree 2.0, it is assumed that only non-conductive pollutants such as dust are present, except when occasional conductivity caused by condensation.

2.2.2 Electrical Requirements

- The cabinet should be connected to its own dedicated power outlet.
- The power rating for each model is shown in below. Ensure that the outlet is rated accordingly.

Model		Electrical Rating
LHG-3__-F8 LVG-3__-F8	LHS-3__-F8 LVS-3__-F8	230 V AC, 50/60 Hz, 1Ø, 4.7 A
LHG-3__-F9 LVG-3__-F9	LHS-3__-F9 LVS-3__-F9	115 V AC, 50/60 Hz, 1Ø, 7.8 A
LHG-4__-F8 LVG-4__-F8 LVG-4__-S8	LHS-4__-F8 LVS-4__-F8 LVS-4__-S8	230 V AC, 50/60 Hz, 1Ø, 5.5 A
LHG-4__-F9 LVG-4__-F9	LHS-4__-F9 LVS-4__-F9	115 V AC, 50/60 Hz, 1Ø, 8.5 A
LHG-5__-F8 LVG-5__-F8	LHS-5__-F8 LVS-5__-F8	230 V AC, 50/60 Hz, 1Ø, 5.8 A
LHG-5__-F9 LVG-5__-F9	LHS-5__-F9 LVS-5__-F9	115 V AC, 50/60 Hz, 1Ø, 8.7 A
LHG-6__-F8 LVG-6__-F8 LVG-6__-S8	LHS-6__-F8 LVS-6__-F8 LVS-6__-S8	230 V AC, 50/60 Hz, 1Ø, 6 A
LHG-6__-F9 LVG-6__-F9	LHS-6__-F9 LVS-6__-F9	115 V AC, 50/60 Hz, 1Ø, 9.2 A
---	LHS-8__-F8 LVS-8__-F8	230 V AC, 50/60 Hz, 1Ø, 11 A
---	LHS-8__-F9 LVS-8__-F9	115 V AC, 50/60 Hz, 1Ø, 20 A

- The power cable is located on the right hand side of the cabinet and is 3 m long. When preparing the installation site, try to ensure that the outlet is located to the right of the cabinet for ease of access.
- The cabinet's maximum voltage fluctuation is $\pm 2\%$ of the nominal voltage. Therefore, where voltage fluctuation is higher, equipment such as voltage regulator or UPS with appropriate features is recommended.
- Surge protection and UPS are strongly recommended for better protection. Uninterruptible Power Supply (UPS) with power stabilization function could also be used to eliminate or minimize the voltage fluctuation seen by the cabinet. Where UPS is installed, it is recommended that the UPS is sized to enable the clean bench to operate for about 20 minutes to provide ample time for the personnel to react to the power failure.



A reliable protective earth connection is recommended for better operation and safety.

2.3 Unpacking and Moving your Cabinet

1. Check the packing labels and delivery note before unpacking to ensure the correct consignment has been delivered.
2. Put all packing materials to one side and retain. You may need them to repack the unit if anything untoward is discovered with the unit during installation.
3. Cut the retaining straps and remove the stand (if purchased). This will require at least two persons.
4. Remove the shrink wrap from the outside of the main carton and dispose of carefully.
5. If you have not yet done so, remove the retaining straps. Then remove the top of the packing carton.
6. Remove the cardboard outer from the unit by sliding it up and over the unit.
7. Remove the shrink wrap from around the polystyrene panels and dispose of carefully.
8. Remove all of the polystyrene, cardboard and Styrofoam packing materials and place with the outer carton.

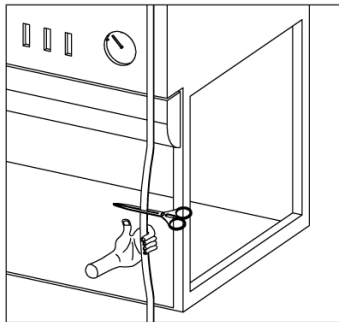


Figure 2.3.1. Removing the strapping

9. Cut the inner shipping straps and remove the large protective plastic bag.
10. There is a protector on the top of the unit (cardboard sheet), remove this before activating the unit.
11. Take out the carton containing the manuals, toolkit and retrofit kits and place safely to one side.
12. Remove any available work area shipping supports.
13. Remove the protective plastic coating on the stainless steel work top.
14. The cabinet may be fastened to the skids on the shipping pallet with two braces – one on each side. If so, unbolt the cabinet from the skids and remove the braces.
15. At this point you should assemble the stand as per the instructions provided.
16. Using a mechanical lifting truck, the cabinet should be positioned over the stand, the stand offered up to the cabinet and the bolts fastened. After the stand is fastened to the cabinet, lower the assembly to the ground slowly.

17. Check that all bolts have been fastened adequately and then move the cabinet to its final position. The side panel should not be used to lift the cabinet as it is not a structural component. Please lift by means of the back edge and corners.

2.3.1 Safety and Warning Labels on the Cabinet



Anyone using the clean bench should familiarize themselves with the various labels displayed in and on the cabinet. It is very important that users are familiar with the meanings of the labels before attempting to use the unit.

2.3.2 Preliminary Cleaning

Wipe the interior and exterior of the clean bench with water or a mild household detergent. The compatibility of the cleaning agent should be verified.



When the cabinet has been used for work, other suitable interior cleaning and disinfection method should be applied.

2.4 Performance Validation/Certification

After having installed the cabinet but before starting to use it, cabinet performance must be validated and certified to factory standards. It is recommended that this validation and certification be performed only by a qualified personnel who is familiar with the methods and procedures for certifying laminar flow clean benches.

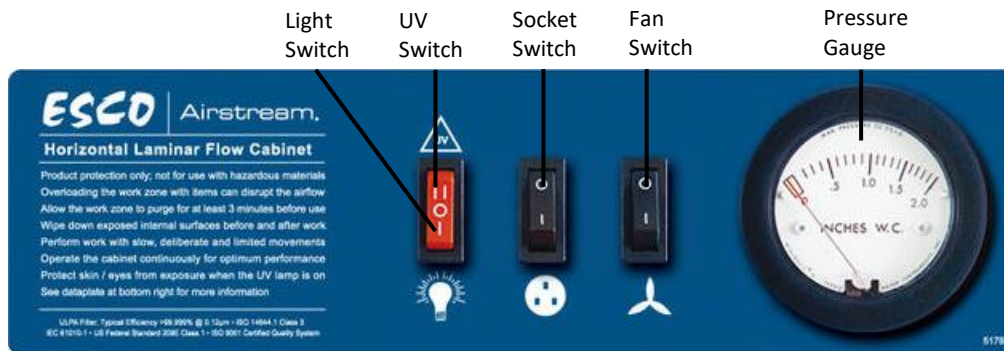
The testing methods and equipment needed for carrying out the tests are specified on the test report accompanying your cabinet.

2.4.1 Disclaimer

The performance and safety of all Esco clean bench are rigorously evaluated at our factory. Regular field certification is important to ensure factory standards are maintained.

Chapter 3 – Switch Control System

3.1 Switch Control System



Fan Switch

- Turns the fan on or off.

Light Switch

- Turns the fluorescent light on or off.

Socket Switch

- Turns the power outlets in the clean bench on or off (retrofit kit)

UV Light Switch

- Turns the UV lamp in the clean bench on or off (retrofit kit)
 - The UV lamp can only be activated when both the fan and light are off.
- Please install the front cover (retrofit kit) if you want to use this feature. Since the front cover is capable of filtering UV rays, users are protected from the harmful UV rays.



- Eyes and skin should not be exposed to direct UV light.
- UV light should not be relied upon as the sole decontaminating agent.
- Check the UV interlock regularly for correct operation.

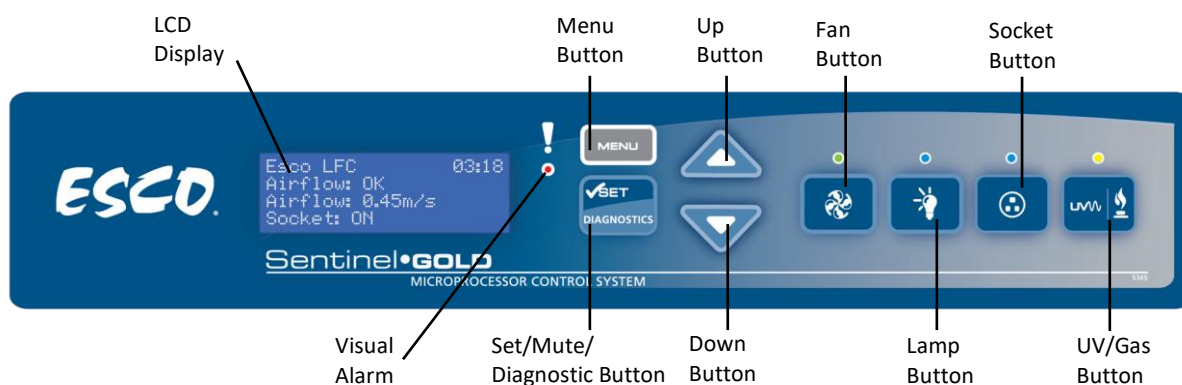
Note: The use of UV lamp voids cULus listing.

Pressure Gauge

- Indicates the pressure drop across the filter, to maintain nominal airflow velocity as specified inside the test report. The gauge is calibrated in *inches of water column* (inch WC) pressure. The initial pressure reading will be approximately 0.50 ± 0.05 inch WC depending on altitude from sea level. The clean bench should be checked for proper airflow at each 0.1-inch increase in static pressure.

Chapter 4 – Sentinel™ Control System

4.1 Sentinel™ Gold Control System



1. Fan Button
 - Turns the fan on or off.
 - To activate standby mode.
2. Lamp Button
 - Turns the fluorescent lamps on or off.
3. Socket Button
 - Turns on or off the electrical socket (*retrofit kit*).
 - The maximum rating of all the outlets in the cabinet is 5 A. If there is an overload, the fuse will blow.
4. UV/Gas Button
 - Turns on or off the UV lamp (*optional retrofit kit*).
 - Please install front cover (*retrofit kit*) if you want to use this feature. Since the front cover is capable of filtering UV rays, users are protected from the harmful UV rays.
5. Up (▲) and Down (▼) Arrow Button
 - Moves the menu options upwards and downwards.
 - Increases and decreases corresponding value inside one of the menu options.
 - For accessing the stopwatch and experiment timer function.
6. Set/Mute/Diagnostic Button
 - To proceed to the next step, level or sequence inside the menu options.
 - Mutes the air fail alarm sound (during normal mode).
 - Enters diagnostic mode.
7. Menu Button

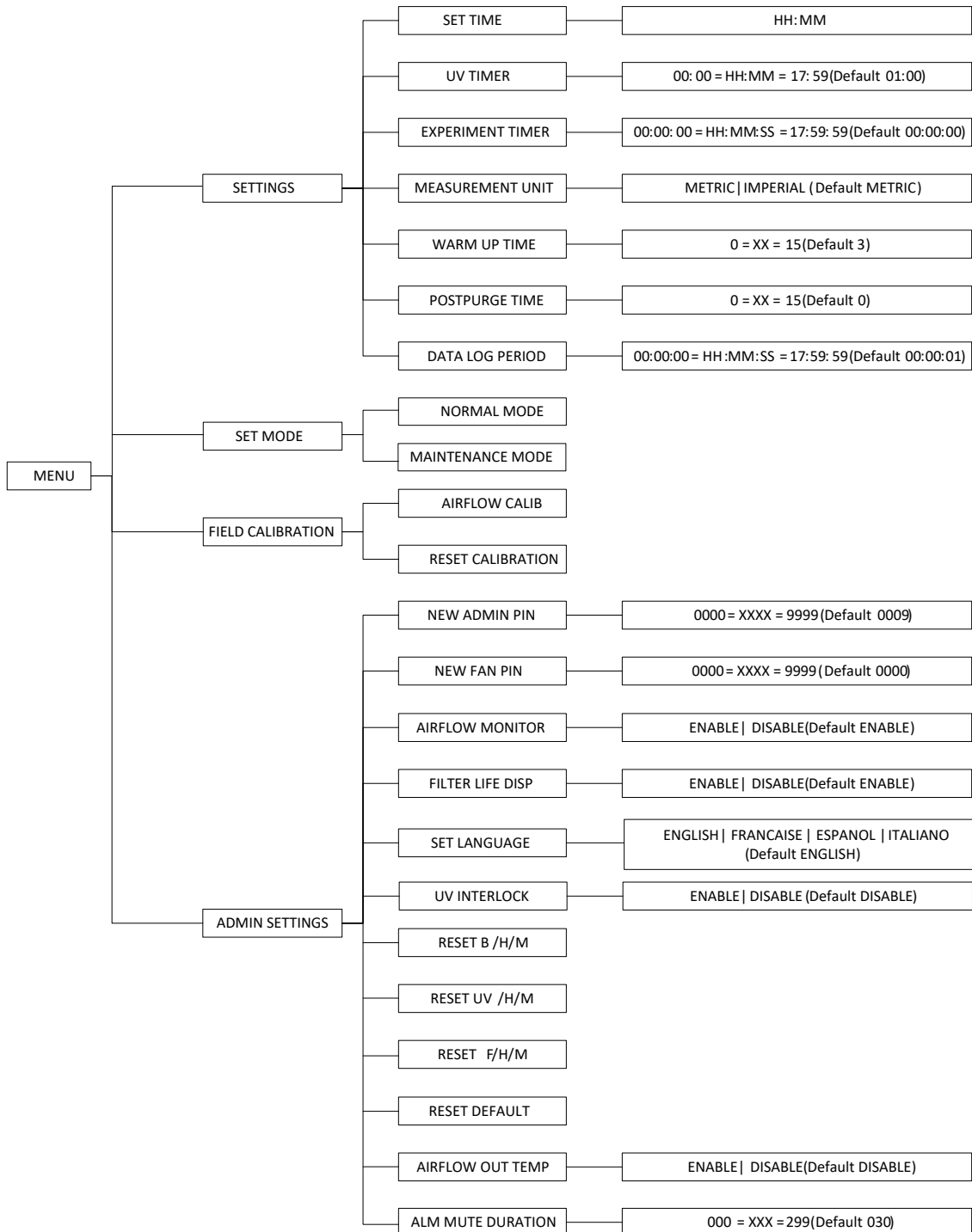


When you are entering menu options, the alarm will sound to indicate that the microprocessor is not monitoring the operation of the cabinet. No further warnings will be given.

- To enter and exit from the menu options.
- To go back to the previous level of the menu options.
- To access maintenance mode from error condition.

4.2 Menu Options

Please refer to the following diagram for complete reference to all menu options available.



4.2.1 Settings

Users may use the settings menu function to customize the operation of the LFC to meet specific application requirements. The settings menu can be entered using either FAN PIN or ADMIN PIN.

4.2.1.1 Set Clock (Time)

Users can set the time by increasing/decreasing the hour and minute values. The correct time will be maintained even after the unit is turned off.

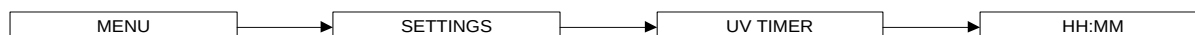


To Set the Time:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose SET TIME. Press SET button to confirm.
6. The time is set in HH:MM format. Use UP/DOWN buttons to choose the hour (HH). Press SET button to confirm. Do the same for the minute
7. The display will show TIME SET for a few second and then return to SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.1.2 UV Timer (If UV is present)

UV timer can be used to switch off the UV lamp automatically after a fixed period. The UV timer can be set up to 18 hours. By default, the timer is set to 60 minutes. Esco does not recommend leaving the UV lamp on for more than 60 minutes per decontamination cycle as it shortens the lifespan of the UV lamp. Unless the UV timer is activated, the lamp has to be switched off manually.



To Set the UV Timer:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose UV TIMER. Press SET button to confirm.
6. The time is set in HH:MM format. Use UP/DOWN buttons to choose the hour (HH). Press SET button to confirm. Do the same for the minute.
7. The display will show UV TIMER SET for a few second and then return to SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.1.3 Experiment Timer

Experiment timer is a countdown timer that can be used for critical experiment. Experiment timer can be set between “00:00:00” and “17:59:59”.



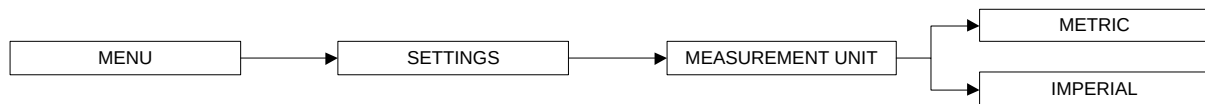
To Set the Experiment Timer:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.

3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose EXPERIMENT TIMER. Press SET button to confirm.
6. The time is set in HH:MM:SS format. Use UP/DOWN buttons to choose the hour (HH). Press SET button to confirm. Do the same for the minute and second.
7. The display will show EXPERIMENT TIMER SET for a few second and then return to SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.1.4 Measurement Unit

Using this option, the user can select the unit in which air velocity is measured and displayed. The user can choose between metric (m/s) and imperial (fpm) units.



To Set Measurement Unit:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose MEASUREMENT UNIT. Press SET button to confirm.
6. Use UP/DOWN buttons to choose between METRIC and IMPERIAL. Press SET button to confirm.
7. The display will return to SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.1.5 Warm Up Time

There will be a warm up period before the LFC is fully functioning upon activation of the unit. This is to ensure that the sensors, the blower, and the control system are stabilized, as well as to ensure the work zone is purged of contaminants. The default setting is 3 minutes and the user can set it between 3 to 15 minutes.

During the warm up period, the user can use the FAN button to turn off the blower, LIGHT button to turn on and off the fluorescent lamp and MENU button. However, to be able to access the menu, the user needs to input ADMIN PIN and even then, some sections of the menu (WARM UP and all FIELD CALIBRATION) are still not accessible for the user. Entering the menu during this time will put the warm up period on pause.



To Set Warm Up Time:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose WARM UP TIME. Press SET button to confirm.
6. The time is set in MM format. Use UP/DOWN buttons to set the warm-up period. Press SET button to confirm.
7. The display will show WARM UP SET for a few second and then return to SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.1.6 Post Purge Time

After the user switches off the LFC blower, there will be a post-purge period, to ensure that all contaminants are purged from the work zone. The default setting is zero minute (disabled) and user can set from 0 up to 15 minutes. It is recommended that LFC is purged for a minimum of 3 minutes after the work is complete.



To Set Post Purge Time:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose POSTPURGE TIME. Press SET button to confirm.
6. The time is set in MM format. Use UP/DOWN buttons to set the post purge period. Press SET button to confirm.
7. The display will show POST PURGE SET for a few second and then return to SETTINGS
8. Press MENU button twice to return to the main display.

4.2.1.7 Data Log Period

Using RS232 communication port, the LFC can send data on the condition of the LFC to a PC. Data Log Period option allows the user to control the amount of time within the LFC data sending procedure.

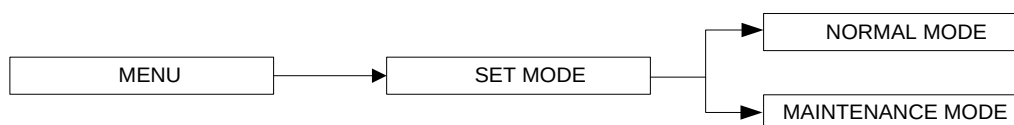


To Set Data Log Period:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose DATA LOG PERIOD. Press SET button to confirm.
6. The time is set in HH:MM:SS format. Use UP/DOWN buttons to choose the hour (HH). Press SET button to confirm. Do the same for the minute and second.
7. The display will show DATA LOG PERIOD SET for a few second and then return to SETTINGS
8. Press MENU button twice to return to the main display.

4.2.2 Setting Mode

The LFC has normal mode can be used in daily activity, and maintenance mode is for the use of qualified personnel during maintenance.



To Set the Mode:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN and ADMIN PIN, then it will ask for the appropriate PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.

4. Use UP/DOWN buttons to choose SET MODE. Press SET button to confirm.
5. Use UP/DOWN buttons to choose between NORMAL MODE, QUICKSTART MODE and MAINTENANCE MODE. Press SET button to confirm.
6. The display will return to SET MODE.
7. Press MENU button twice to return to the main display.

4.2.2.1 Normal Mode

The normal mode is activated by factory default. In this mode, all alarms and interlocks are enabled.

4.2.2.2 Maintenance Mode

Maintenance mode should only be accessed by qualified personnel during maintenance. In this mode, all alarms are disabled and all interlocks are defeated.

4.2.3 Field Calibration



The purpose of calibration is to ensure the accuracy of the airflow display and alarm (if present). This involves measuring airflow with reference instrumentation and establishing reference between airflow sensor(s) on the LFC to the standard reference. Calibration should only be carried out by qualified personnel. This section presents a brief overview of the calibration menu function. For more information, refer to test report.



4.2.3.1 Airflow Calibration

This option allows proper calibration and operation of the airflow sensor alarm. There will be two points to be calibrated, namely airflow fail point and airflow nominal point.

4.2.3.2 Reset Calibration

This option allows the user to reset all values calibrated in the field and return it to the values obtained during factory calibration.

4.2.4 Admin Settings

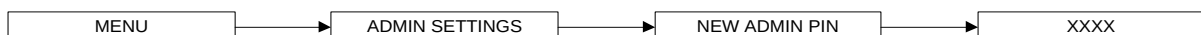
The admin menu allows you to change both FAN and ADMIN PIN. The reset blower, filter and UV hour meter (if present) functions are usually used after the blower, filter or UV lamp is changed as they can easily serve as an indication to the user on when the LFC needs maintenance. The reset default function will return the options in the settings menu to their factory settings.

4.2.4.1 New ADMIN PIN (Default 0009)

ADMIN PIN restricts access to some of the more delicate menu functions, namely admin and field calibration, which should only be accessed by qualified personnel. User must enter a four-digit ADMIN PIN before accessing these menus.



ADMIN PIN can also be used to switch to maintenance mode from error condition.



To Set New ADMIN PIN:

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.

4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose NEW ADMIN PIN. Press SET button to confirm.
6. The FAN PIN consists of 4 numerical digits (XXXX) that the user must enter one by one. Use UP/DOWN buttons to choose the first digit (X). Press SET button to confirm. Do the same for the following three digits
7. The display will show CONFIRM PIN? Press SET button to confirm.
8. The display will return to ADMIN SETTINGS.
9. Press MENU button twice to return to the main display.

4.2.4.2 New FAN PIN (Default 0000 - DISABLED)

FAN PIN restricts access to fan control and some parts of the menu, settings and set mode. User must enter the four-digit PIN before switching the fan on or off. This feature prevents unauthorized personnel from accessing critical control sections. It will also prevent unauthorized shutdown of the LFC when continuous operation is required.

It is recommended that the FAN PIN be issued only to personnel authorized to use the LFC. With FAN PIN, the user can access admin and set mode parts of the menu.

Setting the PIN to 0000 will disable this feature. The FAN PIN is disabled by default. When the FAN PIN is disabled, the LFC can be turned on and off without requiring PIN. However, to access the menu, the user is still required to enter the FAN PIN (0000).



To Set New FAN PIN:

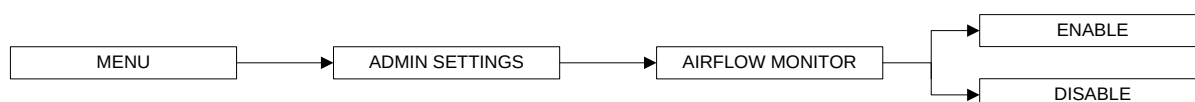
1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose NEW FAN PIN. Press SET button to confirm.
6. The FAN PIN consists of 4 numerical digits (XXXX) that the user must enter one by one. Use UP/DOWN buttons to choose the first digit (X). Press SET button to confirm. Do the same for the following three digits
7. The display will show CONFIRM PIN? Press SET button to confirm.
8. The display will return to ADMIN SETTINGS.
9. Press MENU button twice to return to the main display.

4.2.4.3 Airflow Monitor

Whenever the air velocity falls below the fail point, the air fail alarm will be triggered. This option is used to enable/disable alarm. The alarm is enabled by default.

When the Airflow Monitor is disabled, the warm up period is removed but the airflow will not be displayed for the first three minutes.

If the ambient temperature is outside of 18-30°C (cabinet working temperature), the Airflow Monitor is automatically disabled.



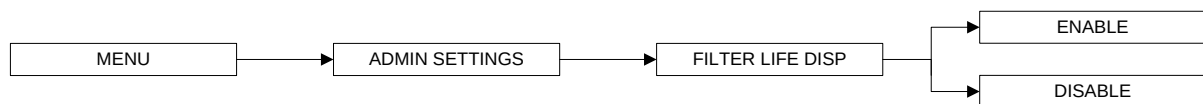
To Set A/F Monitor:

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose AIRFLOW MONITOR. Press SET button to confirm.
6. Use UP/DOWN buttons to choose between ENABLE and DISABLE. Press SET button to confirm.
7. The display will return to ADMIN SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.4.4 Filter Life Display

Using this option, the user can select whether the filter life is displayed or not.

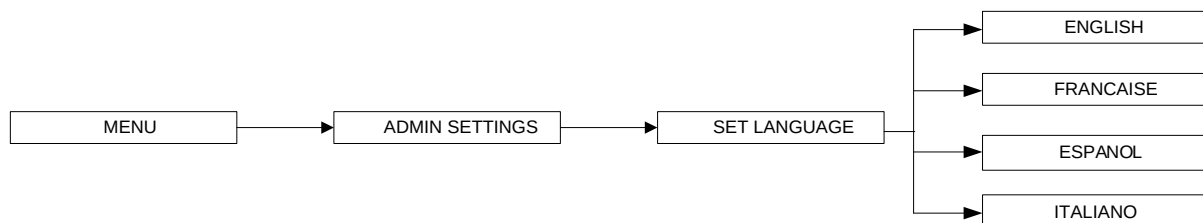
Filter life is calculated based on the filter hour meter (F/H/M). The filter life display will count down according to the amount of hours left in the filter hour meter with respect to filter life expectancy of 10,000 hours. When the filter is changed, the F/H/M must be reset (*please see section 4.2.4.9 to reset the F/H/M*). Please note that the life of the filter is dependent on multiple factors which include environmental air cleanliness. A dirty / dusty environment will load the filter fast.

**To Set Filter Life Display:**

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose FILTER LIFE DISP. Press SET button to confirm.
6. Use UP/DOWN buttons to choose between ENABLE and DISABLE. Press SET button to confirm.
7. The display will return to ADMIN SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.4.5 Set Language

Using this option, the user can select the language of messages displayed on the LCD.



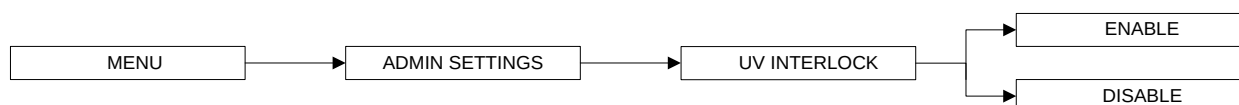
Airstream® Gen 3 Laminar Flow Clean Bench comes with English as the default language but user has an option to choose from Italian, Spanish or French language. This option must be specified when ordering the cabinet.

To Set Language:

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose SET LANGUAGE. Press SET button to confirm.
6. Use UP/DOWN buttons to choose between ENGLISH, FRANÇAISE or ESPAÑOL (depending on the option the user has requested). Press SET button to confirm.
7. The display will return to ADMIN SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.4.6 UV Interlock

When the UV Interlock is enabled, this means that the UV can't be switched on when the front cover is not installed. When it is disabled, the UV can be switched on without the front cover (*only if the fan and light are switched off*). After the UV is switched on, the fan and light can't be switched on. By default this option is disabled.

**To Set UV Interlock:**

1. Press MENU button to enter the menu display – if the LFC is secured by a FAN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the FAN PIN or ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose UV INTERLOCK. Press SET button to confirm.
6. Use UP/DOWN buttons to ENABLE or DISABLE. Press SET button to confirm.
7. The display will return to ADMIN SETTINGS.
8. Press MENU button twice to return to the main display.



It is recommended that the UV Interlock be enabled

4.2.4.7 Reset B/H/M

This option is used to reset the blower hour meter. The blower hour meter indicates how long the blower has been in operation. There is no maximum value in blower hour meter. The counter value can be checked in the diagnostic mode. The value can also provide some help in setting up maintenance schedule.

**To Reset B/H/M:**

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose RESET B/H/M. Press SET button to confirm.
6. The display will show READ MANUAL and PRESS SET. Press SET button to confirm.

7. The display will show CONFIRM?. Press SET button to confirm.
8. The display will return to ADMIN SETTINGS.
9. Press MENU button twice to return to the main display.

4.2.4.8 Reset UV/H/M (If UV is present)

This option is used to reset the UV lamp hour meter. The UV lamp hour meter indicates how long the UV lamp has been in operation. Maximum counter is set at 2,000 hours (100%). The counter value can be checked while in the diagnostic mode.



Please reset the UV lamp hour meter after each UV lamp replacement.



To Reset UV/H/M:

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose RESET UV/H/M. Press SET button to confirm.
6. The display will show READ MANUAL and PRESS SET. Press SET button to confirm.
7. The display will show CONFIRM?. Press SET button to confirm.
8. The display will return to ADMIN SETTINGS.
9. Press MENU button twice to return to the main display.

4.2.4.9 Reset F/H/M

This option is used to reset the filter hour meter. The filter hour meter indicates how long the filter has been in operation. Maximum counter is set at 10,000 hours (100%). The counter value can be checked while in the diagnostic mode.



Please reset the filter hour meter after each filter replacement.



To Reset F/H/M:

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose RESET F/H/M. Press SET button to confirm.
6. The display will show READ MANUAL and PRESS SET. Press SET button to confirm.
7. The display will show CONFIRM?. Press SET button to confirm.
8. The display will return to ADMIN SETTINGS.
9. Press MENU button twice to return to the main display.

4.2.4.10 Reset Default

User can reset default settings by choosing this option. The features being reset are warm up period (3 minutes), post-purge period (0 minute), UV timer (60 minutes) if present, measurement unit (metric), airflow monitor (enabled), ADMIN PIN (0009), filter life display (disabled) and FAN PIN (0000).

Note that the calibration settings cannot be reset as it may cause the LFC to operate in an unsafe manner. The hour meters cannot be reset using this function either.

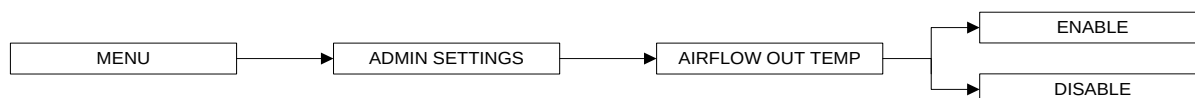


To Reset Default:

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose RESET DEFAULT. Press SET button to confirm.
6. The display will show READ MANUAL and PRESS SET. Press SET button to confirm.
7. The display will show CONFIRM?. Press SET button to confirm.
8. The display will return to ADMIN SETTINGS.
9. Press MENU button twice to return to the main display.

4.2.4.11 Airflow out Temp

Using this option, the user can select whether or not to display the airflow when the ambient temperature is out of the optimum temperature range, below 18°C (65°F) or above 30°C (86°F).

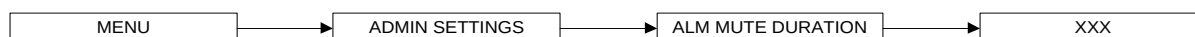


To Set Airflow Out Temp:

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose AIRFLOW OUT TEMP. Press SET button to confirm.
6. Use UP/DOWN buttons to choose between ENABLE and DISABLE. Press SET button to confirm.
7. The display will return to ADMIN SETTINGS.
8. Press MENU button twice to return to the main display.

4.2.4.12 Alarm Mute Duration

To mute *Airfail* alarm for a certain period. The mute period can be set from 0 up to 299 seconds; the default value is 30 seconds. Alarm will be activated when the airflow velocity is below the value prescribed by the standard the cabinet is designed or certified to.



To Set Alarm Mute Duration:

1. Press MENU button to enter the menu display – if the LFC is secured by an ADMIN PIN, then it will ask for the PIN, otherwise go to step 3.
2. Use UP/DOWN button to enter the ADMIN PIN digit by digit. Press SET button to confirm.
3. The alarm buzzer will sound.
4. Use UP/DOWN buttons to choose ADMIN SETTINGS. Press SET button to confirm.
5. Use UP/DOWN buttons to choose ALM MUTE DURATION. Press SET button to confirm.
6. Use UP/DOWN buttons to choose the mute period. Press SET button to confirm.
7. The display will show MUTE DURATION SET for 2 seconds and return to ADMIN SETTINGS.
8. Press MENU button twice to return to the main display.

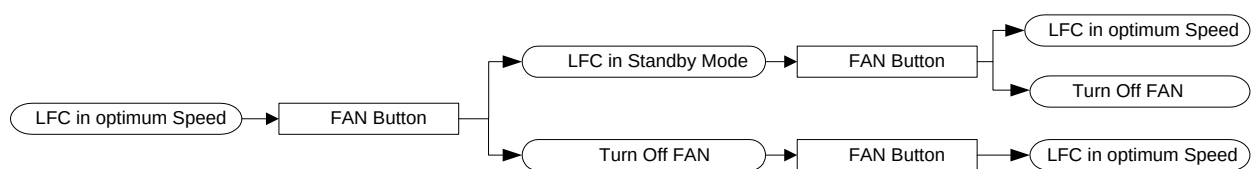
4.3 Stopwatch and Experiment Timer

- The stopwatch function can be started by pressing the UP button while the cabinet is in operation. Pressing UP button again while the stopwatch function is activated will stop or resume the timer. Use the DOWN button to exit the stopwatch function and reset the timer. The timer in the stopwatch function is counting up and is shown using the HH:MM:SS format.
- The experiment timer can be started by pressing the DOWN button while the cabinet is in operation. Pressing the DOWN button while the experiment timer function is activated will stop or resume the timer. Use the UP button to exit the experiment timer function and reset the timer. The timer in the experiment timer function is counting down and is shown using the HH:MM:SS format. Operator can use the SETTINGS → EXPERIMENT TIMER menu (*refer to section 4.2.1.3*) to set the experiment timer.

4.4 Standby Mode

The LFC supports standby mode, where the blower speed is greatly reduced resulting in less power consumption. This mode is typically used during the night to maintain basic level of containment where the cabinet is not used by the operator.

The mode can be accessed by pressing the FAN button when the cabinet is in operation. In standby mode, the airflow monitor is disabled and only the FAN button is operational, while other buttons are interlocked.

**4.5 Alarms and Warnings**

A LFC uses alarms to indicate that the condition inside the LFC is not safe for the products, so check the LCD display to understand the cause of these alarms.

Other alarms that indicate a failure or an error in the LFC system:

- **AIRFLOW: NO!** Will be displayed if there is an airflow failure.
- **SENSOR UNCALIBRATED** will be displayed if the airflow velocity sensor is not yet calibrated.
- **WARNING: LOW AIRFLOW!** when the velocity is below fail point airflow



If the message "Call Service for re-certification" is displayed, it means that the LFC certification has expired. Call service or Esco's local distributor for re-certification.

4.6 Diagnostic Mode

Diagnostic mode can be accessed by pressing the SET button. The diagnostic mode allows the user to know the condition of the LFC or help the service engineer during maintenance and troubleshooting.

On Screen	Explanation
MODE	Shows which mode is active: NORMAL MODE or MAINTENANCE MODE
VERSION	Shows the version of the software; <i>e.g.</i> CP104D V 1.0
TEMPERATURE	Shows the temperature inside the cabinet.
B/H/M	Blower Hour Meter – increase by the hour.
FILTER LIFE	Shows percentage of filter life (based on Filter Hour Meter) and expected filter life of 10000 hours.
AF OUT TEMP	Velocity display status when optimum temperature is out of range
UV INTERLOCK	UV interlock status enabled or disabled.
UV LIFE	Shows percentage of UV lamp life (based on UV Lamp Hour Meter).
UV TIMER	Shows the UV timer value – default is 60 minutes. Maximum value is 00 minutes (infinite on).
MUTE TIMER	Shows the mute timer value – default is 30 seconds. Maximum value is 299 seconds.
ADC AFF	ADC for Fail Point Airflow – calculated using offset based on Airflow Nominal Point.
ADC AFN	ADC for Nominal Point Airflow – based on field calibration.
ADC AFA	ADC for Actual Airflow – showing real time sensor reading.
ADC AF0	ADC for factory calibrated Zero Point Airflow (no airflow).
ADC AF1	ADC for factory calibrated Fail Point Airflow.
ADC AF2	ADC for factory calibrated Nominal Point Airflow.
CONSTANT	Airflow sensor constant. This value is needed when ordering a new sensor.
CALIB TEMP	Temperature when the factory calibration was performed.
ADC TEMP	ADC value for TEMPERATURE.
M_SWITCH1	Shows the condition of magnetic switch 1 – front cover

Chapter 5 – Basic Cabinet Operation

5.1 Working in the Clean Bench

1. Thoroughly surface-decontaminate the work surface, side walls and back wall using 70% ethanol (or some other disinfectant depending on materials used in the clean bench). Surface-decontaminate the UV lamp and electrical outlets as well. Do not use any disinfectant containing chlorine-based substances as this may cause corrosion of the stainless steel surfaces.
2. Minimize room activity (personnel movements, closing and opening of doors, etc.) since these external airflow disturbances may adversely affect the clean bench's airflow, thereby possibly impairing optimum sample protection.
3. Allow only essential items in the work zone. Items/instruments should not be placed between the filter and any area where the clean environment is needed. New items introduced into the work zone should be placed downstream of items already in the work zone for several minutes to allow contaminants to flush off.
4. Work in the clean bench in a slow and controlled manner. While putting items inside / removing items from the work zone, move your hands in and out of the work zone opening slowly and in a direction perpendicular to the plane of the work zone opening. Rapid movement of arms in a sweeping motion may disrupt the air barrier, thereby allowing contaminants to enter the clean bench.
5. Exercise particular care when placing equipment within the work space. Where possible, instruments should be placed on perforated platforms to allow air movement under as well as around the object.
6. Perform all work with the operator's hand or head downstream of the critical process points. Keep unnecessary movement within the work zone to a minimum.
7. Do not use Bunsen burner and aerosol-generating instruments whenever possible as they interfere with airflow.
8. Wipe surfaces clean by applying appropriate disinfecting solution to clean bench surface. Most surface disinfectants require a specific contact time. After the specified contact time, wipe the excess disinfectant.
9. Leave the unit running for 5-30 minutes before continuing work to allow the system to purge itself.

5.2 Working Ergonomics

On most occasions, you would most likely be operating the clean bench in a sitting rather than a standing position. There are some obvious advantages of the sitting position:

- The physiological energy cost and fatigue involved in sitting are relatively less
- Sitting position provides the body with a stable support

However, the sitting position has some drawbacks too:

- The working area available is fairly limited
- There is a potential risk of being constrained in the same posture for a long time
- Sitting posture is one of the most stressful postures for one's back



Therefore you should pay careful attention to the following guidelines in order to achieve comfortable and healthy working conditions:

- Always ensure that your legs have enough legroom.
- Keep your lower back comfortably supported by your chair. Adjust the chair or use a pillow behind your back whenever necessary.
- You should place your feet flat on the floor or on a footrest. Don't dangle your feet and compress your thighs.
- You should keep varying your sitting position throughout the day at regular intervals so that you are never in the same posture for too long.
- Observe the following precautions with respect to your eyes:
 - Give your eyes frequent breaks. Periodically look away from the work area and focus at a distant point.

- Keep your glasses clean.
- Arrange the items/apparatus frequently used in your work in such a way that you can minimize the physical strain involved in handling them.
- Exercise regularly

The clean bench's noise emission has been tested and found to be in compliance with EN 12469, ISO 4871 and NSF/ANSI 49 which is important to ensure health and comfort for the operator.

Ergonomic accessories available with Esco include:

- Armrest padding
- Lab chair
- Footrest

Please contact your local distributor or Esco for more information.

5.3 UV Lamps

Shortwave UV (UVC) with wavelength at 253.7 nm is considered as germicidal and virucidal. Even at the minimum acceptable irradiance in a clean bench – 40 $\mu\text{W}/\text{cm}^2$ (US Department of Health and Human Services et. al., 2000), it would only take 12.5 minutes to reach 30,000 $\mu\text{W}/\text{cm}^2$ (1 W = 1 J/sec), which has been listed as germicidal for spore forming organism. Unlike many other type of decontamination agent, UV light doesn't leave any residue. The decontamination action stops upon de-energizing of the lamp.

However, due to the short wavelength, the UV light does not penetrate well. Thus UV light can only be used to effectively disinfect the work area of an empty clean bench. For any container stored inside the clean bench's work area, the UV radiation will only disinfect the outer surface of the material, leaving the inner surface and the content inside the material untouched.

- UV light decontamination method may be used before and after working with vegetative organisms and viruses. However, it should not be the sole decontamination agent; the use of chemical decontamination agent is still encouraged.
- Minimize the material inside the clean bench's work area during the process of UV light decontamination. A direct interaction with UV light can degenerate plastic- or rubber-based material and can cause other hazard.



Make sure that front cover is placed and the interlock is working properly before activating the UV lamp. Avoid direct contact with skin and eyes as UV light can cause direct eye damage and erythema of skin.

- The UV lamp should be activated for around 60 minutes to work effectively. Use the UV timer feature to easily control the period of decontamination (UV timer is disabled by default). Leaving the UV lamp on for over 60 minutes or even overnight is not recommended because it shortens the lifespan of the lamp. The UV lamps used in Esco clean bench have a lifespan of 2,000 hours.
- The UV lamp should be cleaned of any dust and dirt weekly and changed annually to ensure its effectiveness.



The use of UV lamp in clean benches have been explicitly discouraged in all major international standards and recommendations. You can read more on this in our technical paper:

http://www.escoqlobal.com/resources/pdf/white-papers/UV_lamps.pdf

Chapter 6 – Maintenance

6.1 Scheduled Maintenance

Proper and timely maintenance is crucial for a trouble-free operation of any device and your Esco clean bench is no exception to this rule. We strongly recommend that you follow the maintenance schedule suggested hereunder in order to obtain optimal performance from your Esco clean bench.

No.	Description of Task to Perform	Maintenance to be carried out every					
		Day	Week	Month	Quarter	1 Year	2 Years
1	Surface decontaminate the work zone	√					
2	Clean the exterior surfaces of the laminar hood			√			
3	Check all service fixtures (where present) for proper operation			√			
4	Replace Pre-Filter				√		
5	Inspect the work bench for any abnormalities or malfunction				√		
6	Clean up stainless steel surfaces with MEK				√		
7	Re-certification					√	
8	Change UV Lamp (where present)					√	
9	Change the fluorescent lamps						√

6.1.1 Cleaning Procedures

- Clean the work surface and walls with appropriate disinfectant agent and soapy water.
- Use a damp cloth to clean the exterior surface of the clean bench, particularly on the front and top in order to remove dust that accumulated there.
- Use clean water to finish the cleaning and wash away any residue of disinfectant agent, soapy water and glass cleaner.
- For removing stubborn stains or spots on the stainless steel surface, make use of MEK (Methyl-Ethyl-Ketone). In such cases, make sure that you wash the steel surface immediately afterwards with clean water and some liquid detergent. Use a polyurethane cloth or sponge for washing. Regularly cleaning the stainless steel surface can help you retain the attractive factory finish.

6.1.2 Check the cabinet's functionality



- Check the clean bench's electrical functionality; *e.g.* fluorescent lamp – replace if necessary
- Check the clean bench for any defect, repair immediately 

6.1.3 Re-certification



*All clean benches must be re-certified annually by a certified engineer.
See test report for re-certification procedure.*

6.1 Maintenance/Service Log

It is good practice (and in some cases regulatory requirement) to maintain a log of all maintenance work carried out on your cabinet.

Chapter 7 – Re-Certification by Service Personnel

Esco products generally provide years of trouble-free operation. However, like any other equipment, they require maintenance and service. Maintenance and service should be carried out by trained personnel. Esco offers training courses to equip service providers with the latest skills, information and tools to successfully maintain and service Esco products. For more information on the nearest training course, please contact Esco.

Service providers should familiarize themselves with the basic operating principles of products before working on them. Suitable references include information in the User Section of this Manual or brochures which may be downloaded from the Esco web site.

Laminar Flow Clean Benches generally require:

- **Re-certification, when:**

- The clean bench is re-located
- Decline of clean bench performance is suspected
- After filter or blower replacement
- At least once a year

During recertification:

- Clean bench airflow velocities and flow patterns are verified against the manufacturer's specifications and relevant international standards
- Filters are scan-tested to ensure they do not leak
- Operator comfort tests may be performed
- If airflow velocities are found to be off the setpoint, adjustments are made as part of the certification process before final values are recorded.

- **Airflow alarm calibration, when:**

- The cause of the airflow alarm cannot be determined
- Re-certification indicates the displayed airflow deviates by $>0.02\text{m/s}$ (4fpm) from actual measured velocities. *(Only for products with airflow velocity displays)*

- **Filter replacement, when:**

- The filters are clogged and the fan(s) is/are already adjusted to maximum setting
- Filter leaks which cannot be repaired are found during scan-testing

- Fan replacement (typically rare) if failure occurs.

- **Routine maintenance also includes:**

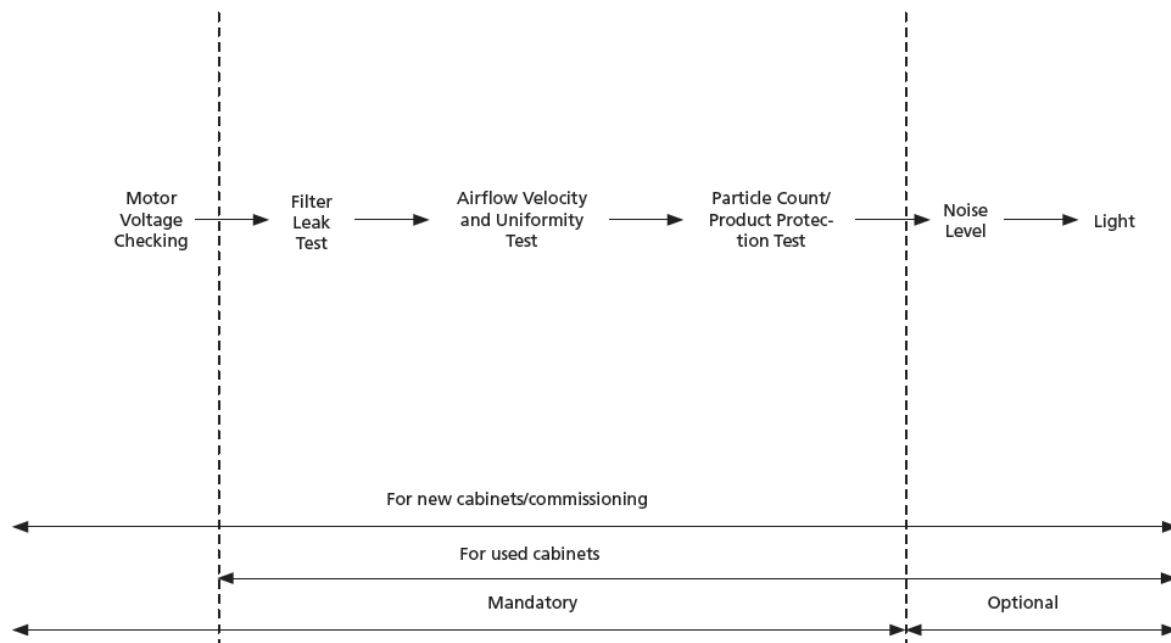
- Fluorescent lamp(s) replacement - typically once every 2 years.
- UV lamp replacement - typically once every 1 year.

7.1 Cabinet Field Calibration

Cabinet field certification must be performed on:

- New installation.
- Relocation of cabinet.
- Activation of airflow alarm with undetermined cause.
- When cabinet performance is suspect.
- At least once in a year or based on your risk assessment.

The steps to perform field certification:



7.2 Performing Certification

7.2.1 Setting the Cabinet to Maintenance Mode

To start the certification, ensure that the fan and lamps are turned on. The cabinet should then be put to maintenance mode. In Maintenance Mode all alarms are defeated and the window can be moved without the light being switched off.

1. Press MENU button. Enter the ADMIN PIN (Default ADMIN PIN is 0009)
2. If SETTINGS is already displayed, the ADMIN code has been removed. Proceed to step 3
 - a. Press SET with 0 flashing
 - b. Press SET with 00 last digit flashing
 - c. Press SET with 000 last digit flashing
 - d. Press SET with 0000 last digit flashing
 - e. Press (▼) button to key in 9
 - f. Press SET

- Alarm will sound, wait until the alarm stops. Cabinet should read.

MENU OPTIONS	10:27
→ SETTINGS	
SET MODE	
FIELD CALIB	

- Use the (▼) button until the LCD displays SET MODE. Press the SET button and LCD would show :

MENU OPTIONS	10:27
→ NORMAL MODE	
MAINTENANCE MODE	

- Use the (▼) button until the LCD displays MAINTENANCE message at the topmost line of the LCD. Press the SET button. The cabinet is now in maintenance mode and the word "MAINTENANCE" will blink continuously to indicate unsafe condition.
- Press the MENU button twice to exit from the menu tree.
- The word "MAINTENANCE" is displayed on the LCD as shown below.

MAINTENANCE	10:28
Airflow: 0.45 m/s	

OR

MAINTENANCE	10:28
Airflow: 90 fpm	

7.2.2 Locating the Speed Controller

Prepare for certification by locating the ECM speed controller (ECMS) which you would need to use for changing the motor speed.

- Remove the pre-filter on top of cabinet.
- The speed control is located on electrical panel below the mesh guard.
- Use a screwdriver to adjust the speed controller.
- Set back the pre-filter after adjusting the speed controller.

7.2.3 HEPA/ULPA Filter PAO Leak Test

The filter(s) should be challenged with PAO aerosol of fixed concentration (23 psi). The aerosol should be evenly distributed throughout the supply cabinet plenum. An aerosol photometer should be used to monitor aerosol penetration downstream of filter(s) and to scan the presence of leaks. Place the probe at distance of 0.5 to 1 cm from the area to be scanned, and move the probe with traverse speed of 5 cm/sec (2 in/sec) along the filter media. When scanning the filter media, move the photometer probe with slightly overlapping manner to ensure leak detection (if present).

Use the following table to assess the filter performance:

PAO Concentration	_____ µg/L	Aerosol concentration calculative method $= \frac{13,500 \times N}{TotalAirflowCfm}$ See next table for the total airflow.
Airflow Filter		
Leaks detected in media	Yes / No	
Leaks detected in gasket	Yes / No	
Particle penetration	_____ %	

Where, N = Number of Laskin nozzle(s) opened

ACCEPTANCE CRITERIA: Maximum particle penetration of **0.01%** for filter(s); no filter(s) leaks present.

RESULT: **PASS / FAIL**



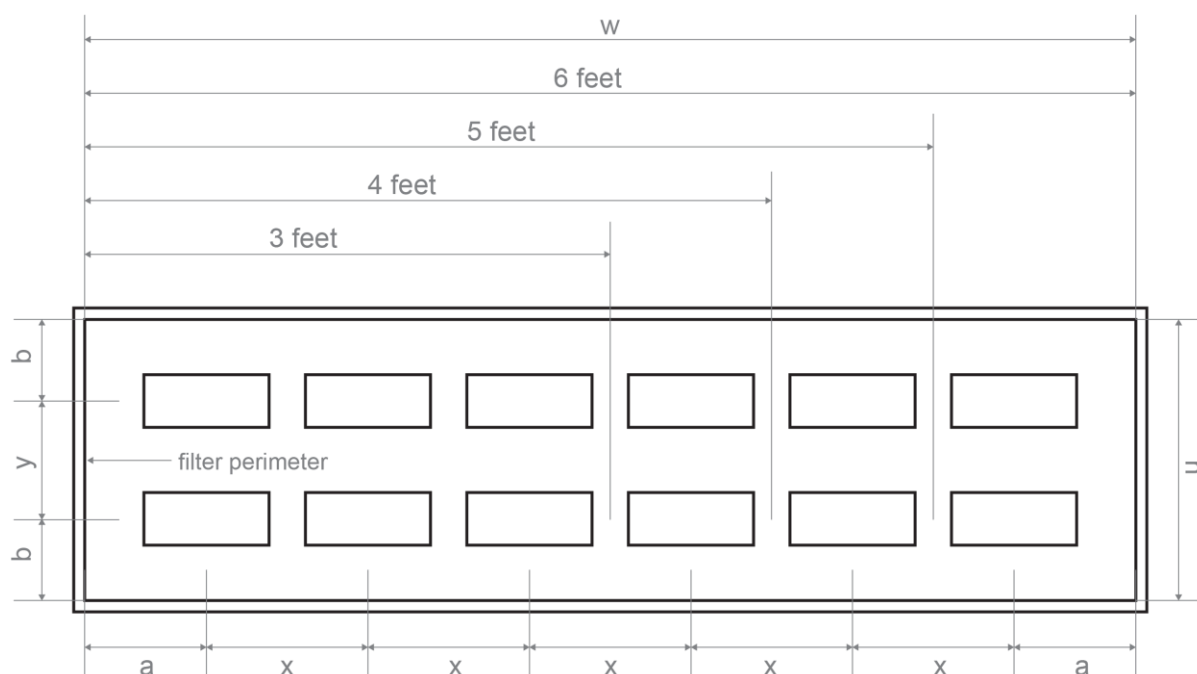
- Always check background aerosol concentration in room, make sure it is not too high.
- For field testing: to avoid accessing contaminated sampling tube, use calculative method shown in the table above.

Cabinet Model LV-A, C / LH-A	3 feet	4 feet	5 feet	6 feet
Total Airflow (cfm)	532	709	886	1064
CABINET MODEL LH-B	3 feet	4 feet	5 feet	6 feet
Total Airflow (cfm)	617	824	1037	1244
CABINET MODEL LH-C	3 feet	4 feet	5 feet	6 feet
Total Airflow (cfm)	755	1006	1267	1520

7.2.4 Nominal Airflow Velocity and Uniformity Test

The units below are in (mm/inches):

For Cabinet Models LV-A/C and LH-A/B

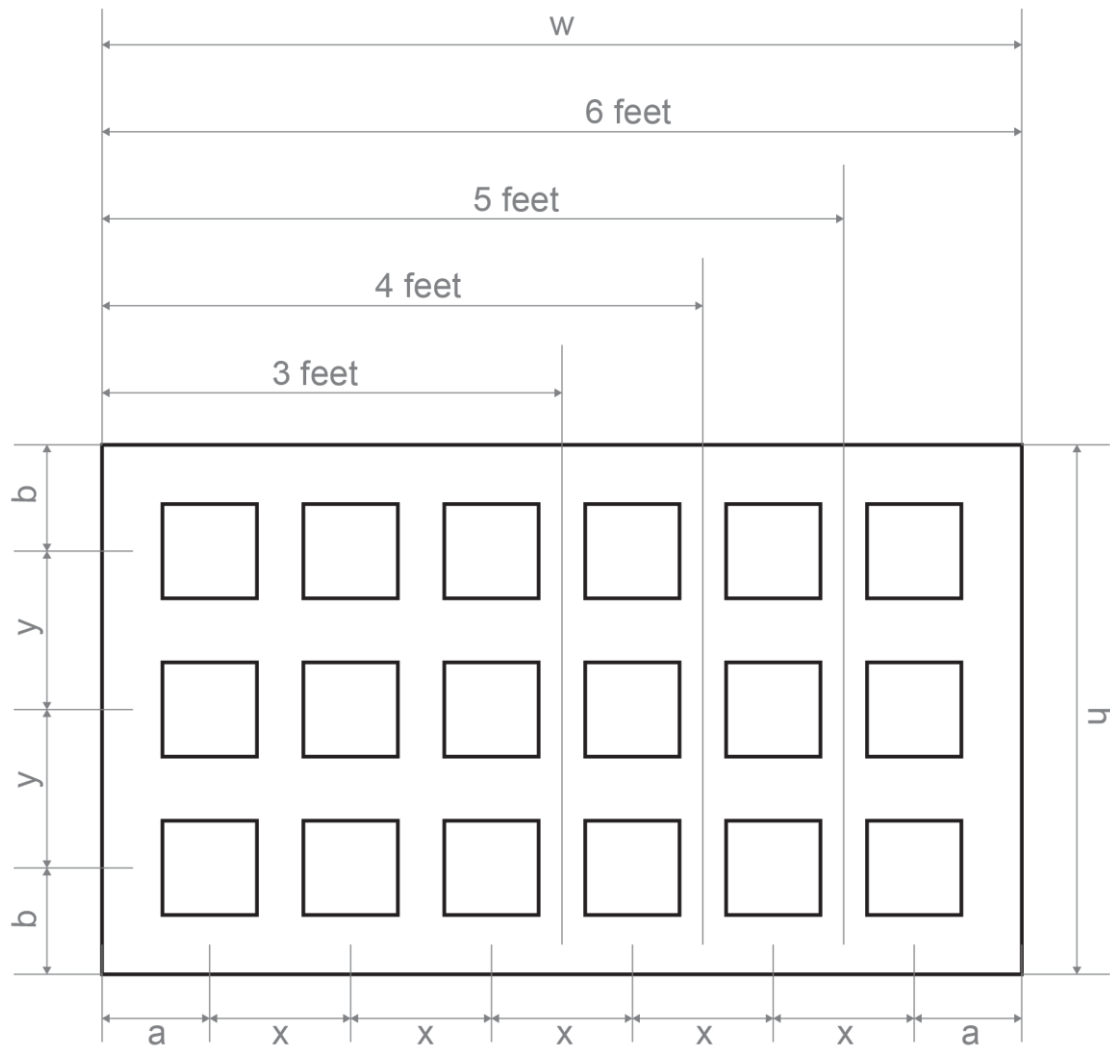


Airflow plane is 15 cm (6 inches) in front of filter face on LHG/LHS and 15 cm (6 inches) in front of diffuser face for LVG/LVS.

Cabinet Models LV-A, C / LH-A		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		914 (36)	1219 (48)	1524 (60)	1829 (72)
Height/Depth = (mm) (in)		610 (24)	610 (24)	610 (24)	610 (24)
Left to Right	<i>Distance from filter perimeter = a (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = x (mm) (in)</i>	307 (12.1)	306 (12)	306 (12)	306 (12)
Front to Back	<i>Distance from filter perimeter/table = b (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = y (mm) (in)</i>	310 (12.2)	310 (12.2)	310 (12.2)	310 (12.2)

Cabinet Models LH-B		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		900 (35.4)	1200 (47.2)	1510 (59.2)	1812 (71.3)
Height/Depth = (mm) (in)		720 (28.3)	720 (28.3)	720 (28.3)	720 (28.3)
Left to Right	<i>Distance from filter perimeter = a (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = x (mm) (in)</i>	300 (11.8)	300 (11.8)	303 (11.9)	302 (11.9)
Front to Back	<i>Distance from filter perimeter/table = b (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = y (mm) (in)</i>	420 (16.2)	420 (16.2)	420 (16.2)	420 (16.2)

For Cabinet Models LH-C



Airflow Plane is 150 mm (6 inch) below the filter face for LH

Cabinet Model LH-C		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		900 (35.4)	1200 (47.2)	1510(59.2)	1812 (71.3)
Height = h (mm) (in)		880 (34.6)	880 (34.6)	880 (34.6)	880 (34.6)
Left to Right	<i>Distance from walls = a (mm) (in)</i>	150 (5.9)	150 (5.9)	150 (5.9)	150 (5.9)
	<i>Distance apart = x (mm) (in)</i>	300 (11.8)	300 (11.8)	302 (11.9)	302 (11.9)
Front to Back	<i>Distance from walls/table = b (mm) (in)</i>	150 (5.9)	150 (5.9)	150 (5.9)	150 (5.9)
	<i>Distance apart = y (mm) (in)</i>	290 (11.4)	290 (11.4)	290 (11.4)	290 (11.4)

AVERAGE VELOCITY: _____ m/s (fpm)

MAXIMUM DEVIATION: _____ %

ACCEPTANCE CRITERIA: Average filter face velocity of **0.45 ± 0.025 m/s (90 fpm)** with no single value deviating more than 20% from the calculated average.

RESULT: **PASS / FAIL**

7.2.5 Fail Airflow Velocity Measurement

Using the grid in the previous section, measure the fail airflow velocity



Note: Only applicable for cabinets with microprocessor control

AVERAGE AIRFLOW VELOCITY: _____ m/s (fpm)

MOTOR DUTY CYCLE: _____ %

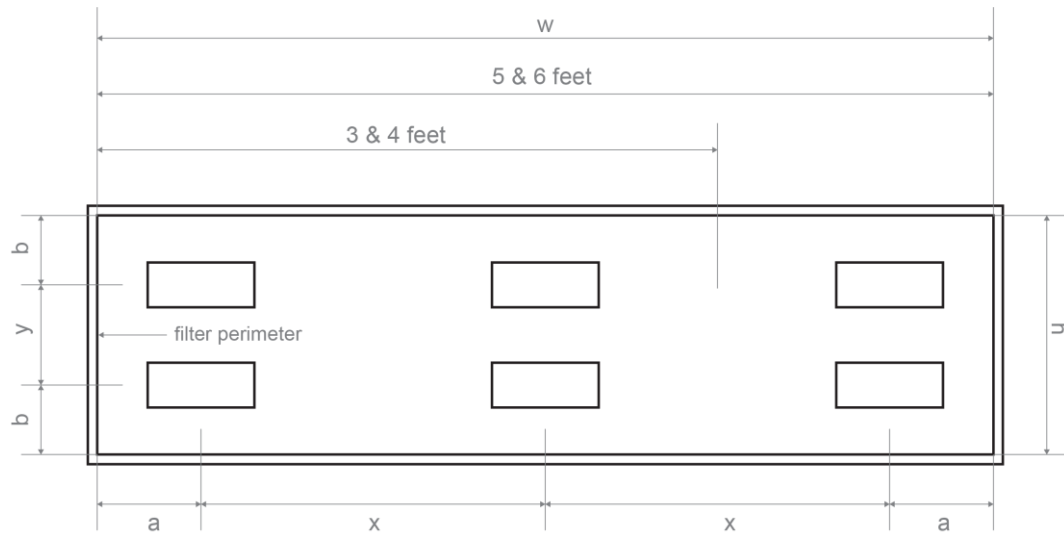
ACCEPTANCE CRITERIA: Average fail airflow velocity of **0.30 ± 0.025 m/s (60 fpm)**

Testing was performed in accordance with the following IEST (Institute of Environmental Sciences and Technology) contamination control document: IEST-RP-CC002.2 "Recommended Practice for Unidirectional Flow Clean Air Devices" Section 6.1.

7.2.6 Particle Count Test

The units below are in (mm/inches):

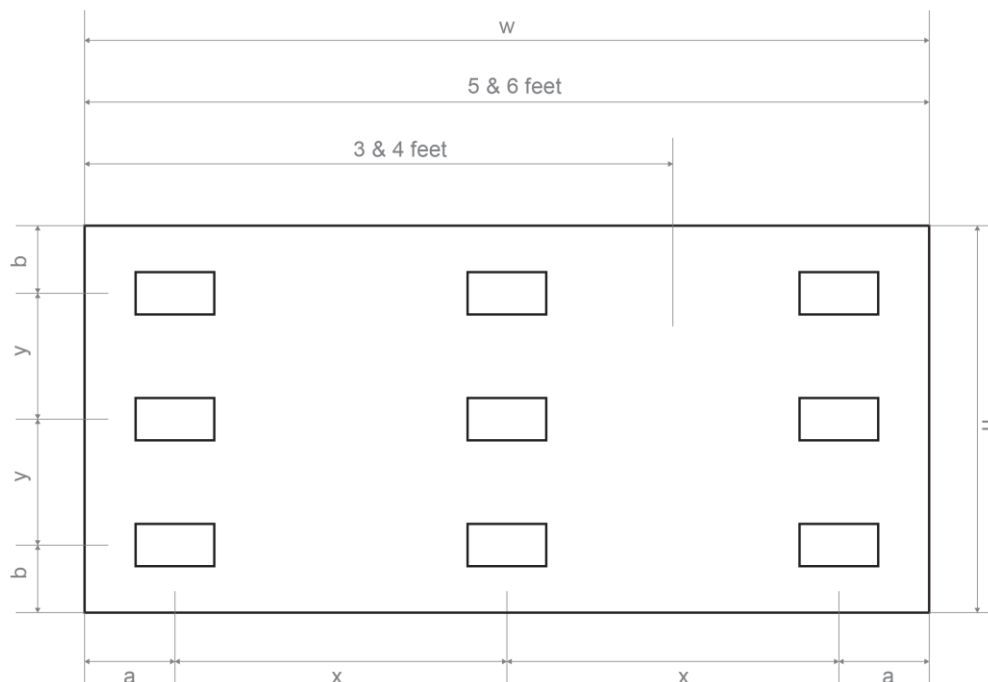
For Cabinet Models LV-A/C and LH-A/B



Plane is 300 mm in front of the filter/diffuser for LHG/LHS and on the work surface for LVG/LVS.

Cabinet Models LV-A, C / LH-A		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		914 (36)	1219 (48)	1524 (60)	1829 (72)
Height/Depth = h (mm) (in)		610 (24)	610 (24)	610 (24)	610 (24)
Left to Right	<i>Distance from filter perimeter = a (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = x (mm) (in)</i>	307 (12.1)	306 (12)	306 (12)	306 (12)
Front to Back	<i>Distance from filter perimeter/table = b (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = y (mm) (in)</i>	310 (12.2)	310 (12.2)	310 (12.2)	310 (12.2)

Cabinet Models LH-B		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		900 (35.4)	1200 (47.2)	1510 (59.2)	1812 (71.3)
Height/Depth = h (mm) (in)		720 (28.3)	720 (28.3)	720 (28.3)	720 (28.3)
Left to Right	<i>Distance from filter perimeter = a (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = x (mm) (in)</i>	300 (11.8)	300 (11.8)	303 (11.9)	302 (11.9)
Front to Back	<i>Distance from filter perimeter/table = b (mm) (in)</i>	150 (6)	150 (6)	150 (6)	150 (6)
	<i>Distance apart = y (mm) (in)</i>	420 (16.2)	420 (16.2)	420 (16.2)	420 (16.2)

For Cabinet Models LH-C

Plane is 300 mm (11.8 in) in front of the filter for LHG/LHS

Cabinet Models LH-C		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		900 (35.4)	1200 (47.2)	1510 (59.2)	1812 (71.3)
Height/Depth = h (mm) (in)		880 (34.6)	880 (34.6)	880 (34.6)	880 (34.6)
Left to Right	<i>Distance from filter perimeter = a (mm) (in)</i>	300 (11.8)	300 (11.8)	300 (11.8)	300 (11.8)
	<i>Distance apart = x (mm) (in)</i>	300 (11.8)	600 (23.6)	455 (17.9)	606 (23.9)
Front to Back	<i>Distance from filter perimeter/table = b (mm) (in)</i>	150 (5.9)	150 (5.9)	150 (5.9)	150 (5.9)
	<i>Distance apart = y (mm) (in)</i>	290 (11.4)	290 (11.4)	290 (11.4)	290 (11.4)

AVERAGE COUNT: _____ for particle size of **0.5 micron and larger** samples in 0.028m³ (1 ft³) of air



The particle count done in the factory is at the “as-built” condition.

If 0.5 micron counter is used on the field, there shall be no more than 1 particles of 0.5 µm samples in 1 m³ (35.3 ft³) of air to reach ISO Class 3.

CLEANLINESS RATING: **ISO Class 3** per ISO 14644.1 (≤ 35 count for particles equal to and larger than 0.5 µm per 1 m³ or 35.3 ft³ of air).

RESULTS: **PASS / FAIL**

7.2.7 Noise Level Test (Optional for field testing)

With the cabinet running under regular parameters, obtain the noise level of the cabinet during normal operation using a calibrated noise level meter at 300 mm (12 in.) in front of the work surface area lower edge and 380 mm (15 in) above the recessed work surface area.

ACCEPTANCE: Maximum corrected noise level of **67 dBA**.

RESULTS: **PASS / FAIL**

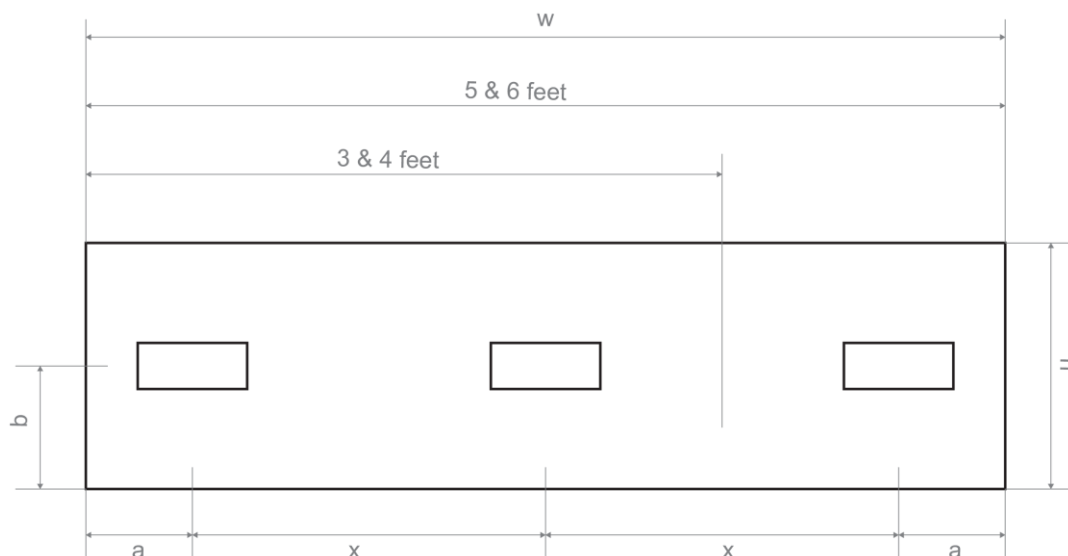
DISCLAIMER:

1. This noise test was done in the factory that is susceptible to reflective, reverberant, and background noise effect. This noise figure may be different than the brochure value that is obtained from an anechoic chamber.
2. If the difference between cabinet and ambient noise is less than 10 dBA, and a correction chart was used, which combined with transient ambient noise, the corrected noise level may be varied by ± 1.5 dBA.

Testing was performed in accordance with the following IEST (Institute of Environmental Sciences and Technology) contamination control document: IEST-RP-CC002.2 "Recommended Practice for Unidirectional Flow Clean Air Devices" Section 6.11.

7.2.8 Light Intensity Test (Optional for field testing)

A light intensity meter was used to measure light intensity at front to back centerline work surface level of the cabinet.



Cabinet Models LV-A, C / LH-A, B		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		914 (36)	1219 (48)	1524 (60)	1829 (72)
Height/Depth = h (mm) (in)		610 (24)	300 (11.8)	300 (11.8)	300 (11.8)
Left to Right	<i>Distance from filter perimeter = a (mm) (in)</i>	300 (11.8)	300 (11.8)	300 (11.8)	300 (11.8)
	<i>Distance apart = x (mm) (in)</i>	314 (12.4)	619 (24.4)	462 (18.2)	615 (24.2)
Front to Back	<i>Distance from filter perimeter/table = b (mm) (in)</i>	305 (12)	305 (12)	305 (12)	305 (12)

Cabinet Models LH-C		3 feet	4 feet	5 feet	6 feet
Width = w (mm) (in)		900 (35.4)	1200 (47.2)	1510 (59.2)	1812 (71.3)
Height/Depth = h (mm) (in)		640 (25)	640 (25)	640 (25)	640 (25)
Left to Right	<i>Distance from filter perimeter = a (mm) (in)</i>	300 (11.8)	300 (11.8)	300 (11.8)	300 (11.8)
	<i>Distance apart = x (mm) (in)</i>	314 (12.4)	619 (24.4)	462 (18.2)	614 (24.2)
Front to Back	<i>Distance from filter perimeter/table = b (mm) (in)</i>	319 (12.6)	319 (12.6)	319 (12.6)	319 (12.6)

AVERAGE LIGHT READING: _____ lux (ft.-candles)

ACCEPTANCE CRITERIA: _____ Average light reading should be at least 800 lux (74 ft.-candles)

RESULTS: **PASS / FAIL**

7.3 Calibration of Sentinel™ Gold Control

The purpose of calibrating the controller is to let the controller record particular output voltages from the airflow sensor and correspond them to fail and nominal points of air velocity. Once the calibration has been completed, the control will be able to show the air velocity whenever the fan is in operation. All Esco cabinets are factory-tested and calibrated before being shipped.



Calibration is only necessary when the air velocity display is inaccurate or faulty alarm setpoints need to be corrected.

7.3.1 Calibrating the Airflow Nominal Set Point – 0.45 ± 0.025 m/s (90 fpm)

- Press the MENU button. If the cabinet is in Maintenance Mode, no PIN will be asked. If the cabinet is in Normal Mode, enter PIN 0009. Press the ▼ button until the display shows “FIELD CALIB”, then press SET.
- The pointer will be at “AirFlow Calib”. Press SET button.
- A “Read Manual” message is shown to discourage unauthorized person. Press SET to continue.
- Ensure that the Nominal Airflow Set Point is **0.45 ± 0.025 m/s (90 ± 5 fpm)**. The airflow volume is blinking—indicating that it can be changed. Select your obtained Airflow using ▲ and ▼ buttons. Press SET button to continue.
- Wait for 3 minutes for the software to record this sensor ADC point. Do not change the Motor Duty Cycle or disrupt airflow/ sensor during this time.
- After the 3 minute countdown has been completed, a beep sound will be heard, then the display will show “CALIBRATION DONE”. The screen will go back to MENU options shown in Step 2.



Note: *If the Calibration Error message is displayed, please ensure that airflow is within ± 0.025 m/s from nominal, and repeat the calibration.*

- Return the cabinet from Maintenance Mode to Normal Mode.
- Microprocessor calibration is finished

7.3.2 Setting the Standby Mode Airflow

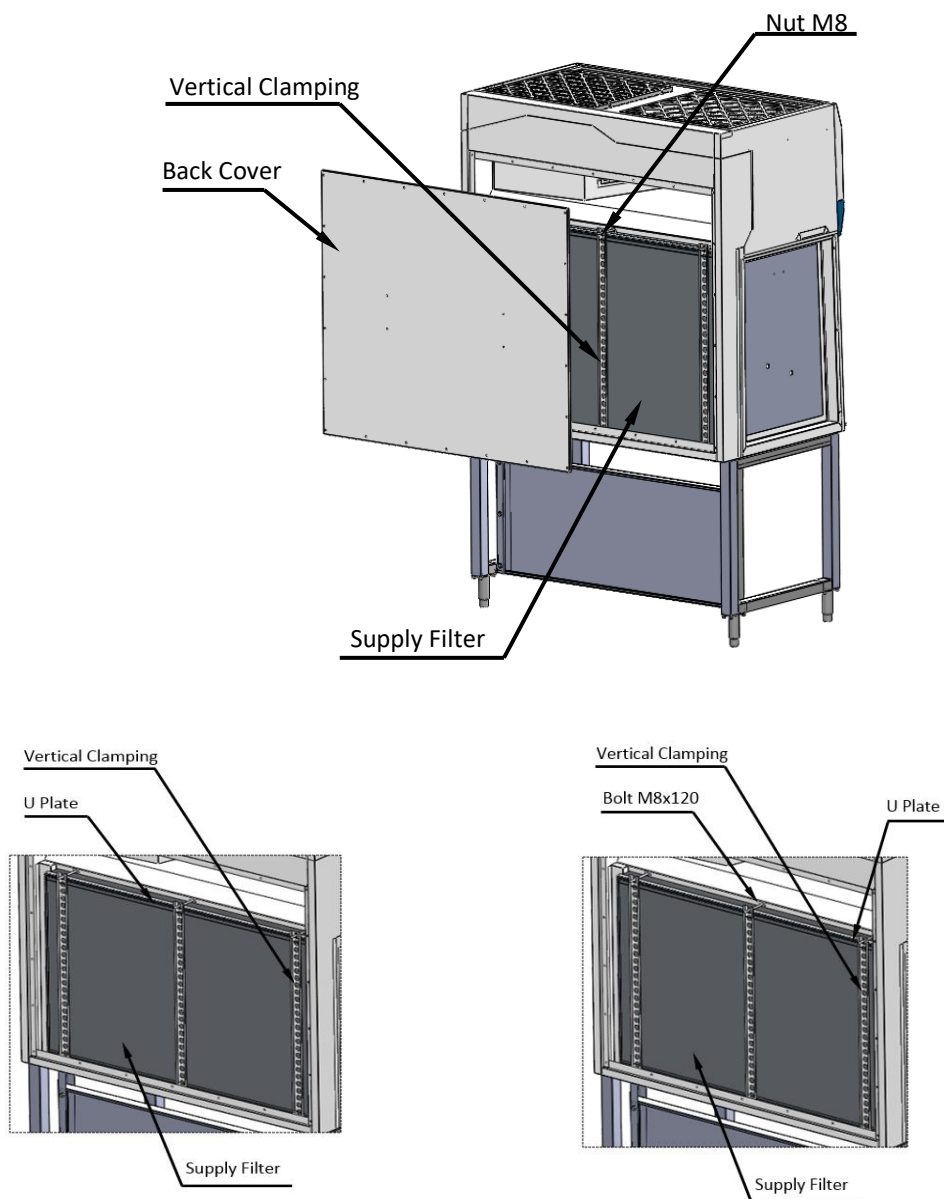
- Set the cabinet to Standby Mode. Press black bottom on the ECMS Controller for few seconds until reading is “0”.
- Adjust the Standby Mode duty cycle to get the Standby Mode Airflow **0.32 m/s (63 fpm)**.
Duty Cycle: _____%
- Press black bottom on the ECMS Controller a few second until reading “dn”.
- Set the Duty Cycle to Normal Speed Mode.

7.4 Replacement of Filter

7.4.1 LHG/LHS

7.4.1.1 Filter Replacement Instructions

1. Remove the back cover, which are held in position by Phillips pan head screws.
2. Remove the nuts located on the 2 sides (top and bottom) of the vertical clamping.

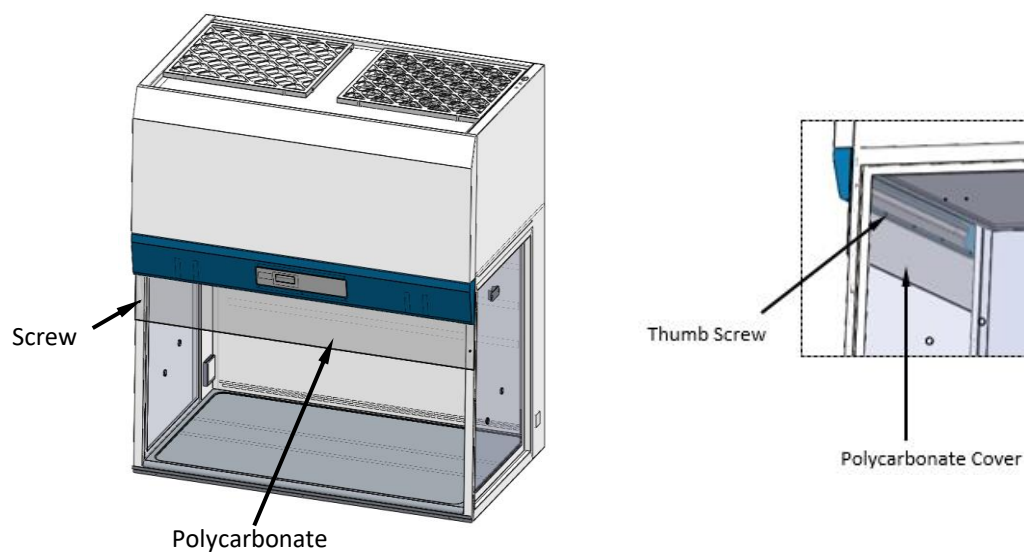


3. Remove all the vertical clamping along with the u-plates, then carefully remove the supply filter.
4. Replace new filter by reversing the above steps.

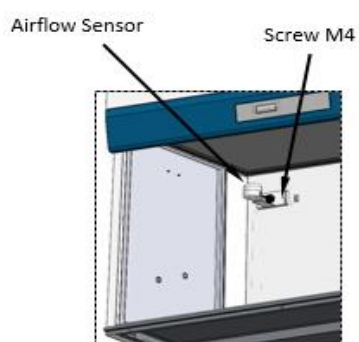
7.4.2 LVG/LVS

Filter replacement instructions

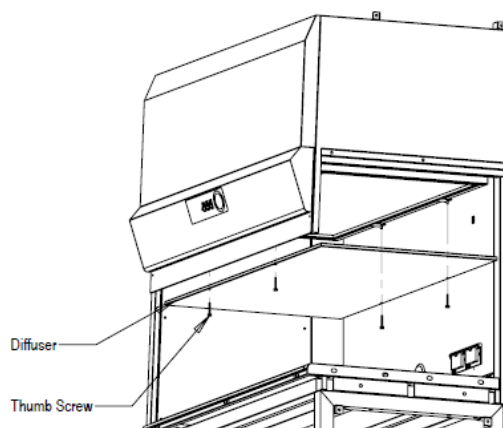
1. Remove the polycarbonate cover, which are held in position by screws M4 in front and thumb screws at the back.



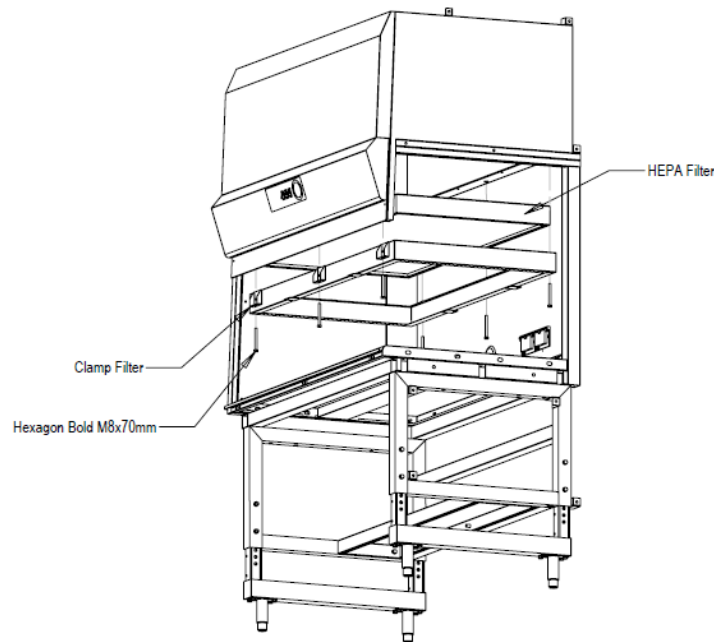
2. Remove the airflow sensor, which are held in position by screws M4.



3. Remove the diffuser



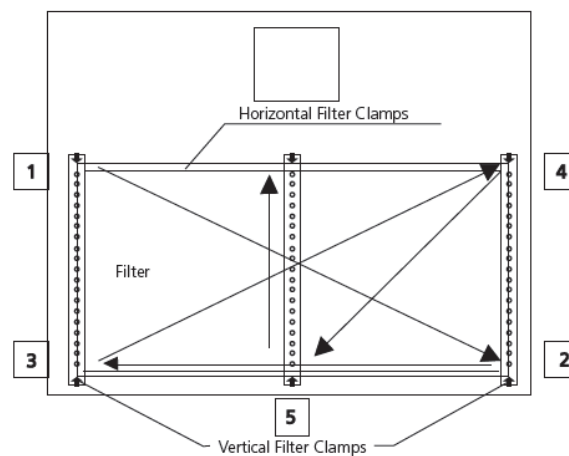
4. Remove the Hexagon Bold M8 screw and then the Clamp Filter. Carefully remove the HEPA filter.



5. Replace new filter. Reverse the above steps for the assembly.

Note: The following applies for both Horizontal and Vertical Laminar Flow Clean Benches.

The nuts that are used to clamp the filter have to be tightened until 50% compressed with alternating pattern after the new filter is installed. The nuts must be equally tightened to give uniform compression. This is to prevent over-compression on one side of the filter that can cause filter leakage. The nut tightening pattern is shown below:



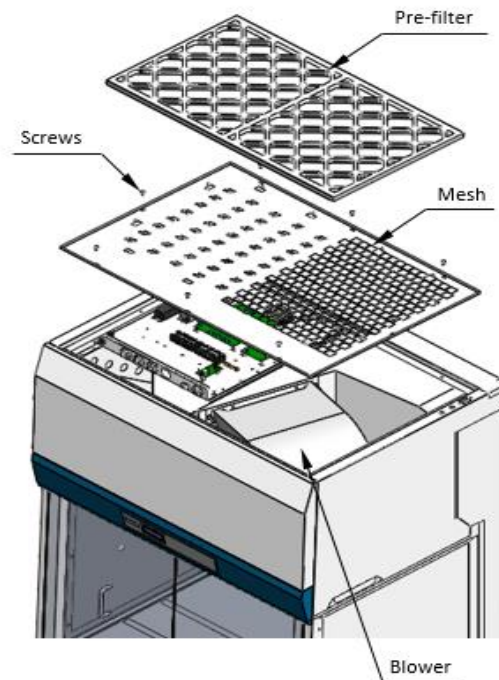
Note: The cabinet must undergo re-certification after changing the filter.

7.5 Replacement of Blower

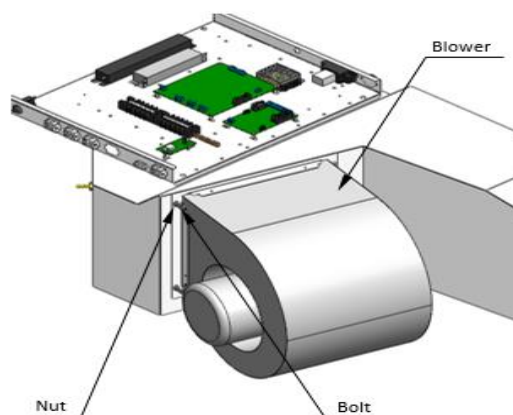
7.5.1 LHG/LHS

Blower Replacement Instruction

1. Pre-filter and mesh must be removed first before proceeding to the blower replacement instruction.
Disconnect electrical connections as necessary to free blower.



2. Remove the nuts, which clamp the blower. Pull out and lift up the blower carefully.

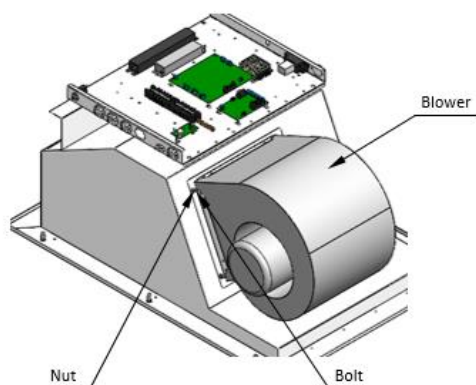
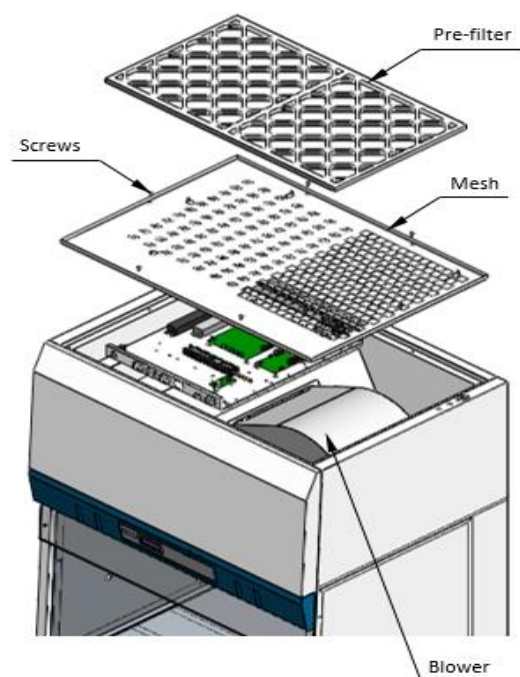


3. Repeat this procedure in reverse order with the new blower to complete the replacement.

7.5.2 LVG/LVS

Blower Replacement Instruction

1. Pre-filter and mesh must be removed first before proceeding to the blower replacement instruction.
Disconnect electrical connections as necessary to free blower.
2. Remove the nuts, which clamp the blower. Pull out and lift up the blower carefully.
3. Replace new blower by reversing the above steps.



7.6 Replacement of Fluorescent Lamp and UV Lamp



7.6.1 Replacement of Fluorescent Lamp

- Before changing the fluorescent bulbs, ensure that the clean bench is powered down and disconnected from the electrical supply.
- Locate the bulbs.
- Remove the power clips at the ends of the bulbs by gently pulling whilst holding the bulb steady.
- Remove the bulbs from the mounting clips and replace with new ones.
- Replace the power clips on the ends of the new bulbs and ensure they are firmly seated.
- Reconnect the clean bench to the electrical supply and test the bulbs for proper operation.



7.6.2 Replacement of UV Lamp

- Disconnect the electrical connections.
- Take out the old UV lamp by rotating 90° counter-clockwise and pull downward.
- Install the new UV lamp by pushing it upward and rotate 90° clockwise.

7.7 Replacement of Tray

- Dismantle the back cover plate and frame using Screwdriver M5. (Fig. 1.7.1)
- Disconnect the magnetic switch cable (white) from the cable jumper and de-attached cable ties from back wall (Fig. 1.7.2). Magnetic switch can be found at the bottom of the tray (Fig. 1.7.3).
- From the bottom of tray (Fig. 1.7.4), find the grounding cable (green-yellow). Open the nut clamping the cable using Socket Wrench 8. Insert the grounding cable into the cable hole to prevent the cable from being stuck and over-stretched/broken while removing the tray.
- Open the hex-bolt clamping the tray using Socket Wrench 13.
- Pull down the tray carefully.
- Reverse above steps to re-install the tray

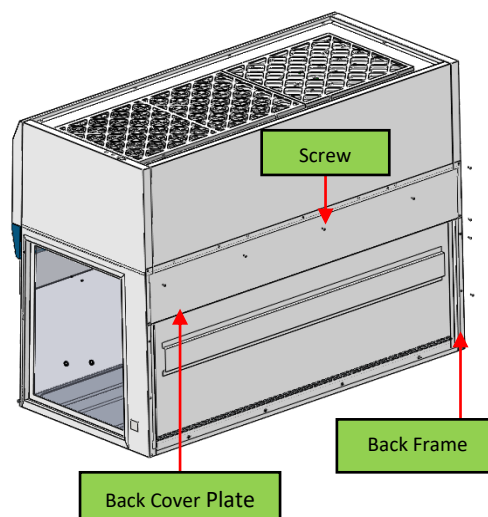


Fig. 1.7.1

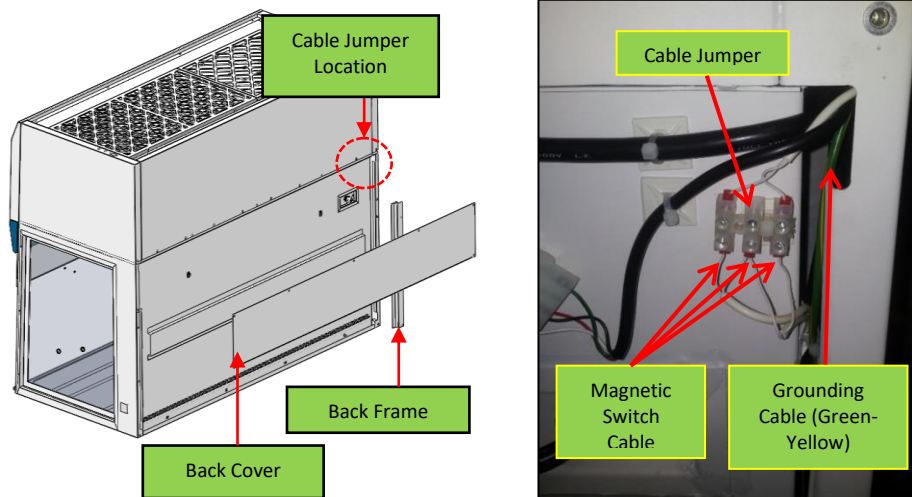


Fig 1.7.2

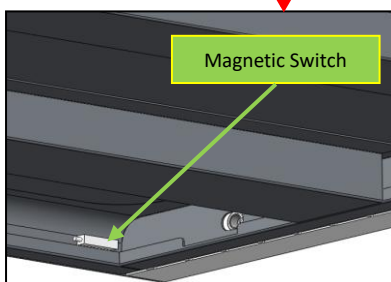
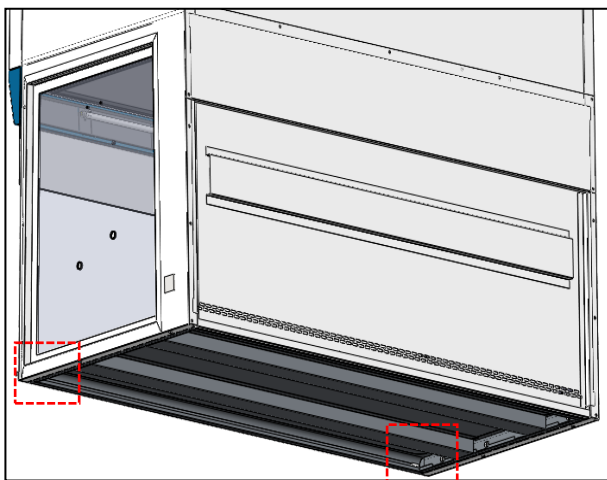


Fig. 1.7.3

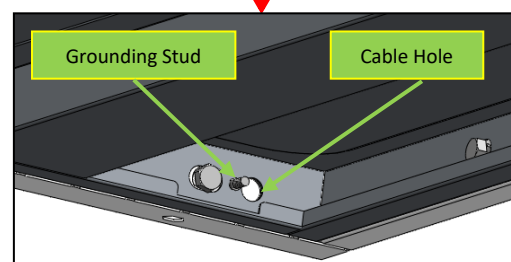
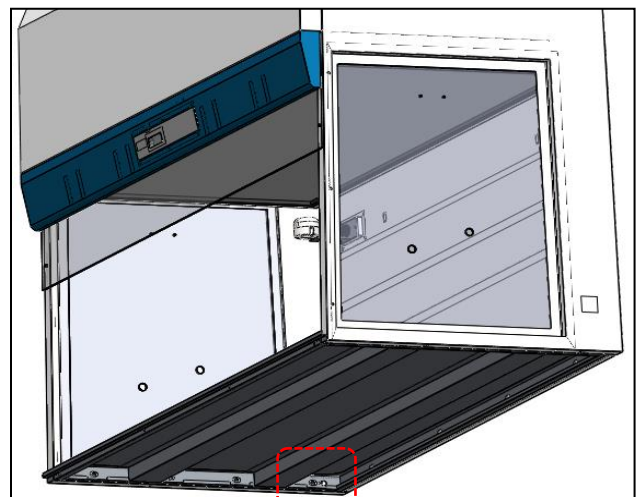


Fig. 1.7.4

Chapter 8 – Troubleshooting

This section helps you troubleshoot some of the common problems you might face while operating this clean bench. Should you have any queries left unanswered here, please feel free to contact Esco.

PROBLEM 1

Cabinet does not start (LCD Display, Button, Fan, Light, UV and socket are all inoperative)

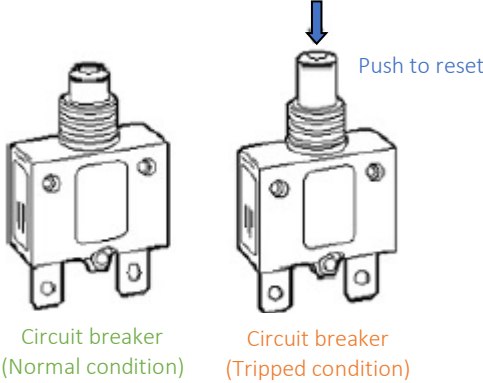
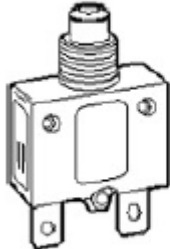
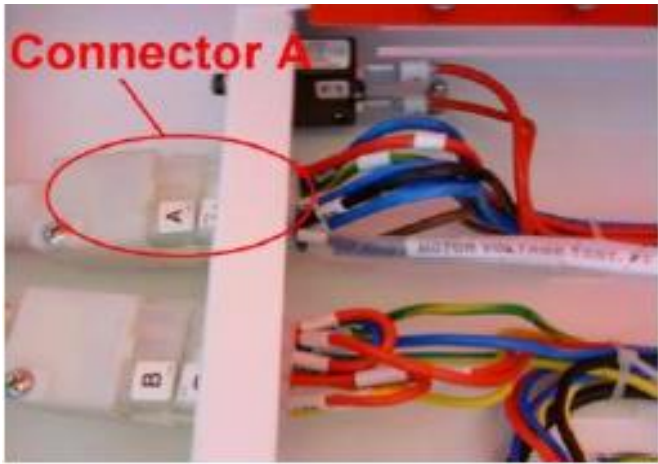
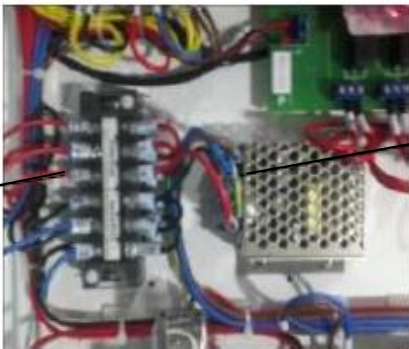
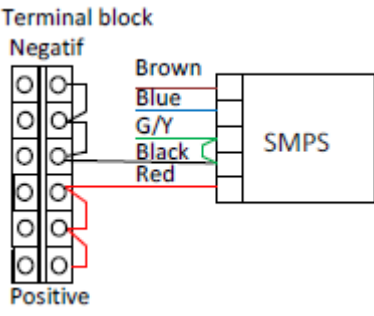
CAUSE	CORRECTIVE ACTION
<i>Power failure</i>	- Check wall outlet for power. - If the outlet has a switch, make sure it is turned ON.
<i>Power cord is not connected properly or is faulty</i>	- Ensure that the power cord is connected properly to the wall outlet.  Note: Some cabinets have 2 cord sets. - Disconnect the power cord at the cabinet and measure the AC voltage between the live and the neutral terminal of the cord. - If the voltage is not within $\pm 2\%$ of the wall socket voltage, replace the cord set; otherwise, proceed to the next step.
<i>Circuit breaker has tripped</i>	- Check the circuit breaker inside the electrical panel.  Note: If circuit breaker has tripped, do not reset the breaker before checking all electrical components and wiring connections (see Figure 1.1) - Does the cabinet operate correctly after resetting the circuit breaker? If it does not, proceed to the next step.   Circuit breaker (Normal condition)  Circuit breaker (Tripped condition)

Fig. 1.1

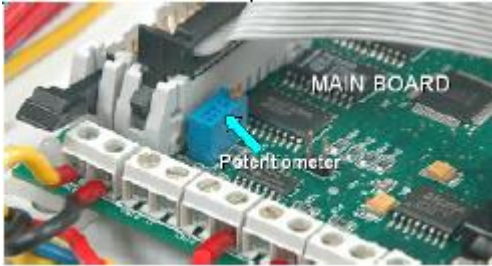
CAUSE	CORRECTIVE ACTION
<i>Improper connection</i>	• Measure AC voltage between LIVE (RED) and NEUTRAL (BLUE) terminal blocks inside electrical box (see Fig. 1.2). • Voltage must be within $\pm 10\%$ of the rated voltage. • Refer to cabinet Serial Number Tag for required voltage. • If the voltage is out the range of there is no voltage, check cable connection on both sides of Connector A pin 1 and 2. Pin 3 is ground (see Fig. 1.3). • If there is voltage at Connector A but no voltage is measure as the RED (LIVE) and BLUE (NEUTRAL) terminal blocks, the problem is between connector A and the terminal strips. • If the voltage is not present at Connector A, but voltage is measured at the female connector for the line cord at the top of the cabinet that the power cord is plugged into, the problem is between the female connector and the pins on Connector A.  Fig. 1.2  Fig. 1.3

CAUSE	CORRECTIVE ACTION
<i>Defective Power Supply – Switch Mode Power Supply (SMPS)</i>	• Locate the SMPS (Power Supply) and Terminal block. • On the side of terminal block, measure the DC voltage between RED (pin 3) and BLACK (pin 4) (see Fig. 1.4). Measure on the side away from the SMPS. • The voltage should be in range of $\pm 12 \text{ VDC} \pm 10\%$. • If you do not have voltage, measure the other side of terminal block. • If out of range or no voltage, check incoming power to the SMPS – check terminal where cord is connected. The input of SMPS should be $\pm 10\%$ of the cabinet's incoming voltage. • If input voltage exists but there is no output voltage, please replace the SMPS. • If input and output voltage is correct, please proceed to the next step.   Fig. 1.4
<i>Connection problem to Main Board of from Main Board to Display (Only for cabinets with microprocessor control)</i>	• Measure the incoming voltage on the main board at terminal 13 (Note polarity, BLUE cable closest to edge is negative -). See Fig. 1.5. • Voltage should be between 10.8-13.2 VDC, • If voltage is out of range or no voltage, ensure adequate connection between SMPS and main board. • Ensure adequate connection between flat ribbon cables from main board to LCD/keypad membrane. Note: There are two ribbon cables from the main board and the display and keypad. The large flat cable is for the Display. The smaller flat cable is for the LEDs/keypad membrane. • If the LCD Display is on, but not the LEDs in keypad membrane, check connection on the smaller Flat ribbon cable. (continued on next page)

CAUSE	CORRECTIVE ACTION
	See Fig. 1.5. If the incoming voltage at J13 is of right voltage and flat ribbon cable are connected adequately but the LEDs on the keypad membrane and LCD display are still not turned on, please proceed to the next step.  Fig. 1.5
<i>Defective Main Board (Only for cabinets with microprocessor control)</i>	- The main board is defective if the main board incoming supply is between 10.8-13.2 VDC, if 12 VDC is installed and: - All LEDs on the control panel are off - The LCD is blank - No buzzer sound - If these conditions exist, replace the Main Board. Otherwise, proceed to the next step.  Note: When replacing main board, reconnect all wires correctly. Any wrong wiring may result in damage.

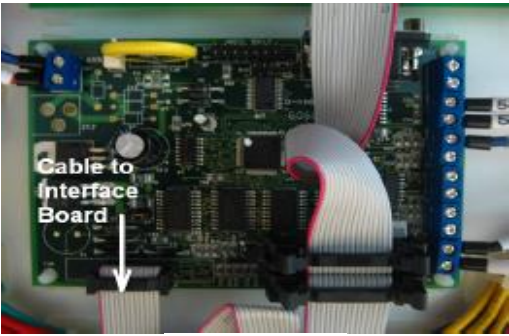
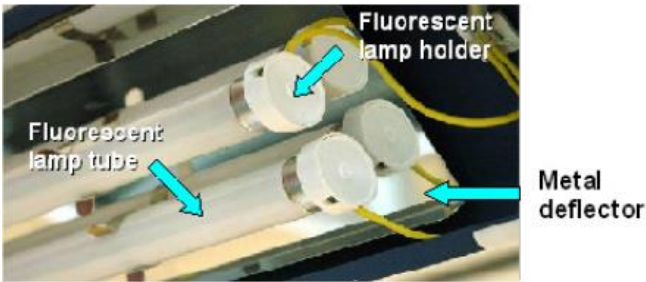
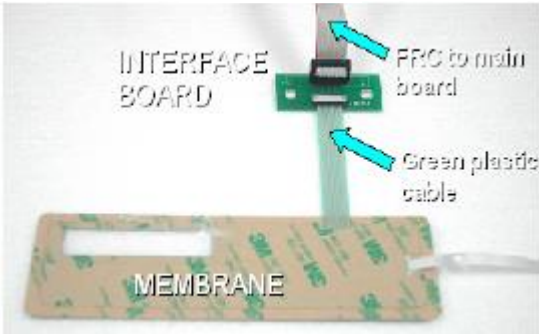
Problem 2

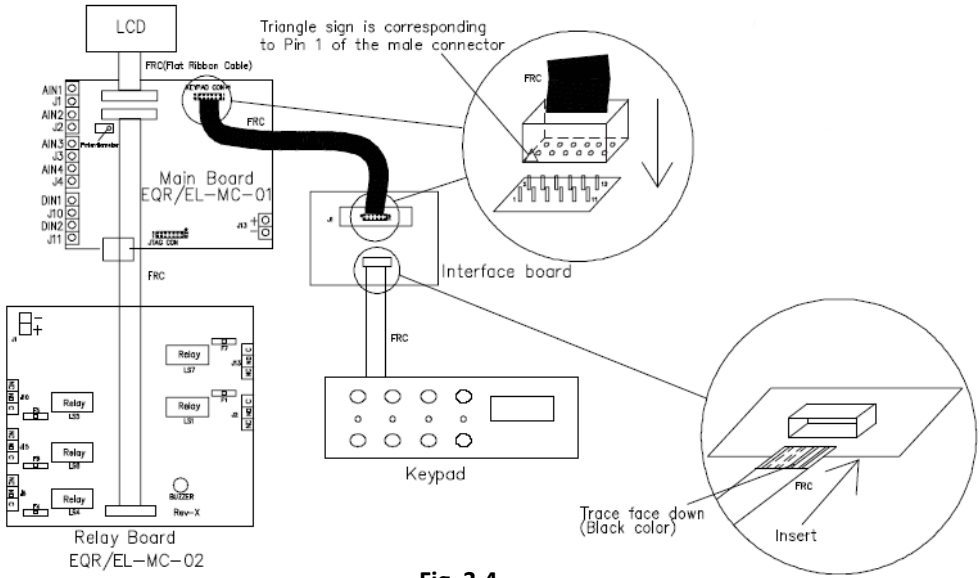
Blank LCD (Only for cabinets with microprocessor control)

CAUSE	CORRECTIVE ACTION
<i>Connection problem</i>	• Ensure that the LCD large Flat Ribbon Cable is adequately connected to the main board with the display at the main board end. • Ensure that the cable has been inserted adequately into its socket on the main board. This is a keyed connector with locking arms. The arms should point to the same direction as the cable (<i>see Fig. 2.1</i>). • If the LCD display is not on, check power to main board as detailed in Problem 1.  Fig. 2.1
<i>Contrast problem</i>	• Adjust the potentiometer on main board by rotating the top metal part using flat screw driver to achieve the best LCD contrast. Rotating the part in the counter clockwise direction will increase the contrast (<i>see Fig. 2.2</i>). • If the LCD remains blank, replace it.  Fig. 2.2
<i>Defective LCD</i>	• Connect a new LCD to the LCD port on main board. • If the new LCD functions properly, this indicates that the old LCD panel is defective. • If the new LCD does not work, replace the large Flat Ribbon Cable. • If the LCD still does not work, replace the main board.

Problem 3

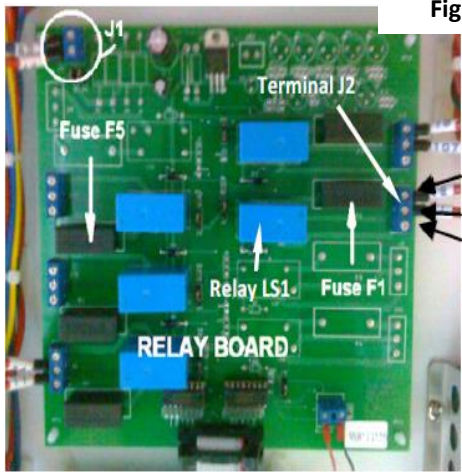
Inoperative buttons

CAUSE	CORRECTIVE ACTIONS
Connection problem	- Does the LCD work? If it does not, please refer to Problem 1 and ensure that the main board is powered. - Ensure that the small Flat Ribbon Cable is connected to the main board (<i>see Fig. 3.1</i>). The triangle sign on the female connector of the cable indicated PIN number 1. - Interface board and membrane/keypad are located behind the blue panel, underneath the metal light deflector. - Turn off the cabinet and then remove the bulbs and the clips that hold the bulbs (<i>see Fig. 3.2</i>). - Ensure adequate connections between the cable from the main board and the Interface board (<i>see Fig. 3.3</i>). - Ensure adequate connections between the green plastic cable from the Interface board and the Display board (<i>see Fig. 3.4</i>).  Fig. 3.1  Fig. 3.2  Fig. 3.3

CAUSE	CORRECTIVE ACTIONS
	 Triangle sign is corresponding to Pin 1 of the male connector FRC (Flat Ribbon Cable) Main Board EQR/EL-MC-01 Interface board Keypad Relay Board EQR/EL-MC-02 Trace face down (Black color) Insert Fig. 3.4
<i>Defective cable and/or Interface Board and/or keypad</i>	• Replace them one by one to check which one(s) among them is/are defective. • Replace the defective part(s).

Problem 4

Blower does not function

CAUSE	CORRECTIVE ACTIONS
<i>Motor Blower Fan is not running</i>	• Switch on the fan by pressing Fan button on the control panel • Enter the fan PIN number if required (default is 0001). • The LED for fan should illuminate and the fan should start. • If the fan does not operate, proceed to the next step. • If the LED on the cabinet does not illuminate, please refer to Problem 1.
<i>DC voltage connection problem to relay board</i>	• Measure the incoming voltage on the Relay Board at terminal J1 (note polarity, BLUE cable closest to edge is negative). <i>See Figure 4.1.</i> • Voltage should be between 6.75-8.25 VDC, if 7.5 VDC SMPS is used. • Voltage should be between 10.8-13.2 VDC, if 12 VDC SMPS is used. • If voltage is out of range or no voltage, check connection between SMPS (power supply) and relay board. • If voltage is correct, proceed to the next step.
<i>Fuse F1 is blown or circuit is not energized</i>	• Check that the AC voltage is supplied to the Relay board. • Ensure the Motor Blower FAN button and LED is energized. • Locate LS1 circuit and measure AC voltage between large BLUE (Neutral) terminal block and the end terminal (with RED cable) on the three terminal strips (J2 on relay board). <i>See Figure 4.1.</i> • If there is voltage to the end terminal (Terminal with cable), check the central terminal (Normally Open Circuit) to the BLUE (Neutral) terminal block. • Should there be no voltage detected at the central terminal, the fuse could have been blown or the circuit is not energized. • Check Fuse F1 on relay board (<i>see Fig. 4.1</i>). • If fuse F1 is blown, as temporary solution, use F5 (spare) to replace F1. • If fuse F1 is okay, check the voltage between the BLUE (Neutral) terminal block and the end terminal on the LS1 circuit (J2 on relay board) • Detection of voltage at the end terminal with no cable indicates that the F1 fuse is good and the circuit is not energized. Fig. 4.1 
<i>Faulty switch (applicable for cabinets with simple switch</i>	• Fan switch is a black color switch on control panel. • Switch the unit off and detach the blue front panel of the unit. Unfasten the screws behind and at the inner side of the panel. • Behind the blue panel, ensure fan switch condition is good at ON and OFF

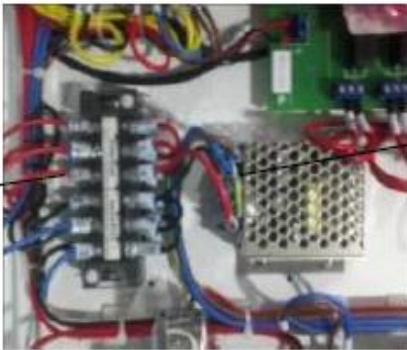
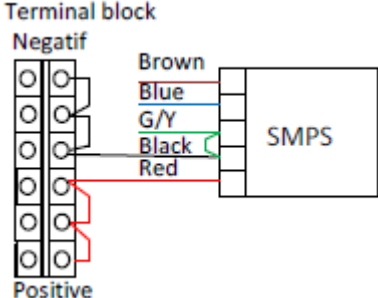
CAUSE	CORRECTIVE ACTIONS
<i>control only)</i>	position.
<i>Defective motor speed controller</i>	• Switch on the blower on the cabinet. • If the blower does not work, switch off the cabinet. • Refer to Problem 11: Motor Speed Controller Not Functioning troubleshooting
<i>Motor failure</i>	• Check for physical damage. • Does the motor rotate? • Check wiring. • Replace the blower.

Problem 5

Airflow Failure (AIRFAIL) – Alarm is triggered

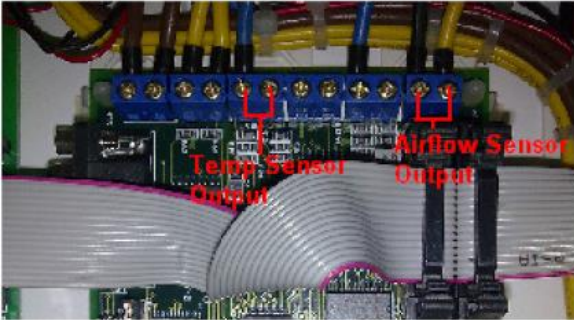
CAUSE	CORRECTIVE ACTIONS
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CAUSE	CORRECTIVE ACTIONS
<i>External air interference</i>	• Ensure that there are no external sources of airflow disturbance close to the sensor. Check vents, lights, ceiling distance, etc. • If there is no external air interference, proceed to the next step.
<i>Low building supply voltage (if new cabinet, check this first)</i>	• For a new cabinet, if the motor voltage is lower than the Nominal Voltage value recorded in the factory test report, adjust the speed controller to get output voltage to blower as specified in the Test Report – Test Conditions Documentation Sheet. • For a cabinet that has been used for more than a year, refer to last motor Nominal Voltage from last certification report or if not available, measure the actual airflow in the cabinet and increase motor voltage accordingly to achieve nominal airflow velocity. • If supply voltage is same as Nominal voltage, proceed to the next step.
<i>Defective airflow Sensor power supply – Switch Mode Power supply (SMPS)</i>	• Cabinet must be turned on to perform this test. • Locate the 5 pin connector. • On the BACK side of the 5 pin connector measure the DC voltage between GREEN (pin 5) and BLACK (pin 4). Measure on the side, away from the SMPS. See Figures 1.4 and 1.5. • The voltage should be in range of +12VDC $\pm$ 10%. • Voltage on the other side of the 5 pin connector does not need to be measured. • If out of range, check incoming power to the SMPS Molded cord – check the terminal where the cord is connected. The input should be $\pm$10% of the cabinet voltage • If you have an input voltage but no output voltage, replace the SMPS.
<i>Defective Airflow Sensor power supply – Switch Mode Power supply (SMPS) (Only when SMPS (item code 1080945) is installed)</i>	• Cabinet must be turned on to perform this test. • Locate the terminal block. • On the side of the terminal block, measure the DC voltage between RED and BLACK pins. • Measure on the side, away from the SMPS (see Figure 5.1 and 5.2). • The voltage should be in range of +12VDC $\pm$ 10%. • The voltage has to be measured on the other side of terminal block and SMPS. • If out of range, check incoming power to the SMPS – check terminal where the cord is connected. The input of SMPS should be $\pm$10% of the cabinet incoming voltage. • If input voltage is measured, but not output voltage, then replace the SMPS.

CAUSE	CORRECTIVE ACTIONS
	SMPS output RED- BLACK/BLUE is 12 VDC  SMPS input BROWN- BLUE-G/Y is 230 VAC or 115 VAC Fig. 5.1  Fig. 5.2
<i>Cabinet requires the airflow to be set</i>	Refer to the Test Report for this model and set the air flow value to the required value — 0.45 m/s (90 fpm). This information is also contained on the label at the bottom of the blue front panel.
<i>Filter loaded or wrong speed controller setting</i>	- Plug in the voltmeter to the Motor Voltage Sampling Port. - Measure the actual airflow velocity using a digital anemometer or flow hood. If actual airflow is normal but LCD still shows AIRFAIL, Proceed to the next step. If actual airflow is not acceptable, please adjust motor speed controller accordingly until alarm is turned off and the LCD shows nominal airflow reading. After this, if LCD still shows AIRFAIL, Proceed to the next step. - The sensor will take about 2-3 minutes before it can register a new airflow setting. Note: The cabinet must be certified at least annually - If the blower is already operating at maximum voltage and the airflow alarm is still triggered due to the filter being loaded, it's time to change the filter. - If filter is not loaded, proceed to the next step.

CAUSE	CORRECTIVE ACTIONS
Faulty/inaccurate calibration	- Re-calibrate the microprocessor. Follow the calibration procedure in the test report. Warning: To be carried out by an authorized person or service engineer CHECKING CALIBRATION DATA - After calibration is done, go to main display by pressing MENU button twice. - Press SET/Diagnostic button. It will display the mode and software version that is currently being used on the cabinet. Example: DIAGNOSTIC 17:23 MODE: NORMAL CP104D VERSION 1.5 - Press ▼ button until you find the following: TEMPERATURE: Shows actual ambient temperature ADC AFF: ADC value for Fail Point Airflow – calculated using offset based on Airflow Nominal Point. ADC AFN: ADC value for Nominal Point Airflow – set during field calibration. ADC AFA: ADC value for Actual Airflow – showing real time sensor reading. ADC AF0: ADC value for factory calibrated Zero Point Airflow (no airflow) set during zero calibration. ADC AF1: ADC value for factory calibrated Fail Point Airflow. ADC AF2: ADC value for factory calibrated Nominal Point Airflow. CONSTANT: Airflow sensor constant. This value is needed when ordering a new sensor CALIB TEMP: Temperature when the factory calibration was performed. ADC TEMP: ADC value for temperature. Note: Depending on preset unit at SETTINGS>MEASUREMENT UNIT> Metric or Imperial, the units for temperature and velocity will change accordingly. For Metric — °C, m/s while for Imperial — °F, fpm. Default factory setting is Metric. - In order to prevent any AIRFAIL or unstable velocity displayed on LCD problem, please ensure that the following are done after the calibration: - CONSTANT value is correct. Check it with constant written on airflow sensor body or inside Replaceable Component Record List which comes with the test report. - TEMPERATURE shows room temperature. If not, refer to next possible cause, Temperature Sensor Failure, Part B of next situation.

CAUSE	CORRECTIVE ACTIONS
	c. ADC AF2 - ADC AF1 should be at least above 80 points (PASS) and ideally above 100 points (GOOD). If not please redo calibration. If CALIBRATION ERROR is encountered, please refer to next possible cause, part A. Just after calibration, ADC AFA value should be close to ADC AFN value. If not, please adjust the speed controller to make both values closer by at least ± 20 units.
Sensor failure/ misalignment	A. Air Flow Sensor failure/misalignment Important to check: 1. Ensure that the sensor area is not blocked and the through-holes are perpendicular to the filter's surface. 2. If sensor box is used, there should be no gap between the sensor box and filter surface. If doughnut housing is used, the installation direction is in accordance with the airflow direction. 3. Sensor box/housing is installed tightly to the cabinet. - Turn off the cabinet. - Remove the pre-filter and mesh guard. Locate the electrical panel behind it. - Refer to the Component Layout at the beginning of this section to locate the main board in the electrical box. - Using a multimeter, check the exhaust sensor's output voltage (DC) at channel AIN1 (+) and J1 (-) on main board (See Fig. 5.3). - The airflow sensor voltage should increase if airflow is increased and decrease if airflow is decreased. - If your observations are different, check airflow sensor connection on top of the unit (see Fig. 5.3) and at connector D pin 9, 10, 11, 12 on electrical panel for any bad/loose connection. - If connections are all right but sensor is still not working properly, replace it. - If airflow sensor is working properly, proceed to the next step. B. Temperature Sensor Failure - Temperature sensor is located together with the airflow sensor (see Fig. 5.4). - Using a multimeter, check the temperature sensor's output voltage (DC) at channel AIN4 (-) and J4 (+) on main board (see Fig. 5.3). 0.01VDC should represent one degree Celsius. For example, if output is 0.25VDC, it indicates temperature of 25°C. Tolerance of $\pm 1^\circ\text{C}$ is acceptable. - Replace temperature sensor if the reading is wrong. - If temperature sensor is working properly, proceed to the next step.

CAUSE	CORRECTIVE ACTIONS
	 Fig. 5.3  Temperature Sensor Fig. 5.4

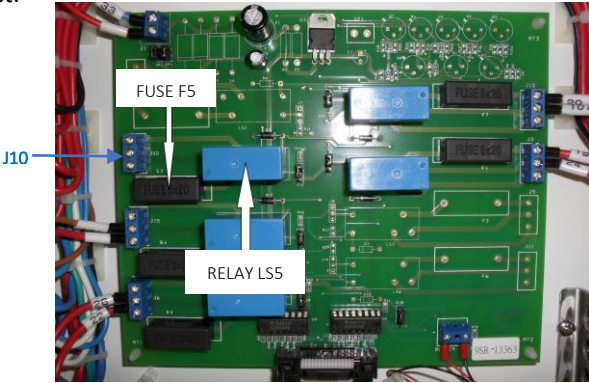
Problem 6

Excessive blower noise

CAUSE	CORRECTIVE ACTION
<i>Resonance</i>	Note: Requires a Certifier or qualified personnel • Locate the motor speed controller. • Measure the motor voltage and record this number. • Increase the motor speed by 5-10 VAC. Does the noise change or go away? • If the noise is gone, measure the actual airflow velocity. This requires a Certifier. ○ If the velocity is still in the acceptable range, maintain it. ○ If velocity is out of acceptable range, or if noise persists, go to next step.
<i>Loose motor or impeller wheel mounting</i>	• Open the blower access panel and check if the blower mounting bolts are tight. • Does the motor rotate freely without noise? • If motor is physically damaged, replace it.

Problem 7

Light(s) always OFF

CAUSE	CORRECTIVE ACTIONS
<i>Burned out fluorescent tube</i>	• Replace the faulty fluorescent tube. • Fluorescent tubes are located inside the blue panel (Vertical-type models) or at the top portion of the work zone (Horizontal-type models). • The white lamp holder can be pulled off from the tubes.
<i>Defective fluorescent ballast or Relay Board issues</i>	• Check incoming voltage to the Relay board. • Locate LS5 circuit and measure AC voltage between BLUE (NEUTRAL) terminal block and the end terminal (with RED cable) on the three terminal strip (J10 on relay board). See Fig. 7.1. • If there is voltage to the end terminal (Terminal with cable), check the central terminal (Normally Open Circuit) to the BLUE (NEUTRAL) terminal block. No voltage at the central terminal indicates a blown fuse or the circuit is not energized. • Check Fuse F5 on relay board (see Fig. 7.1). If fuse F5 is blown, replace F5. • If fuse F5 is okay, check the voltage between the BLUE (NEUTRAL) terminal block and the end terminal on the LS5 circuit (J10 on relay board). • Voltage at the end terminal with no cable indicates that F5 fuse is good and the circuit is not energized. • Make sure that the LIGHT LED is on. • If the ballast has power and the bulbs are new, the problem lies either in the connections or the ballast. • Check Connector C connections. If there are no loose terminals, replace the ballast.  Fig. 7.1
<i>Connection problem</i>	• Locate Connector C and D and check male and female side for any bad connection. • If the fluorescent lamp still does not light with the LIGHT LED ON return to start of Problem 7 and repeat the troubleshooting process.
<i>Faulty switch (applicable for cabinet with simple switch only)</i>	• Fluorescent lamp switch is a RED-colored switch on the control panel. • Switch the unit off and detach the blue front panel of the unit. Unfasten the screws behind and at the inner side of the panel. • Behind the blue panel, ensure UV switch condition is good at ON/OFF position

Problem 8

UV Light always OFF

CAUSE	CORRECTIVE ACTIONS
<i>Faulty UV tube</i>	Replace the faulty UV tube.
<i>Faulty UV ballast</i>	- Check incoming voltage to the Relay board. - Locate LS8 (J15 Terminal) circuit and measure AC voltage between large BLUE (Neutral) terminal block and the end terminal (with RED cable) on the three terminal strip (J15 on relay board). - If there is no incoming voltage to the end terminal (Terminal with cable), refer to the next step NO power to UV Relay circuit LS8. - If voltage to the end terminal (Terminal with cable) exists, check the central terminal (Normally Open Circuit) to the BLUE (Neutral) terminal block. - Absence of voltage at the central terminal indicates a blown fuse or the circuit is not energized. - Check Fuse F8 on relay board (see Fig. 8.1). - If fuse F8 is blown, as temporary solution, use F5 (spare) to replace F8. - If fuse F8 is okay, check the voltage between the BLUE (NEUTRAL) terminal block and the end terminal on the LS8 circuit (on J15 terminal of relay board). - Voltage at the end terminal with no cable indicates that the F8 fuse is good and the circuit is not energized. - Make sure the UV LED is on. - If the ballast has power and the UV lamp is new, the problem is either the cable connections or the UV ballast. - Check Connector B cable connections. If there are no loose terminals, replace the UV ballast.
<i>Faulty switch (applicable for cabinet with simple switch only)</i>	- UV switch is a RED-colored switch on the control panel. - Switch the unit off and detach the blue front panel of the unit. Unfasten the screws behind and at the inner side of the panel. - Behind the blue panel, ensure UV switch condition is good at ON/OFF position



Fig. 8.1

<i>Connection Problem</i>	TURN OFF Warning: Before checking connection, be sure to disconnect main plug. • Turn the cabinet off. • Open the big electrical box and locate the UV ballast and connector E inside. • Check for any loose or bad connection between UV ballast and female connector C. • For connector C, disconnect the male side from electrical box, then check male and female side for any bad connection. • Check for any loose or bad connection between UV switch to UV ballast. Use fingers to pull and check cable tightness. • Check also connection on the UV lamp holder.
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Problem 9

Electrical socket(s) always OFF

CAUSE	CORRECTIVE ACTIONS
<i>Connection problem</i>	• Is the OUTLET LED Illuminated? If not, press the Outlet button at the keypad membrane in the front panel. • Check voltage coming to the Relay board. • Locate LS4 (J6 Terminal) circuit and measure AC voltage between Large BLUE (Neutral) terminal block and the end terminal (with RED cable-COM) on the three terminal strip (J6). • If there is voltage to the end terminal (with RED cable-COM), check the central terminal (Normally Open circuit) to the BLUE (Neutral) terminal block. • No voltage at the central terminal (Normally Open circuit) indicates a blown fuse or the circuit is not energized. • Check Fuse F4 on relay board. <i>See Fig. 9.1 below.</i> • If fuse F4 is blown, replace with F5 fuse. • If fuse F4 is OK, check the voltage between the BLUE (NEUTRAL) terminal block and the end terminal (COM) on the LS4 circuit (J6 on relay board). • Voltage at the end terminal (Normally Closed circuit) with no cable indicates the F4 fuse is good and the circuit is not energized. Make sure the OUTLET LED is on. • If there is power to the relay and it is energized Check voltage across Connector B pin 1 to 2 and pin 4 to 5. • If there is power through Connector B pins, remove socket outlets from wall. • Check incoming power to the plug in the cable to the outlet • No power indicates a wiring issue. Recheck voltage from Connector B pins. • Check for loose or faulty connection between the power cord, IEC inlet and Connector B. Fig. 9.1
<i>Faulty electrical socket</i>	• Socket(s) is/are located inside the work zone. • Check electrical socket connection. • If connection is proper but socket has no output, replace electrical socket.

CAUSE	CORRECTIVE ACTIONS
<i>Faulty switch (applicable for cabinet with simple switch only)</i>	• Socket switch is a BLACK-colored switch on the control panel. • Switch the unit off and detach the blue front panel of the unit. Unfasten the screws behind and at the inner side of the panel. • Behind the blue panel, ensure socket switch condition is good at ON and OFF position.

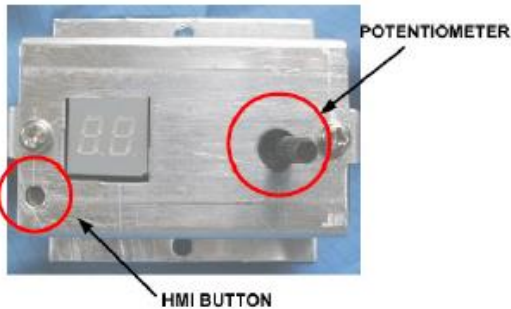
Problem 10

Contaminated sample/lab

CAUSE	CORRECTIVE ACTIONS
<i>Cabinet overloaded with too many items</i>	 Note: Contaminated samples are normally the result of any of the following: ○ Improper technique of the user or lack of training. ○ Samples being contaminated in the incubator or somewhere other than the laminar flow cabinet. Items to check on the laminar flow cabinet: • Make sure that items inside the work zone are properly placed i.e. no large object is obstructing the airflow or items are placed downstream of another item (in Horizontal LFC). • Remove all equipment and other items from the cabinet and surface decontaminate the work top. • Surface decontaminate or better yet, sterilize everything that is returned to the laminar flow cabinet. • If the contamination problem persists, check for HEPA filter leaks and recertify the cabinet.

Problem 11

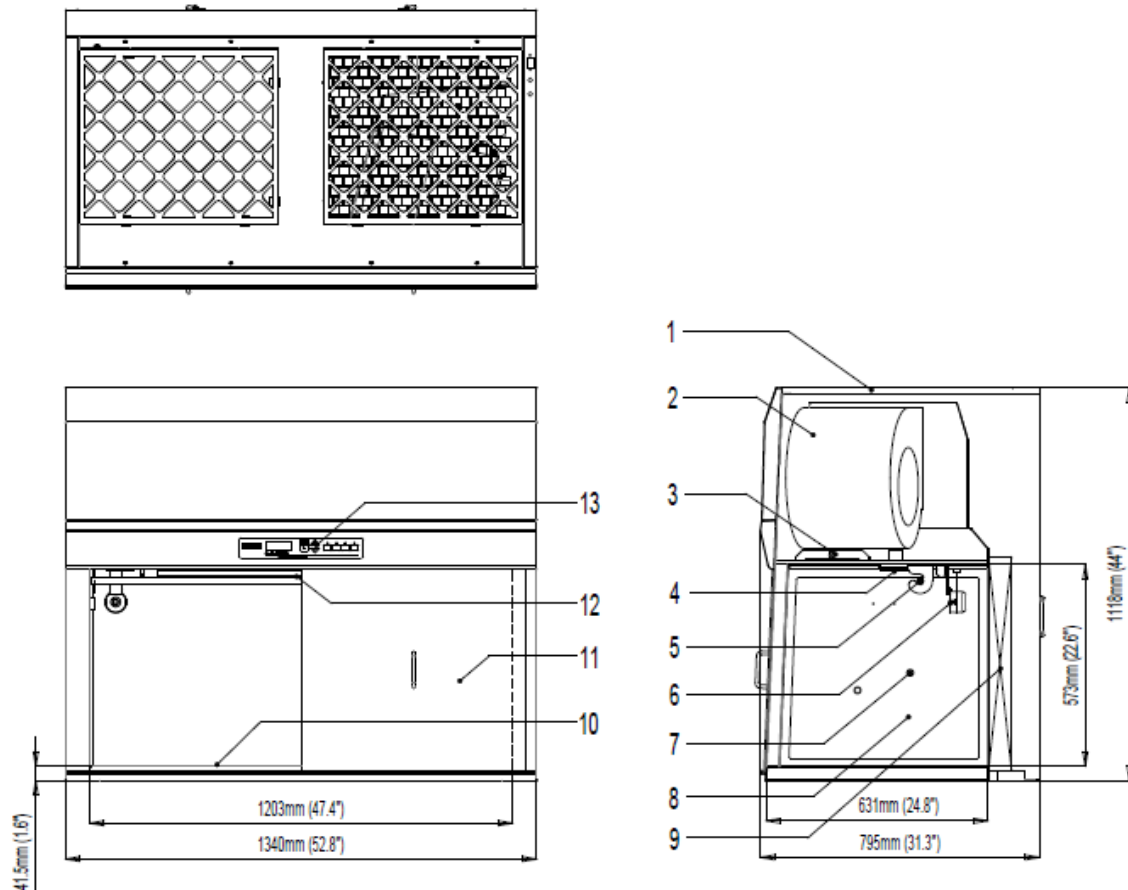
Motor Speed Controller (ECMS) not functioning

CAUSE	CORRECTIVE ACTIONS
<i>No output; the power indicator LEDs are OFF</i>	• Turn the system on by supplying 12 VDC to the ECMS • Replace the ECMS with a new one (see Fig. 11.1)  Fig. 11.1
<i>No output, but the power indicator LEDs are ON</i>	• Make sure that the correct DC line voltage is connected properly and turned on. • Replace the ECMS with a new one.  Fig. 11.2 ECMS with power indicator
<i>The output voltage cannot be adjusted by turning the potentiometer</i>	• Replace the ECMS with a new one  Fig. 11.3 ECMS showing HMI button and potentiometer

CAUSE	CORRECTIVE ACTIONS
<i>The half speed mode cannot be activated</i>	• Make sure both of the digital input has been shorted. You can check the continuity of the digital input by using a Multimeter. • Replace the ECMS with a new one.

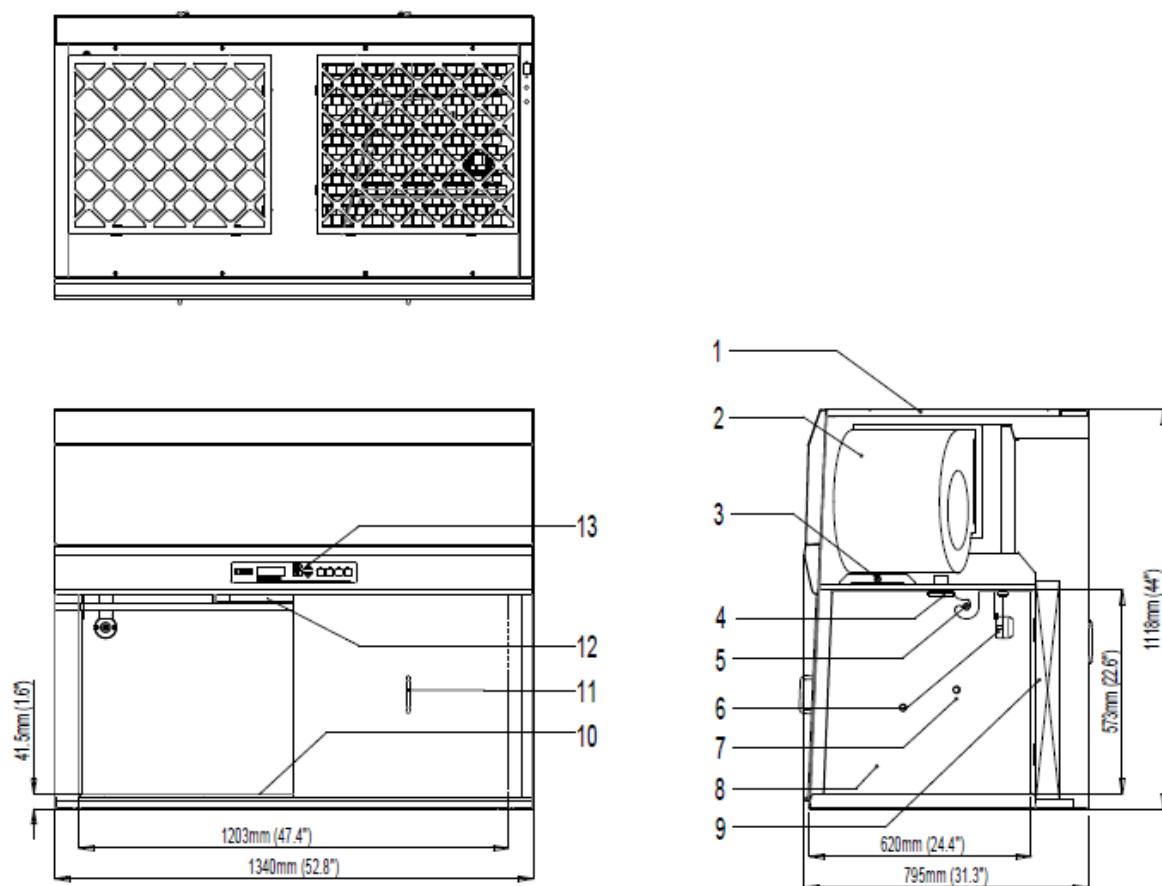
Chapter 9 – Product Specifications

9.1 LHG Engineering Details



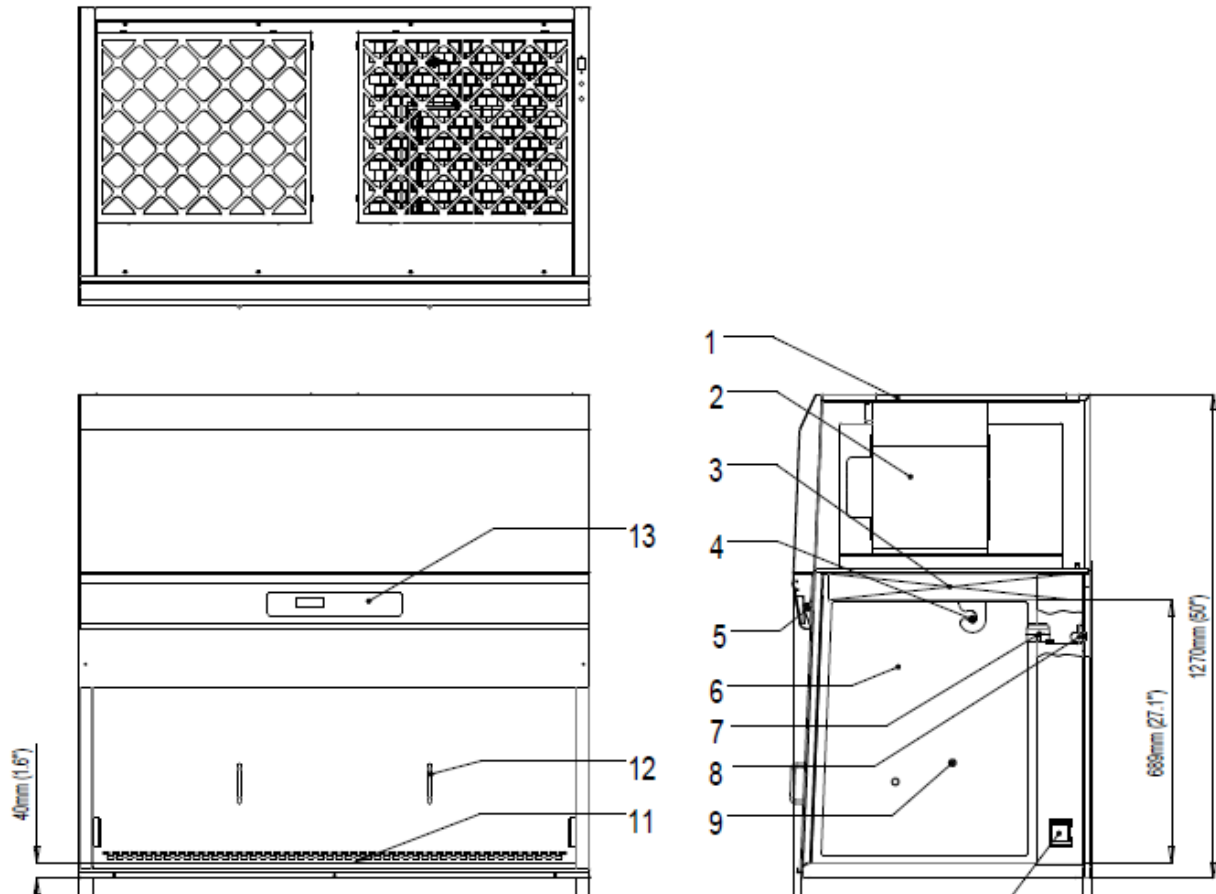
1. Pre-filter
2. DC ECM Blower (Self-compensating and lower noise)
3. Fluorescent Lamp
4. Electrical Outlet Retrofit Kit Provision (maximum of 2)
5. IV Bar Retrofit Kit Provision
6. Airflow Sensor
7. Service Fixture Retrofit Kit Provision (2 penetrations on each side)
8. Tempered Glass Side Panels
9. ULPA Filter
10. Spill-retaining Stainless Steel Work Top
11. Front Cover (optional)
12. UV Light Retrofit Kit Provision
13. Esco Sentinel™ Gold Microprocessor Control System

9.2 LHS Engineering Details



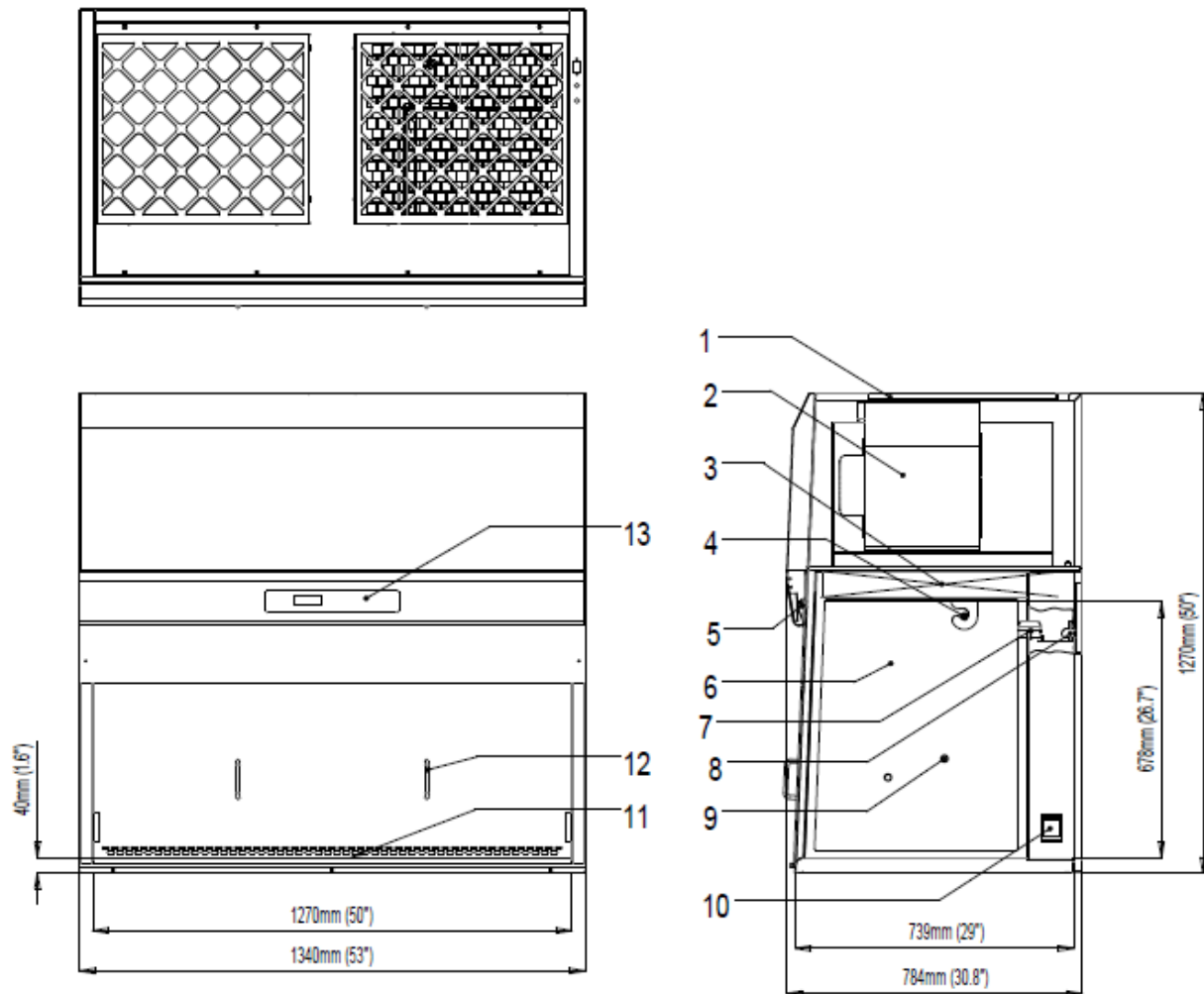
1. Pre-filter
2. DC ECM Blower (Self-compensating and lower noise)
3. Fluorescent Lamp
4. Electrical Outlet Retrofit Kit Provision (maximum of 2)
5. IV Bar Retrofit Kit Provision
6. Airflow Sensor
7. Service Fixture Retrofit Kit Provision (2 penetrations on each side)
8. Stainless Steel Side Panels
9. ULPA Filter
10. Spill-retaining Stainless Work Top
11. Front Cover (optional)
12. UV Light Retrofit Kit Provision
13. Esco Sentinel™ Gold Microprocessor Control System

9.3 LVG Engineering Details



1. Pre-filter
2. ECM Blower (Self-compensating and lower noise)
3. ULPA Filter
4. IV Bar Retrofit Kit Provision
5. Fluorescent Lamp
6. Tempered Glass Side Panels
7. Airflow Sensor
8. UV Lamp
9. Service Fixture Retrofit Kit Provision (2 holes on each side)
10. Electrical Outlet Retrofit Kit Provision (1 outlet on each side)
11. Spill-retaining Stainless Steel Work Top
12. Front Cover (optional)
13. Esco Sentinel™ Gold Microprocessor Control System

9.4 LVS Engineering Details



1. Pre-filter
2. ECM Blower (Self-compensating and lower noise)
3. ULPA Filter
4. IV Bar Retrofit Kit Provision
5. Fluorescent Lamp
6. Stainless Steel Side Panels
7. Airflow Sensor
8. UV Lamp
9. Service Fixture Retrofit Kit Provision (2 holes on each side)
10. Electrical Outlet Retrofit Kit Provision (1 outlet on each side)
11. Spill-retaining Stainless Steel Work Top
12. Front Cover (optional)
13. Esco Sentinel™ Gold Microprocessor Control System

9.5 Technical Specification Summary Table

9.5.1 LHG/LHS A-Height (Interior Height: 2 ft. / 0.6 m)

Airstream® Horizontal Laminar Flow Clean Benches, A-Height (interior Height: 2ft. / 0.6 m)					
Stainless Steel Sides		LHS-3AG-F8 2120377	LHS-4AG-F8 2120378	LHS-5AG-F8 2120379	LHS-6AG-F8 2120380
		LHS-3AG-F9 2120425	LHS-4AG-F9 2120427	LHS-5AG-F9 2120429	LHS-6AG-F9 2120431
Glass Sides		LHG-3AG-F8 2120387	LHG-4AG-F8 2120368	LHG-5AG-F8 2120372	LHG-6AG-F8 2120373
		LHG-3AG-F9 2120417	LHG-4AG-F9 2120419	LHG-5AG-F9 2120421	LHG-6AG-F9 2120423
Nominal Size		0.9 meter (3')	1.2 meter (4')	1.5 meter (5')	1.8 meter (6')
External Dimensions (W x D x H)	Without Base Stand	1035 x 795 x 1118 mm (40.8" x 31.3" x 44.0")	1340 x 795 x 1118 mm (52.8" x 31.3" x 44.0")	1645 x 795 x 1118 mm (64.8" x 31.3" x 44.0")	1955 x 795 x 1118 mm (76.8" x 31.3" x 44.0")
Internal Work Area, Dimensions (W x D x H)	LHG Models	898 x 631 x 573 mm (35.4" x 24.8" x 22.5")	1203 x 631 x 573 mm (47.4" x 24.8" x 22.5")	1508 x 631 x 573 mm (59.4" x 24.8" x 22.5")	1813 x 631 x 573 mm (71.4" x 24.8" x 22.5")
	LHS Models	898 x 620 x 573 mm (35.4" x 24.4" x 22.5")	1203 x 620 x 573 mm (47.4" x 24.4" x 22.5")	1508 x 620 x 573 mm (59.4" x 24.4" x 22.5")	1813 x 620 x 573 mm (71.4" x 24.4" x 22.5")
Internal Work Area, Space		0.5 m ² (5.4 ft ²)	0.7 m ² (7.5 ft ²)	0.9 m ² (9.6 ft ²)	1.0 m ² (10.8 ft ²)
Average Airflow Velocity		0.45 m/s (90 fpm) at initial set point			
Air Volume		834 m ³ /hr. (491 cfm)	1117 m ³ /hr. (657 cfm)	1400 m ³ /hr. (824 cfm)	1683 m ³ /hr. (911 cfm)
ULPA Filter Typical Efficiency		>99.999% at particle size between 0.1 to 0.3µm			
Sound Emission per IEST-RP-CC002.2*		53.2 dBA	55.8 dBA	58.4 dBA	60.0 dBA
Fluorescent Lamp Intensity at Zero Ambient		1045 Lux (97 foot candles)	1139 Lux (106 foot candles)	984 Lux (91 foot candles)	1221 Lux (113 foot candles)
Cabinet Construction	Main Body	1.2 mm (0.05") 18-gauge electro-galvanized steel with white, oven baked epoxy-polyester powder-coated finish			
	Work Zone	1.2 mm (0.05") 18-gauge stainless steel, grade 304, with 4B finish			
	Side Walls	LHS Models: 1.2 mm (0.05") 18 gauge stainless steel, grade 304, with 4B finish LHG Models: UV absorbing tempered glass, 5 mm (0.2"), colorless and transparent			
Electrical Power Rating 8: 220-240 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	7 A	7.3 A	7.6 A	8 A
	Optional Outlets (FLA)	5 A			
	Cabinet Nominal Power (W)	140	171	211	249
	Heat Rejected, BTU per Hour	478	583	720	850
Electrical Power Rating 9: 110-130 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	11.3 A	11.4 A	11.4 A	11.5 A
	Optional Outlets (FLA)	5 A			
	Cabinet Nominal Power (W)	144	175	216	255
	Heat Rejected, BTU per Hour	491	597	737	870
Net Weight**		100 Kg (220 lbs.)	145 Kg (320 lbs.)	167 Kg (368 lbs.)	212 Kg (467 lbs.)
Shipping Weight**		132 Kg (291 lbs.)	200 Kg (440 lbs.)	224 Kg (494 lbs.)	277 Kg (611 lbs.)
Shipping Dimensions, Maximum (W x D x H)**		1120 x 900 x 1590 mm (44" x 35" x 62")	1400 x 900 x 1590 mm (55" x 35" x 62")	1720 x 900 x 1590 mm (68" x 35" x 62")	2200 x 900 x 1590 mm (87" x 35" x 62")
Shipping Volume, Maximum**		1.6 m ³ (56.6 ft ³)	2.0 m ³ (70.6 ft ³)	2.5 m ³ (88.2 ft ³)	3.3 m ³ (116.5 ft ³)

Specifications are subject to change without notice.

*Noise reading in open field condition/ anechoic chamber. Noise reading in normal room varies by room size, layout, and background noise, but may reach roughly 3 dBA above these values.

**Cabinet only, excludes optional stand.

9.5.2 LVG/LVS A-Height with Fixed Sash (Interior Height: 2.25 ft. / 0.7 m)

Airstream® Vertical Laminar Flow Clean Benches, A-Height (interior Height: 2.25 ft. / 0.7 m)					
Stainless Steel Sides		LVS-3AG-F8 2120381	LVS-4AG-F8 2120382	LVS-5AG-F8 2120383	LVS-6AG-F8 2120384
		LVS-3AG-F9 2120443	LVS-4AG-F9 2120445	LVS-5AG-F9 2120447	LVS-6AG-F9 2120449
Glass Sides		LVG-3AG-F8 2120374	LVG-4AG-F8 2120369	LVG-5AG-F8 2120375	LVG-6AG-F8 2120407
		LVG-3AG-F9 2120435	LVG-4AG-F9 2120437	LVG-5AG-F9 2120439	LVG-6AG-F9 2120441
Nominal Size		0.9 meter (3')	1.2 meter (4')	1.5 meter (5')	1.8 meter (6')
External Dimensions (W x D x H)	Without Base Stand	1035 x 784 x 1270 mm (40.8" x 30.8" x 50.0")	1340 x 784 x 1270 mm (52.8" x 30.8" x 50.0")	1645 x 784 x 1270 mm (64.8" x 30.8" x 50.0")	1950 x 784 x 1270 mm (76.8" x 30.8" x 50.0")
Internal Work Area, Dimensions (W x D x H)	LVG Models	965 x 739 x 689 mm (38.0" x 29.1" x 27.1")	1270 x 739 x 689 mm (50.0" x 29.1" x 27.1")	1575 x 739 x 689 mm (62.0" x 29.1" x 27.1")	1880 x 739 x 689 mm (74.0" x 29.1" x 27.1")
	LVS Models	965 x 739 x 678 mm (38.0" x 29.1" x 26.7")	1270 x 739 x 678 mm (50.0" x 29.1" x 26.7")	1575 x 739 x 678 mm (62.0" x 29.1" x 26.7")	1880 x 739 x 678 mm (74.0" x 29.1" x 26.7")
Internal Work Area, Space		0.6 m ² (6.5 ft ²)	0.8 m ² (8.6 ft ²)	1.0 m ² (10.7 ft ²)	1.3 m ² (14.0 ft ²)
Average Airflow Velocity		0.45 m/s (90 fpm) at initial set point			
Air Volume		1117 m ³ /hr. (657 cfm)	1471 m ³ /hr. (866 cfm)	1824 m ³ /hr. (1074 cfm)	2177 m ³ /hr. (1281 cfm)
ULPA Filter Typical Efficiency		>99.999% at particle size between 0.1 to 0.3µm			
Sound Emission per IEST-RP-CC002.2*		51.6 dBA	52.4 dBA	55.6 dBA	57.6 dBA
Fluorescent Lamp Intensity at Zero Ambient		980 Lux (94 foot candles)	904 Lux (84 foot candles)	894 Lux (83 foot candles)	1062 Lux (99 foot candles)
Cabinet Construction	Main Body	1.2 mm (0.05") 18-gauge electro-galvanized steel with white oven baked epoxy-polyester powder-coated finish			
	Work Zone	1.2 mm (0.05") 18-gauge stainless steel, grade 304, with 4B finish			
	Side Walls	LVS Models: 1.2 mm (0.05") 18 gauge stainless steel, grade 304, with 4B finish LVG Models: UV absorbing tempered glass, 5 mm (0.2"), colorless and transparent			
Electrical Power Rating 8: 220-240 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	7.3 A	7.5 A	7.8 A	8.5 A
	Optional Outlets (FLA)	5 A			
	Cabinet Nominal Power (W)	129	151	199	258
	Heat Rejected, BTU per Hour	440	515	679	880
Electrical Power Rating 9: 110-130 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	12 A	12.5 A	13.3 A	13.5 A
	Optional Outlets (FLA)	5 A			
	Cabinet Nominal Power (W)	132	155	204	264
	Heat Rejected, BTU per Hour	450	529	696	908
Net Weight**		135 Kg (298 lbs.)	158 Kg (348 lbs.)	199 Kg (438 lbs.)	208 Kg (459 lbs.)
Shipping Weight**		167 Kg (368 lbs.)	202 Kg (445 lbs.)	256 Kg (564 lbs.)	273 Kg (602 lbs.)
Shipping Dimensions, Maximum (W x D x H)**		1120 x 900 x 1590 mm (44" x 35" x 62")	1400 x 900 x 1590 mm (55" x 35" x 62")	1720 x 900 x 1590 mm (68" x 35" x 62")	2200 x 900 x 1590 mm (87" x 35" x 62")
Shipping Volume, Maximum**		1.6 m ³ (56.6 ft ³)	2.0 m ³ (70.6 ft ³)	2.5 m ³ (88.2 ft ³)	3.3 m ³ (116.5 ft ³)

Specifications are subject to change without notice.

*Noise reading in open field condition/ anechoic chamber. Noise reading in normal room varies by room size, layout, and background noise, but may reach roughly 3 dBA above these values.

**Cabinet only, excludes optional stand.

9.5.3 LVG/LVS A-Height with Sliding Sash (Interior Height: 2.25 ft. / 0.7 m)

Airstream® Vertical Laminar Flow Clean Benches, A-Height (interior Height: 2.25 ft. / 0.7 m)			
Stainless Steel Sides		LVS-4AG-S8 2120759	LVS-6AG-S8 2120760
		LVS-4AG-S9 2120762	LVS-6AG-S9 2120764
Glass Sides		LVG-4AG-S8 2120701	LVG-6AG-S8 2120703
		LVG-4AG-S9 2120761	LVG-6AG-S9 2120763
Nominal Size		1.2 meter (4')	1.8 meter (6')
External Dimensions (W x D x H)	Without Base Stand	1340 x 780 x 1270 mm (52.8" x 30.7" x 50.0")	1950 x 784 x 1270 mm (76.8" x 30.8" x 50.0")
Internal Work Area, Dimensions (W x D x H)	LVG Models	1270 x 695 x 689 mm (50.0" x 27" x 27.1")	1880 x 739 x 689 mm (74.0" x 29.1 x 27.1")
	LVS Models	1270 x 695 x 678 mm (50.0" x 27" x 26.7")	1880 x 739 x 678 mm (74.0" x 29.1 x 26.7")
Internal Work Area, Space		0.8 m ² (8.6 ft ²)	1.3 m ² (14.0 ft ²)
Average Airflow Velocity		0.45 m/s (90 fpm) at initial set point	
Air Volume		1471 m ³ /hr. (866 cfm)	2177 m ³ /hr. (1281 cfm)
ULPA Filter Typical Efficiency		>99.999% at particle size between 0.1 to 0.3µm	
Sound Emission per IEST-RP-CC002.2*		52.4 dBA	57.6 dBA
Fluorescent Lamp Intensity at Zero Ambient		904 Lux (84 foot candles)	1062 Lux (99 foot candles)
Cabinet Construction	Main Body	1.2 mm (0.05") 18-gauge electro-galvanized steel with white oven baked epoxy-polyester powder-coated Finish	
	Work Zone	1.2 mm (0.05") 18-gauge stainless steel, grade 304, with 4B finish	
	Side Walls	LVS Models: 1.2 mm (0.05") 18 gauge stainless steel, grade 304, with 4B finish LVG Models: UV resistant tempered glass, 5 mm (0.2"), colorless and transparent	
	Sash Window	Manual Sliding Sash: 5 mm (0.2") UV Resistant Tempered Glass with counter-balance system and ergonomic handle	
Electrical Power Rating 8: 220-240 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	7.5 A	8.5 A
	Optional Outlets (FLA)	5 A	
	Cabinet Nominal Power (W)	151	258
	Heat Rejected, BTU per Hour	515	880
Electrical Power Rating 9: 110-130 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	12.5 A	13.5 A
	Optional Outlets (FLA)	5 A	
	Cabinet Nominal Power (W)	155	264
	Heat Rejected, BTU per Hour	529	908
Net Weight**		158 Kg (348 lbs.)	208 Kg (459 lbs.)
Shipping Weight**		202 Kg (445 lbs.)	273 Kg (602 lbs.)
Shipping Dimensions, Maximum (W x D x H)**		1400 x 900 x 1590 mm (55" x 35" x 62")	2200 x 900 x 1590 mm (87" x 35" x 62")
Shipping Volume, Maximum**		2.0 m ³ (70.6 ft. ³)	3.3 m ³ (116.5 ft. ³)

Specifications are subject to change without notice.

*Noise reading in open field condition/ anechoic chamber. Noise reading in normal room varies by room size, layout, and background noise, but may reach roughly 3 dBA above these values.

**Cabinet only, excludes optional stand.

9.5.4 LHG/LHS B-Height (Interior Height: 2.5 ft. / 0.7 m)

Airstream® Horizontal Laminar Flow Clean Benches, A-Height (interior Height: 2ft. / 0.6 m)					
Stainless Steel Sides		LHS-3BG-F8 2120463	LHS-4BG-F8 2120465	LHS-5BG-F8 2120467	LHS-6BG-F8 2120469
		LHS-3BG-F9 2120503	LHS-4BG-F9 2120505	LHS-5BG-F9 2120507	LHS-6BG-F9 2120509
Glass Sides		LHG-3BG-F8 2120453	LHG-4BG-F8 2120455	LHG-5BG-F8 2120457	LHG-6BG-F8 2120459
		LHG-3BG-F9 2120493	LHG-4BG-F9 2120495	LHG-5BG-F9 2120497	LHG-6BG-F9 2120499
Nominal Size		0.9 meter (3')	1.2 meter (4')	1.5 meter (5')	1.8 meter (6')
External Dimensions (W x D x H)	Without Base Stand	1035 x 788 x 1270 mm (40.8" x 31.0" x 50.0")	1340 x 788 x 1270 mm (52.8" x 31.0" x 50.0")	1645 x 788 x 1270 mm (64.8" x 31.0" x 50.0")	1950 x 788 x 1270 mm (76.8" x 31.0" x 50.0")
Internal Work Area, Dimensions (W x D x H)	LHG Models	898 x 631 x 725 mm (35.4" x 24.8 x 28.5")	1203 x 631 x 725 mm (47.4" x 24.8 x 28.5")	1508 x 631 x 725 mm (59.4" x 24.8 x 28.5")	1813 x 631 x 725 mm (71.4" x 24.8 x 28.5")
	LHS Models	898 x 620 x 725 mm (35.4" x 24.4 x 28.5")	1203 x 620 x 725 mm (47.4" x 24.4 x 28.5")	1508 x 620 x 725 mm (59.4" x 24.4 x 28.5")	1813 x 620 x 725 mm (71.4" x 24.4 x 28.5")
Internal Work Area, Space		0.5 m ² (5.4 ft ²)	0.7 m ² (7.5 ft ²)	0.9 m ² (9.6 ft ²)	1.0 m ² (10.8 ft ²)
Average Airflow Velocity		0.45 m/s (90 fpm) at initial set point			
Air Volume		1055 m ³ /hr. (621 cfm)	1413 m ³ /hr. (832 cfm)	1771 m ³ /hr. (1042 cfm)	2129 m ³ /hr. (1253 cfm)
ULPA Filter Typical Efficiency		>99.999% at particle size between 0.1 to 0.3µm			
Sound Emission per IEST-RP-CC002.2*		52.8 dBA	55.4 dBA	58.0 dBA	59.6 dBA
Fluorescent Lamp Intensity at Zero Ambient		1279 Lux (119 foot candles)	1394 Lux (130 foot candles)	1204 Lux (112 foot candles)	1494 Lux (139 foot candles)
Cabinet Construction	Main Body	1.2 mm (0.05") 18-gauge electro-galvanized steel with white oven baked epoxy-polyester powder-coated finish			
	Work Zone	1.2 mm (0.05") 18-gauge stainless steel, grade 304, with 4B finish			
	Side Walls	LHS Models: 1.2 mm (0.05") 18 gauge stainless steel, grade 304, with 4B finish LHG Models: UV absorbing tempered glass, 5 mm (0.2"), colorless and transparent			
Electrical Power Rating 8: 220-240 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	7 A	7.3 A	7.6 A	8 A (2 Power Inlet)
	Optional Outlets (FLA)	5 A			
	Cabinet Nominal Power (W)	160	207	255	302
	Heat Rejected, BTU per Hour	546	702	870	1030
Electrical Power Rating 9: 110-130 VAC, 50/60Hz, 1 phase	Cabinet Full Load Amps (FLA)	11.3 A	11.4 A	11.4 A	11.5 A and 6.5 A (2 Power Inlet)
	Optional Outlets (FLA)	5 A			
	Cabinet Nominal Power (W)	163	211	260	308
	Heat Rejected, BTU per Hour	556	720	887	1051
Net Weight**		108 Kg (238 lbs.)	156 Kg (344 lbs.)	180 Kg (397 lbs.)	228 Kg (503 lbs.)
Shipping Weight**		140 Kg (308 lbs.)	200 Kg (441 lbs.)	237 Kg (522 lbs.)	293 Kg (646 lbs.)
Shipping Dimensions, Maximum (W x D x H)**		1120 x 900 x 1590 mm (44" x 35" x 62")	1400 x 900 x 1590 mm (55" x 35" x 62")	1720 x 900 x 1590 mm (68" x 35" x 62")	2200 x 900 x 1590 mm (87" x 35" x 62")
Shipping Volume, Maximum**		1.6 m ³ (56.6 ft ³)	2.0 m ³ (70.6 ft ³)	2.5 m ³ (88.2 ft ³)	3.3 m ³ (116.5 ft ³)

Specifications are subject to change without notice.

*Noise reading in open field condition/ anechoic chamber. Noise reading in normal room varies by room size, layout, and background noise, but may reach roughly 3 dBA above these values.

**Cabinet only, excludes optional stand.

9.5.5 LHG/LHS and LVG/LVS C-HEIGHT (Interior Height: 3 ft / 0.9 m)

Airstream® Horizontal and Vertical Laminar Flow Clean Benches, C-Height (Interior Height: 3 ft. / 0.9 m)											
Model	Nominal Size	External Dimension (W x D x H)	Internal Dimension (W x D x H)	Sound Emission per IEST-RP-CC002.2*	Fluorescent Lamp Intensity at Zero Ambient	Cabinet Full Load Amps (FLA)	Cabinet Nominal Power (W)	Heat Rejected, BTU per Hour	Shipping Weight **	Shipping Dimensions, Maximum (W x D x H)**	Shipping Volume, Maximum **
VERTICAL LAMINAR FLOW (Power Rating 8: 220-240 VAC, 50/60 Hz, 1 phase)											
LVS-4CG-F8 2120565	4 ft. (1.2 m)	1340 x 774 x 1645 mm (52.8" x 30.5" x 64.8")	1270 x 739 x 905 mm (50.0" x 29.1" x 35.6")	52.5 dBA	904 Lux (84 fc)	7.5 A	213	727	243 Kg (536 lbs.)	1490 x 900 x 1750 mm (58.7" x 35.4" x 68.9")	2.3 m³ (81.2 ft³)
LVS-6CG-F8 2120569	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1880 x 739 x 905 mm (74.0" x 29.1" x 35.6")	52.5 dBA	1062 Lux (99 fc)	8.5 A	392	1337	330 Kg (661 lbs.)	2070 x 900 x 1750 mm (81.5" x 35.4" x 68.9")	3.3 m³ (116.5 ft³)
LVS-8CG-F8 2120686	8 ft. (2.4 m)	2520 x 774 x 1645 mm (99.2" x 30.5" x 64.8")	2450 x 739 x 905 mm (96.5" x 29.1" x 35.6")	52.5 dBA	1100 Lux (102 fc)	8.5 A and 3.5 A	422	144	445Kg (1003 lbs.)	2720 x 900 x 1750 mm (107" x 35.4" x 68.9")	4.5 m³ (158.9 ft³)
LVG-4CG-F8 2120555	4 ft. (1.2 m)	1340 x 774 x 1645 mm (52.8" x 30.5" x 64.8")	1270 x 739 x 915 mm (50.0" x 29.1" x 36.0")	52.5 dBA	904 Lux (84 fc)	7.5 A	213	727	243 Kg (536 lbs.)	1490 x 900 x 1750 mm (58.7" x 35.4" x 68.9")	2.3 m³ (81.2 ft³)
LVG-6CG-F8 2120559	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1880 x 739 x 915 mm (74.0" x 29.1" x 36.0")	52.5 dBA	1062 Lux (99 fc)	8.5 A	392	1337	330 Kg (661 lbs.)	2070 x 900 x 1750 mm (81.5" x 35.4" x 68.9")	3.3 m³ (116.5 ft³)
VERTICAL LAMINAR FLOW (Power Rating 9: 110-130 VAC, 50/60 Hz, 1 phase)											
LVS-4CG-F9 2120605	4 ft. (1.2 m)	1340 x 774 x 1645 mm (52.8" x 30.5" x 64.8")	1270 x 739 x 905 mm (50.0" x 29.1" x 35.6")	52.5 dBA	904 Lux (84 fc)	11.5 A	217	741	243 Kg (536 lbs.)	1490 x 900 x 1750 mm (58.7" x 35.4" x 68.9")	2.3 m³ (81.2 ft³)
LVS-6CG-F9 2120609	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1880 x 739 x 905 mm (74.0" x 29.1" x 35.6")	52.5 dBA	1062 Lux (99 fc)	13.5 A	400	1365	330 Kg (661 lbs.)	2070 x 900 x 1750 mm (81.5" x 35.4" x 68.9")	3.3 m³ (116.5 ft³)
LVS-8CG-F9 2120687	8 ft. (2.4 m)	2520 x 774 x 1645 mm (99.2" x 30.5" x 64.8")	2450 x 739 x 905 mm (96.5" x 29.1" x 35.6")	52.5 dBA	1100 Lux (102 fc)	14 A and 8 A	434	1481	445Kg (1003 lbs.)	2720 x 900 x 1750 mm (107" x 35.4" x 68.9")	4.5 m³ (158.9 ft³)
LVG-4CG-F9 2120595	4 ft. (1.2 m)	1340 x 774 x 1645 mm (52.8" x 30.5" x 64.8")	1270 x 739 x 915 mm (50.0" x 29.1" x 36.0")	52.5 dBA	904 Lux (84 fc)	11.5 A	217	741	243 Kg (536 lbs.)	1490 x 900 x 1750 mm (58.7" x 35.4" x 68.9")	2.3 m³ (81.2 ft³)
LVG-6CG-F9 2120599	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1880 x 739 x 915 mm (74.0" x 29.1" x 36.0")	52.5 dBA	1062 Lux (99 fc)	13.5 A	400	1365	330 Kg (661 lbs.)	2070 x 900 x 1750 mm (81.5" x 35.4" x 68.9")	3.3 m³ (116.5 ft³)
HORIZONTAL LAMINAR FLOW (Power Rating 8: 220-240 VAC, 50/60 Hz, 1 phase)											
LHS-4CG-F8 2120545	4 ft. (1.2 m)	1340 x 782 x 1422 mm (52.8" x 30.8" x 56.0")	1203 x 620 x 877 mm (47.4" x 24.4" x 34.5")	56.4 dBA	934 Lux (87 fc)	7.3 A	246	840	222 Kg (489 lbs.)	1400 x 900 x 1590 mm (55.1" x 35.4" x 62.5")	2.0 m³ (70.6 ft³)

LHS-6CG-F8 2120549	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1813 x 620 x 877 mm (71.4" x 24.4" x 34.5")	59.4 dBA	1001 Lux (93 fc)	8 A and 3 A	359	1225	301 Kg (664 lbs.)	2200 x 900 x 1590 mm (86.6" x 35.4" x 62.5")	3.3 m ³ (116.5 ft ³)
LHS-8CG-F8 2120704	8 ft. (2.4 m)	2520 x 774 x 1645 mm (99.2" x 30.5" x 64.8")	2383 x 620 x 877 mm (93.8" x 24.4" x 34.5")	62.3 dBA	1136 Lux (106 fc)	8.3 A and 3.3 A	388	1324	397 Kg (875 lbs.)	2720 x 900 x 1590 mm (107.0" x 35.4" x 62.5")	4.1 m ³ (144.8 ft ³)
LHG-3CG-F8 2120533	3 ft. (0.9 m)	1035 X 782 X 1422 mm (40.8" x 30.8" x 56.0")	898 x 631 x 877 mm (35.4" x 24.8" x 34.5")	54.2 dBA	1195 Lux (111 fc)	7:00 AM	190	648	147Kg (324 lbs.)	1120 x 900 x 1590 mm (44.0" x 35.4" x 62.5")	1.6 m ³ (56.5 ft ³)
LHG-4CG-F8 2120535	4 ft. (1.2 m)	1340 x 782 x 1422 mm (52.8" x 30.8" x 56.0")	1203 x 631 x 877 mm (47.4" x 24.8" x 34.5")	56.4 dBA	934 Lux (87 fc)	7.3 A	246	840	222 Kg (489 lbs.)	1400 x 900 x 1590 mm (55.1" x 35.4" x 62.5")	2.0 m ³ (70.6 ft ³)
LHG-6CG-F8 2120539	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1813 x 631 x 877 mm (71.4" x 24.8" x 34.5")	59.4 dBA	1001 Lux (93 fc)	8 A and 3 A	359	1225	301 Kg (664 lbs.)	2200 x 900 x 1590 mm (86.6" x 35.4" x 62.5")	3.3 m ⁴ (116.5 ft ³)
HORIZONTAL LAMINAR FLOW (Power Rating 9: 110-130 VAC, 50/60 Hz, 1 phase)											
LHS-4CG-F9 2120585	4 ft. (1.2 m)	1340 x 782 x 1422 mm (52.8" x 30.8" x 56.0")	1203 x 620 x 877 mm (47.4" x 24.4" x 34.5")	56.4 dBA	934 Lux (87 fc)	11.4 A	250	853	222 Kg (489 lbs.)	1400 x 900 x 1590 mm (55.1" x 35.4" x 62.5")	2.0 m ³ (70.6 ft ³)
LHS-6CG-F9 2120589	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1813 x 620 x 877 mm (71.4" x 24.4" x 34.5")	59.4 dBA	1001 Lux (93 fc)	11.5 A and 6.5 A	365	1245	301 Kg (664 lbs.)	2200 x 900 x 1590 mm (86.6" x 35.4" x 62.5")	3.3 m ³ (116.5 ft ³)
LHS-8CG-F9 2120719	8 ft. (2.4 m)	2520 x 774 x 1645 mm (99.2" x 30.5" x 64.8")	2383 x 620 x 877 mm (93.8" x 24.4" x 34.5")	62.3 dBA	1136 Lux (106 fc)	12 A and 6 A	394	1344	397 Kg (875 lbs.)	2720 x 900 x 1590 mm (107.0" x 35.4" x 62.5")	4.1 m ³ (144.8 ft ³)
LHG-3CG-F9 2120573	3 ft. (0.9 m)	1035 X 782 X 1422 mm (40.8" x 30.8" x 56.0")	898 x 631 x 877 mm (35.4" x 24.8" x 34.5")	54.2 dBA	1195 Lux (111 fc)	11.3 A	193	659	147Kg (324 lbs.)	1120 x 900 x 1590 mm (44.0" x 35.4" x 62.5")	1.6 m ³ (56.5 ft ³)
LHG-4CG-F9 2120575	4 ft. (1.2 m)	1340 x 782 x 1422 mm (52.8" x 30.8" x 56.0")	1203 x 631 x 877 mm (47.4" x 24.8" x 34.5")	56.4 dBA	934 Lux (87 fc)	11.4 A	250	853	222 Kg (489 lbs.)	1400 x 900 x 1590 mm (55.1" x 35.4" x 62.5")	2.0 m ³ (70.6 ft ³)
LHG-6CG-F9 2120579	6 ft. (1.8 m)	1950 x 774 x 1645 mm (76.8" x 30.5" x 64.8")	1813 x 631 x 877 mm (71.4" x 24.8" x 34.5")	59.4 dBA	1001 Lux (93 fc)	11.5 A and 6.5 A	365	1245	301 Kg (664 lbs.)	2200 x 900 x 1590 mm (86.6" x 35.4" x 62.5")	3.3 m ⁴ (116.5 ft ³)

SPECIFICATIONS AVAILABLE FOR ALL MODELS AND SIZES		
Cabinet Construction	Main Body	1.2 mm (0.05") 18-gauge electro-galvanized steel with white oven baked epoxy-polyester powder-coated finish
	Work Zone	1.2 mm (0.05") 18-gauge stainless steel, grade 304, with 4B finish
	Side Walls	LVS-LHS: 1.2 mm (0.05") 18 gauge stainless steel, grade 304, with 4B finish
		LVG-LHG: UV absorbing tempered glass, 5 mm (0.2"), colorless and transparent
Average Airflow Velocity		0.45 m/s (90 fpm) at initial set point
ULPA Filter Typical Efficiency		>99.999% at particle size between 0.1 to 0.3 μm
Electrical		Power Rating 8: 220-240 VAC, 50/60 Hz, 1 phase

	Power Rating 9: 110-130 VAC, 50/60 Hz, 1 phase
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Specifications are subject to change without notice.

**Noise reading in open field condition/ anechoic chamber. Noise reading in normal room varies by room size, layout, and background noise, but may reach roughly 3 dBA above these values.*

***Cabinet only, excludes optional stand.*

APPENDIX

APPENDIX A: LOG RECORD

Cabinet : _____
Serial Number: _____
Person in Charge: _____

1. This log record should be used by the operator to record any new agent/virus/bacteria/germs that has been introduced to the cabinet during its operation, problems encountered, etc.
2. Any decontamination procedure performed by either the user or the technician should be recorded down as well.

Please also record any major maintenance procedure performed by the service technician, for example: parts replacement, recertification, etc.

DATE	EVENT	USER SIGNATURE	SUPERVISOR SIGNATURE

In case of emergencies, please call:

Name: _____
Cell Phone Number: _____
E-mail: _____

APPENDIX B: FURTHER INFORMATION AND REFERENCE MATERIALS

The following documents are available for download at www.escoglobal.com, in the Resources area. These documents have been selected as pertinent to this product and we would recommend that anyone operating the cabinet familiarize themselves with this information.

1. Introduction to Laminar Flow Cabinets
2. Ultraviolet Lamps in Laminar Flow and Biological Safety Cabinets

FILTRATION TECHNOLOGY

- *Introduction to Contamination Control and Cleanroom Technology*. 2000. Matt Ramstorp. Wiley-VCH. Weinheim.
- *Cleanroom Technology. Fundamentals of Design, Testing, and Operation*. 2001. Whyte.W. Wiley, West Sussex, England.
- *IEST-RP-CC001: HEPA and ULPA Filters*.2002. Institute of Environmental Sciences and Technology, Illinois, USA.
- *IEST-RP-CC007: Testing ULPA Filters*.2002. Institute of Environmental Sciences and Technology, Illinois, USA.
- *IEST-RP-CC021: Testing HEPA and ULPA Filter Media*.2002. Institute of Environmental Sciences and Technology, Illinois, USA.
- *IEST-RP-CC034: HEPA and ULPA Filter Leak Tests*.2002. Institute of Environmental Sciences and Technology, Illinois, USA.
- *BS EN 13091: Biotechnology. Performance Criteria for Filter Elements and Filtration Assemblies*. 2000. British Standards, UK.

CLEAN AIR TECHNOLOGY

- *ISO 14644: Cleanrooms and Controlled Environments*.2000. International Organization for Standardization, Switzerland.
- *IEST-G-CC1001: Counting Airborne Particles for Classification and Monitoring of Cleanrooms and Clean Zones*.1999. Institute of Environmental Sciences and Technology, Illinois, USA.
- *IEST-G-CC1002: Determination of the Concentration of Airborne Ultrafine Particles*.1999. Institute of Environmental Sciences and Technology, Illinois, USA.
- *IEST-G-CC1003: Measurement of Airborne Macro Particles*.1999. Institute of Environmental Sciences and Technology, Illinois, USA.
- *IEST-G-CC1004: Sequential Sampling Plan for Use in Classification of the Particulate Cleanliness of Air in Cleanrooms and Clean Zones*.1999. Institute of Environmental Sciences and Technology, Illinois, USA.

WEBSITES

Esco Micro Pte. Ltd.

www.escoglobal.com

International Organization for Standardization (ISO)

www.iso.org

Institute of Environmental Sciences and Technology (IEST)

www.iest.org

NSF International

www.nsf.org

Occupational Safety and Health Administration, USA

www.osha.gov

American Biological Safety Association

www.absa.org